



NATIONAL EDUCATION STRATEGY

FOR IRAQ

2022–2031

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WITH SUPPORT FROM



Foreword

by H.E. the Prime Minister of the Republic of Iraq

This government, through its ambitious approach, aspires to create an educational system consistent with the approved international standards and benefiting from the best practices and global experiences while preserving the authentic Iraqi cultural values and human values of tolerance and coexistence. Through this system, we seek to achieve a number of goals, most notably providing sustainable, innovative and high-quality education and learning opportunities, as well as promoting the building of a knowledge-based society and responding to the strategic needs of the country and the need of the labor market.

The government platform also looks forward to support our educational institutions in order to strengthen community partnership and build systems that promote scientific and creative thinking and ensure the quality of education outputs that contribute to building a knowledge-based and diversified economy that is in line with the requirements of sustainable development plans.

The quality education that we aspire to will constitute the mainstay and the means to achieve the sustainable development goals endorsed by the United Nations until 2030 in order to prepare a generation of effective young talents who are able to take the lead towards comprehensive national development. The government's task will focus on directly supervising the implementation of this national strategy as stated in the 14th item of the government platform for this term, through the sectoral ministries and through a central committee under our supervision and sectoral work teams that are keen to implement the programs and projects adopted by this strategy. The government is also keen to provide the necessary budget to implement the provisions of this strategy, with the need to continue the process of follow-up and evaluation of the implementation items of this strategy in order to find out what has been achieved and address the defects that taint it during the implementation process, while working to overcome the difficulties and challenges it faces through using effective and practical mechanisms to achieve the desired outcomes of this strategy and in accordance with the plan drawn up.

We thank those in charge of preparing this strategy from the participating sectoral ministries, experts and supporting international organizations. In turn, we look forward to more giving and sincerity in work.

May God bless all the benevolent efforts that have accomplished this noble and ambitious work.

And from God is guidance and success.

His Excellency Engineer Mohamed Shiaa Al-Sudani
Prime Minister of the Republic of Iraq

Acknowledgement

by the Supervising Committee for the Development of the Iraq National Education Strategy

The Supreme Committee supervising the development of this strategy, the Prime Ministers Advisory Commission and the Higher Committee for Education Development in Iraq, extend their gratitude and appreciations to everyone who contributed to the completion of this blessed work, starting from the Prime Minister's Office to the relevant ministries (Education, Higher Education and Scientific Research, Planning, and Finance), as well as the working groups and the technical committees emanating from the Supreme Committee. Thanks are also extended to the international organizations that contributed and funded this achievement. In particular, UNESCO, the International Institute for Educational Planning, the World Bank and UNICEF, as well as the international experts who contributed to the transfer of international best practices.

We pray to God that our national government will be able to implement its ambitious program, which in its fourteenth theme clearly emphasized the direct supervision of the implementation of this strategy.

Committee of the Prime Minister Decree No 343
Supervising the Development of INES Strategy

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Abbreviations

GDP	Gross domestic product
GER	Gross enrollment rate
IMF	International Monetary Fund
INES	Iraq National Education Strategy
ISIS	Islamic State in Syria and Iraq
KPI	Key performance indicator
M&E	Monitoring and evaluation
MoE	Ministry of Education
MoHESR	Ministry of Higher Education and Scientific Research
MoLSA	Ministry of Labor and Social Affairs
MoP	Ministry of Planning
SDG	Sustainable Development Goal
TVET	Technical and vocational education and training
TVQF	Technical and Vocational Qualifications Framework
UNESCO	United Nations Educational, Scientific, and Cultural Organization
UNICEF	United Nations Children's Fund
WB	World Bank

Executive Summary

This document, the Iraq National Education Strategy (INES) 2022-2031, reflects on current education sector challenges in Iraq and presents a way forward for the education sector in Iraq.

Chapter 1 introduces the purpose of the INES and describes its development process. Chapter 2 provides information on the Iraqi and education sector context. Chapter 3 provides an analysis of the education sector. Chapter 4 outlines the strategy's vision, mission, and core values. Chapter 5 describes sector priorities and key programs. Chapter 6 describes the cost and financing framework for the INES. Chapter 7 outlines the monitoring and evaluation and coordination and implementation framework of the INES.

The INES 2022-2031 forms part of a government-wide effort to improve the quality and efficiency of public services for all Iraqis. To contribute to this transformation, an ambitious vision has been developed for the education sector in Iraq:

To aspire to an education system that is committed to values; provides sustainable, innovative, high-quality learning opportunities for all; promotes building a knowledge society; and conforms to global standards.

To this end, the mission of the INES is to:

Strengthen educational institutions for community partnership and build systems that promote scientific and creative thinking, ensure the quality of outputs to meet the needs of the labor market, and help build a diversified knowledge economy in line with the requirements of sustainable development.

The INES has been formulated considering Iraq's overall vision of the socioeconomic development of the country; education laws, including provision of compulsory, fee-free basic education; and international commitments in education according to Sustainable Development Goal 4. It emphasizes equitable access to high-quality, inclusive education and provision of skills required in a 21st century knowledge society to enable youth to contribute to the sustainable economic and social development of Iraq.

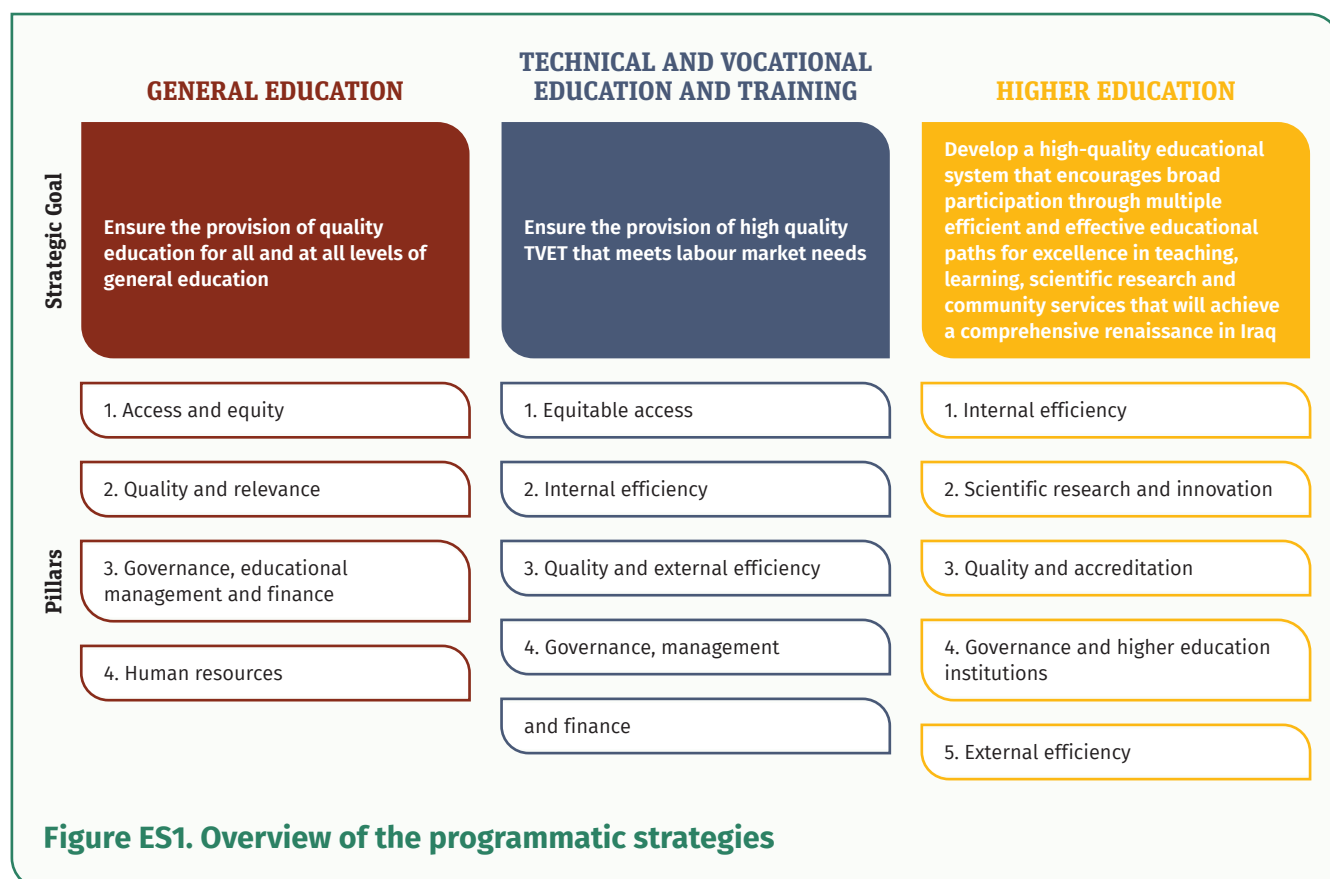
To achieve INES's vision, the objective of the education sector, led by the Ministry of Education (MoE) and the Ministry of Higher Education and Scientific Research (MoHESR), in close collaboration with line ministries, communities, and external partners, is to work toward achievement of strategic goals and related strategies—organized in pillars and specific

to general education, technical and vocational education and training, and higher education (figure ES1).

Key themes across the three sub-sectors are:

- ensuring equitable access to education and training
- improving quality and relevance of education and training and strengthening linkages with the labor market
- improving governance of education, including human resource development, educational planning, and management
- improving human resources and financial management for better service delivery and greater value for money

The identified strategies build upon and are consistent with recent policy, planning, and programming decisions that the MoE and the MoHESR have made. Within the overarching sub-sector strategic goals, the INES 2021-2031 emphasizes various cross-cutting factors, including gender, rural/urban divides, conflict, insecurity, and poverty. Proposed programs address the vulnerabilities of children and youth. The INES proposes establishing programs to address the effects of past and ongoing



crises, including prolonged school closures due to COVID-19, displacement, destruction of education infrastructure, and learners' psychosocial needs. It also identifies concrete avenues for increasing the resilience of the education system by institutionalizing crisis-sensitive planning and response.

The INES 2022-2031 is based on the following assumptions and projections:

Gradual improvement of education system performances dependent on implementation of key sector reforms

It was projected that critical reforms within the INES 2022-2031 would be initiated in 2022, with gradual improvement in education parameters over the first 5 years of the strategy. The assumption is that regulatory and policy frameworks must be implemented before improvements in such areas as enrollment, repetition, and retention can occur. Regulations governing teacher recruitment, efficient teacher allocation, and efficient use of expenditures on education infrastructure should be addressed in the initial phase of the strategy (2022-2025).

Macroeconomic projections

Three parameters can be used to estimate the level of funds available for the education sector during implementation of INES 2022-2031: the evolution of Iraq's gross domestic product (GDP) (with an average increase of 3.3 percent up to 2031); share of government expenditures as a share of GDP (which is expected to remain at 35 percent), and share of education as a percentage of total government spending (which is expected to increase from almost 10 percent to 16 percent by 2031).

Based on these assumptions, public spending for education is expected to increase from approximately 11 trillion dinars in 2019 to 19 trillion dinars in 2031 (in 2019 constant dinars), accounting for 5.6 percent of GDP.

Key education targets

The INES 2022-2031 defines several targets:

- Increase school readiness of pre-primary-aged children by increasing gross enrollment rate (GER) from 10.7 percent in 2018/19 to 30 percent

by 2031, including by increasing the share of private provision of kindergarten to 25 percent

- Increase completion of primary education to 100 percent by 2025 from approximately 83 percent in 2018/19
- Increase completion of intermediate education to 100 percent by 2028 from approximately 62 percent in 2018/19
- Increase GER for preparatory education (academic) from 47 percent in 2018/19 to 68 percent by 2031 for boys and from 46 percent to 66 percent for girls
- Expand vocational education at the preparatory level to so that more students complete grade 12 or access technical training (GER from 7 percent/3 percent in 2018/19 to 14 percent/13 percent by 2031) so that they are prepared to enter the labor market, considering the needs of the national economy
- Ensure controlled expansion of higher education with 82 percent of academic preparatory graduates (compared with 100 percent in 2018/19) and 50 percent of vocational preparatory graduates projected to enter universities and colleges by 2031 while keeping current levels of private provision at 25 percent
- Improve teaching and supervision conditions by decreasing kindergarten and primary education class sizes (from 64 to 30 for kindergarten and 40 to 35 for primary), keeping class sizes at 35 at the secondary level, and increasing class sizes in vocational Schools to 30;
- Increase the number of students participating in non-formal education programs from approximately 70,000 to 300,000; and
- Improve internal inefficiencies by regulating repetition, reducing the teacher-student ratio, applying new teacher allocation regulations, and increasing the number of effective teaching hours

Funding is also provided to establish several programs to reduce disparities in access to education in terms of gender, disability, displacement status, location, and poverty and improving the quality of education:

- Gradual increase in provision of school meals to all public primary school students by 2030 and to the most vulnerable children attending pre-primary education (25 percent of all children attending kindergarten by 2031)
- Provision of school grants as part of the school-based development program to all public

kindergartens and primary and secondary schools from 2022 on¹

- Provision of textbooks in the primary (three per student for grades 1-3, 7 for grades 4-6), intermediate (five per student), and preparatory (five per student) levels and a set of learning materials for vocational education
- Provision of scholarships to 2 percent of all students in preparatory and 5 percent in higher education

Cost of the INES 2022-2031 and funding gap

Based on these assumptions, the total cost of the INES 2022-2031 is approximately 140 trillion dinars (in 2019 constant values) for the duration of implementation of the strategy. The substantial increase in nominal terms takes place in pre-primary and vocational education, reflecting the planned expansion of these sub-sectors. Although operational expenditures

remain the largest portion of education spending, investment expenditures are expected to become an increasingly important part, increasing from 7 percent of total education expenditures in 2022 to 10 percent in 2031, which is in line with current infrastructure needs but also the expected expansion of the system in the coming years.

Because of the unfavorable economic environment that the COVID-19 pandemic has caused, combined with longstanding inefficiencies and the increased demand for resources to accommodate a rapidly increasing number of children and youth, the funding gap is almost 12 trillion dinars (2019 constant prices). By 2026, the funding gap for operational expenditures will be closed, and that for investment expenditures will decrease from 2023 on, although it will remain significant until the end of the strategy period, with 28 percent of total investment expenditures not funded. Major efforts are needed to close the investment gap.

¹ School grants are projected at a unit cost per student of 2,000 dinars for primary and secondary schools and 6,000 dinars for vocational schools. The unit cost is based on number of students per school. School grants do not replace other operational non-salary expenses such as textbooks, in-service teacher training, and school meals.

Introduction

This document, the Iraq National Education Strategy (INES) 2022-2031, outlines the vision of the government of Iraq for its education sector for the next 9 years.

The education sector is one of the three basic pillars of human development (along with health and the economy) and a fundamental pillar of the knowledge economy. Therefore, the current global competition is an educational one set within the information technology revolution, and countries in which education is prioritized and whose economies rely on knowledge and innovation in the production process achieve rapid advances in various fields of comprehensive development. They bring their societies out of ignorance, poverty, backwardness, and corruption and into the ranks of developed countries. Although education is the fourth of the 17 Sustainable Development Goals (SDGs) that the United Nations General Assembly adopted to be achieved by 2030, the other goals can be attained only if this goal is achieved. "Human development through education" also ranked first of five basic pillars in Iraq's priorities for the 2030 SDGs. (The others are good governance, diversified economy, safe society, and a sustainable environment.

Proceeding from the belief of the Government of Iraq that advancing the education sector and improving its level of ambition and quality requires strategic planning so that it can be invested as a tool in achieving comprehensive development. The government developed the first national strategy for education beginning in 2009 with the assistance of the United Nations Educational, Scientific, and Cultural Organization (UNESCO); the United Nations Children's Fund (UNICEF); and the World Bank. It took approximately 3 years to complete and was approved by Cabinet Resolution No. 323 of 2012 and launched in December 2012. The Prime Minister's office formed an executive secretariat for the strategy in November 2013, which prepared a work plan and a timetable for implementing the strategy's projects for 2012 to 2022. The strategy included 178 projects, including 63 for the Ministry of Education (MoE) to implement, 115 for the Ministry of Higher Education and Scientific Research (MoHESR) and 11 for the international organizations. Because of the security situation and financial conditions in Iraq in 2014, the executive secretariat's work was suspended, and the MoE and MoHESR formed their own technical working teams to implement strategic programs and projects.

After more than half of the strategy period had passed, and in response to a proposal from the MoHESR and an initiative of the Prime Minister's Advisory Commission, the Prime Minister's office

issued Order 343 in 2018 to form a committee headed by the Advisory Commission and with the membership of representatives from the MoE, the MoHESR, the Ministry of Planning (MoP), and the Ministry of Finance. The committee was tasked with reviewing and evaluating previous strategy projects, with a focus on challenges faced in their implementation, and preparing a new 10-year strategy, the INES 2022-2031.

The INES takes into account the government's goals of strengthening national identity and decentralization, advancing the private sector, addressing unemployment, linking curricula and programs to the needs of the labor market, strengthening and developing the role of private education, keeping pace with developments in information and communications technology, addressing the remnants of the terrorist Islamic State in Iraq and Syria (ISIS) and its destructive ideas educationally and intellectually, and improving Iraq's financial condition.

The Supreme Committee has taken a series of steps to accomplish the tasks assigned to it:

- Technical working groups were formed in three educational sectors (pre-university, higher education and scientific research, vocational and technical) in coordination with the relevant ministries, a technical team from the MoP, and a committee to draft the strategy themes.
- Contact and discussions with international organizations resulted in financial and technical support from the World Bank, UNESCO, and UNICEF, and a concept note was concluded in which the goals and principles, the general framework for the strategy structure, the roles and responsibilities, and the timetable for achieving the strategic axes were defined. UNESCO was tasked with coordinating between the various parties.
- World Bank assigned a team of experts for each sector under the supervision of the International Institute for Educational Planning to provide technical advice to the sectoral teams.

Four principles were adhered to in preparation of the INES:

- The national authorities will exercise leadership in preparation of the strategy and have the main responsibility for its successful completion. Coordination for this work will reside with the Higher Committee, and teams of staff from the various ministries will perform the technical

work, supported by national and international experts.

- To ensure that the INES is acceptable to everyone, the preparation process is designed in a participatory way, to ensure that stakeholders are consulted and their views considered.
- Successful implementation of this participatory process requires clear definition of the responsibilities and roles of all stakeholders, design of effective consultative mechanisms, and recognition of where final decision-making authority lies.
- The technical support that international and national experts provide in the process of strategy preparation is conceived as a capacity development activity, in a spirit of mutual exchange and collaboration. Its final purpose is to increase the autonomy of the national authorities in designing and implementing a national education strategy.

The mechanism of action adopted in preparation of the strategy is summarized as follows:

1. After agreeing on the roadmap, principles, and content of the strategy with the ministries involved and international organizations supporting preparation of the INES, an education situation analysis was conducted to identify the status of education indicators as a basis for developing strategies to address identified challenges and enhance opportunities.
2. An online workshop was conducted in September 2020 to present and discuss results from the education situation analysis and initiate preparation of problem and objective trees for each sector that would become the basis for policy discussions.
3. Between October 2020 and January 2021, each sector group held regular discussions to identify policy directions and priority programs.
4. The sector groups provided data and prepared a draft of the pillars for each sector to achieve the vision of the concerned ministry.
5. In February 2021, a face-to-face workshop brought together technical staff and policy makers from the MoE, MoHESR, and MoP with partners to discuss identified policy directions and priority programs and strengthen alignment between general education, technical and vocational education and training (TVET), and higher education.
6. The Iraq Simulation Model supported continuing policy discussions to set targets and cost the INES. A policy note summarizing policy targets for each sector and associated costs was shared in July 2021 and formed the basis for final discussions

held during a face-to-face workshop held from September 12-16, 2021, in Erbil.

7. The in-person workshop in Erbil took place in the presence of the Supreme Committee and representatives of the MoE, MoHESR, MoP, international organizations, and the Drafting Committee. The workshop allowed for final policy discussions and finalization of the Iraq Simulation Model. The attending ministries adopted Scenario 3, which outlines an optimistic yet realistic vision for the education sector, as the final scenario to be presented in the INES. Discussions on the monitoring and evaluation (M&E) framework helped finalize a results matrix, which will allow education stakeholders to monitor achievements and progress made toward targets.

Important lessons were drawn from the preparation process, especially from the review of the outgoing INES, which can be summarized as follows:

- Political, security, and administrative stability is one of the most important requirements that must be met to implement future strategic programs and projects.
- Proper planning of resources and financing is crucial to determine the extent of the possibility of implementation. Therefore, reliance on government funding alone may limit the success of any subsequent strategy because of its instability; sources of funding must be diverse, such as international grants and self-financing of some projects, and operating budgets must be relied on instead of investment as much as possible.
- It is appropriate for future strategies to include planning scenarios and contingency plans to be implemented in the event of an emergency or unstable situation, which requires that the strategies be flexible.
- It is preferable that there be a cross-ministerial working team (executive management) specialized in implementation planning, supervision, and follow-up activities and that coordination with higher government agencies be comprehensive and facilitating to ensure implementation of any future strategy.
- Reconsidering some of the prevailing legislation, laws, and instructions that may stand in the way of proper implementation of the strategy.
- The strategy must be realistic and far from subjective ambitions, and the circumstances of the country and available capabilities must be considered.
- Staff responsible for implementation, supervision, and follow-up on planning, implementation, and control processes must be trained.

Chapter 1

Previous Education and Higher Education Strategy (2012-2022): Evaluation of Achievements and Challenges

The INES (2012-2022) was adopted by the Council of Ministers Resolution No. 323 for 2012 in its fortieth session on 9/11/2012, as a national reference document in developing annual plans of the relevant sectoral ministries and those related to the education and higher education sector. The ministries of Finance and Planning shall take the necessary measures regarding the budgets, plans and projects proposed in the document. Despite the adoption of the strategy in accordance with the above-mentioned resolution, efforts made by the Office of the Deputy Prime Minister and PMAC to allocate the required funds (or even an advancement) to work on the 2013 strategy, did not prove any success. In addition, given the circumstances Iraq has experienced since the drop of oil prices on the one hand, and the allocation of most of the public budget to liberate the areas occupied by ISIS, the implementation of many of the strategy projects was not accomplished.

Below are the most important achievements of the ministries of Education and Higher Education since the launch of the previous strategy until 2019 (table 1.1):

Table 1.1:
The Ministry of Education (MoE), Institutional Context

Completed projects	Projects under study and discussion	Incomplete and under-study projects
1. Ministry of Education Law No. (22) of 2011 2. Literacy Law No. (23) of 2011 3. Private and foreign education regulation No. (5) of 2013.	1. Preparing and applying job description. 2. Activating the compulsory Education Law and extending it to include the intermediate stage. 3. Activating the Parents and Teachers Councils Law. 4. The Ministry of Education system. 5. Public examination system.	1. School Shops Law. 2. Kindergarten regulation. 3. Regulation on granting a certificate of educational leadership and administration. 4. Development of electronic management in all areas of management.

Infrastructure achievements

- Constructing 3751 school buildings out of 25307 (15%). Also, 719 school annexes with various classes were built.
- Repairing 241 school buildings out of 5910 (4%).
- Providing 203 laboratories and workshops out of 565 (36%).
- Deficiency of 8093 school buildings.
- 50% of school buildings in Iraq need rehabilitation or reconstruction.
- Schools vary from 89% governmental to 11% private.
- Student/class rates reached 37, 41 and 37 in primary, middle and preparatory schools respectively, while the global indicator is 30 students/class.

Accessibility (enrollment, equality and efficiency) achievements

- The net enrollment rate for kindergartens remained at 11%.
- The net enrollment rate for the primary stage increased from 89% to 94%.
- The net enrollment rate for the intermediate stage increased from 36% to 55%.
- The net enrollment rate for the preparatory stage increased from 19% to 30%.
- The net enrollment rate for vocational education decreased from 1.5% to 1.1%.
- The number of classes within schools covered by special education increased from 1396 to 1547, and the number of students increased from 13005 to 15683.
- Enrollment rates varied according to gender, and were in the primary stage (96% for boys versus 93% for girls), in the Intermediate stage (56% for

boys versus 54% for girls), and at the preparatory stage (26% for boys versus 30% for girls).

- The internal efficiency rates were 76% for the primary stage, 51% for the intermediate stage, 60% for the preparatory stage/scientific path, and 65% for the preparatory stage/literary path.
- Repetition was 13.9% in the primary stage, 27% in the intermediate stage, and 20.6% in the preparatory stage. All rates were far from achieving the national target (3%).

Quality improvements

- A total quality management department has been created in the Ministry's management office and the departments of the General Directorates of Education. Specific criteria and controls have been adopted for schools' self-evaluation, performance appraisal for teachers, supervisors, and school administrators, and setting development plans for schools.
- All curricula are uploaded in the website of the General Directorate of Curricula and are updated annually.
- A complete school package with all its guides were produced for the secondary stage.
- 11543 courses were held to develop the capabilities of teachers and administrators, with 384880 participants, besides other courses and workshops which were held by international organizations, as 7788 teachers were trained on the English language curricula, as well as other courses to obtain an (ICDL) certificate.
- 19368 teachers obtained a bachelor's degree from the Open Educational College and the Colleges of Education.

- The number of schools covered by educational mentoring increased from 4359 to 6638 and the number of mentors increased from 4435 to 6796.
- Computerization in classrooms was done for 3,219 schools (the strategy mentioned 3000 schools).
- The electronic notebook project, the talented tests and acceleration projects were implemented.
- The number of educational supervisors in various specializations increased from 2697 to 4696.

Funding and Spending increases

- The total revenue generated from covering resources amounted to 7.830.912.234 dinars.
- 150,000 dinars were granted per month to the ministry's employees, and suspension on their salaries was lifted.

Scientific Research achievements

- 13327 books, theses and dissertations were indexed electronically, 48 researches were published on the website of the documentary library, and 649 other researches were on their way to publication (see also table 1.2).

The Ministry of Planning, when evaluating what has been achieved from the National Strategy for Education and Higher Education in Iraq (2012-2022), made evaluation comments - about the education sector - about the level of project implementation:

1. There is a large gap in the infrastructure component, as it witnessed a decrease in the rates of building schools to eliminate triple and double shifts, natural growth decline, the demolition and rebuilding of clay dilapidated schools by 16%, and the decrease in the completion rates for the restoration of schools to 4%.
2. There is a gap in the development of scientific training in vocational education, which includes establishing and equipping workshops, as the achievement rate for both is 8% only.
3. Regarding opportunities for efficient education (enrollment and equality), enrollment rates in all stages are still somewhat low, except for the primary and intermediate stages, as they have a remarkable progress in enrollment rates, and there is progress to some extent for the preparatory stage and the kindergarten stage. However, there is a low enrollment rate in some stages of education, including vocational education, special education schools, adolescents' schools, and schools for the talented. Female enrollment rate in public education was acceptable to some extent.
4. Quality - as one of the important components in the strategy - lacked targets in the projects it included, and quantitative and qualitative indicators to measure them.

Table 1.2:
Higher Education and Scientific Research

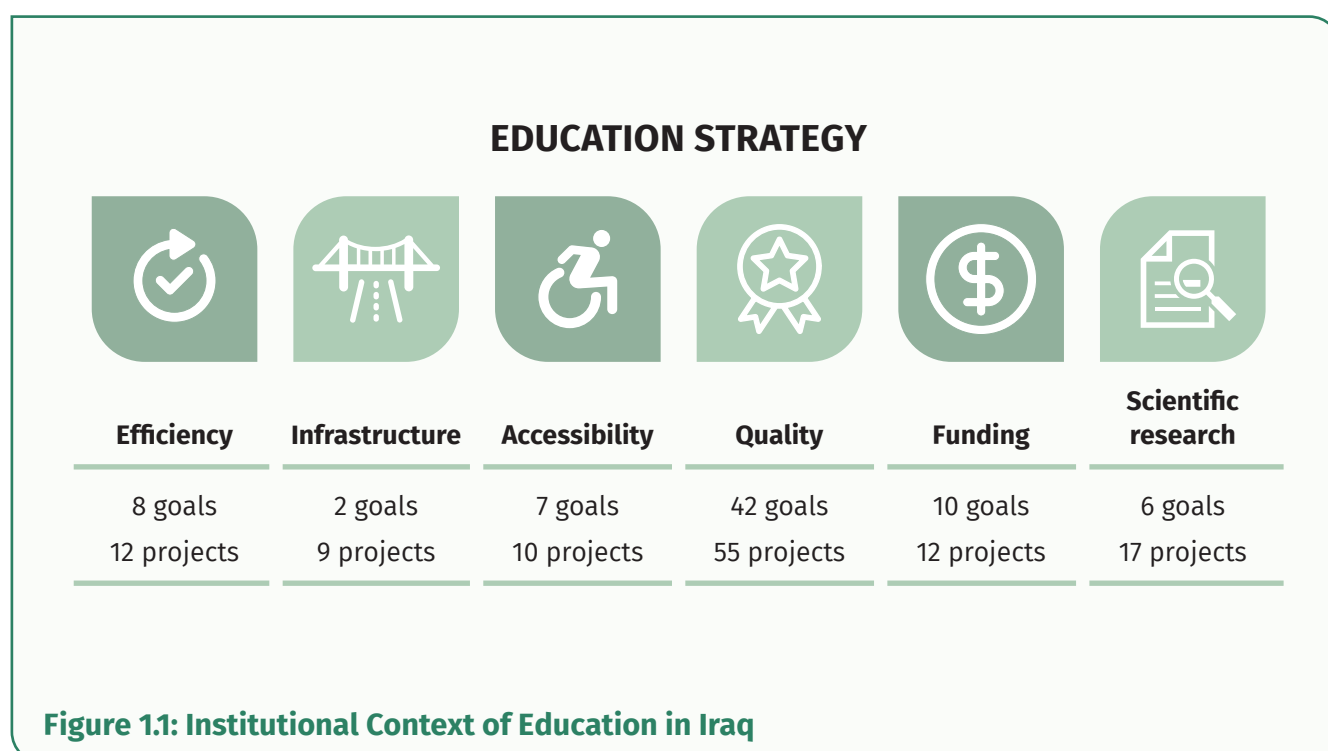
The most important statistical indicators for 2018-2019 were as follows:

NO.	Statistical indicators	Total
1	Number of public universities	35
2	Number of public colleges	369
3	Number of departments in public colleges	1549
4	Number of public technical universities	17
5	Number of public technical institutes	28
6	Number of private universities	14
7	Number of colleges in the private universities	74
8	Number of private university colleges	47

NO.	Statistical indicators	Total
9	Number of departments in private universities and colleges	494
10	Gross enrollment rate in the higher education	19%
11	Number of students enrolled in morning and evening classes – undergraduate study	317379
12	Percentage of females in the students admitted in the undergraduate study	35%
13	Gross enrollment rate	19%
14	Number of students admitted in postgraduate studies	15359
15	Number of published scientific reasearches	12216

The strategy implementation team worked on implementing six main components related to the general context and the institutional context, focusing on legislations, laws and regulations, quality of education, scientific research, and financing the education sector. This required the implementation of 166 projects for the Ministries of Education and Higher Education in all Iraqi governorates. By moving towards developing advanced curricula that enhance Iraq's scientific and economic role, and finding integration and cooperation between the two ministries, the following outline shows projects and objectives distributed over the six components of the strategy for higher education:

The national strategy included 166 main and sub-projects and programs, and sub-projects such as construction and rehabilitation projects were among the main projects of 121 programs and projects as mentioned above. They all are of a complex nature, which will achieve 71 strategic goals in total distributed over the six components in the national strategy document. These goals were measured by 120 indicators, most of which were quantitative. In order to ensure a good presentation of the level of implementation of these projects, a number of indicators were uploaded to cover the main aspects of the strategic objectives, which were listed successively according to a special coding for



each components. It should be noted that the rate of achieving the goal will be included, taking into account the number of indicators for each goal.

A number of goals were selected for each component for the purpose of measuring and discussing their achievement through the use of standard indicators that were prepared for each of those

goals. The following table shows implementation rates of projects and programs from zero to 100%, based on the report of the Follow-up Committee for the Implementation of the National Strategy for Education and Higher Education in Iraq/higher education (table 1.3):

Table 1.3:
Implementation rates of projects and programs

#	Implementation percentage	No. of projects/programs	% total projects/programs
1	0	39	23%
2	Less than 20%	20	12%
3	20 – less than 40%	33	20%
4	40 – less than 60%	25	15%
5	60 – less than 80%	17	10%
6	80 – less than 100%	21	13%
7	100%	11	7%
Total		166	100%

There has been perseverant effort to issue laws, instructions and regulations to support the institutional context as follows:

Table 1.4:
Laws issued to support the institutional context of the education sector

#	Legislation	Type	Amended or updated	Year of amendment/ update
1	Law No. (7) of 2012 on supporting the dormitories in the Ministry of Higher Education	Law	Updated	2012
2	Law No. (32) of 2012, the first amendment to University Service Law No. (23) of 2008	Law	Amended	2012
3	Law No. (63) of 2012 on granting students of Iraqi universities and government institutes	Law	Updated	2012
4	Law No. (1) of 2014 Second Amendment Law to University Service Law No. 23 of 2008	Law	Amended	2014
5	Law No. (2) of 2014 The Eighth Amendment Law of the Ministry of Higher Education and Scientific Research Law No. (40) of 1988	Law	Amended	2014
6	Private Higher Education Law No. (25) of 2016	Law	Updated	2014
7	Law No. (15) of 2016 accepting members of the armed forces, internal security forces, and the two national intelligence and national security agencies in university studies	Law	Updated	2014
8	Employment Law No. (67) for the year 2017	Law	Updated	2017
9	Law No. (59) of 2017 for the Employment of Holders of Higher Degrees	Law	Updated	2017
10	Instructions for Academic Promotions in the Ministry of Higher Education No. (167) for the year 2017	Instructions	Updated	2017
11	Instructions of the Iraqi Council for Medical Specializations No. (168) for the year 2017	Instructions	Updated	2017
12	Instructions No. (169) of /2018 The first amendment to the instructions for student discipline in institutions of higher education No. (160) of /2007	Instructions	Amended	2018
13	Instructions No. (70) for the year / 2018 to facilitate the implementation of the scholarship law for students of Iraqi governmental universities and institutes No. (63) for the year / 2012	Instructions	Updated	2018
14	Instructions No. (171) for the year / 2018 The third amendment to the instructions of the Higher Education Fund No. (122) for the year 1999	Instructions	Amended	2018
15	Instructions No. (172) of 2018 facilitating the implementation of the provisions of the Law on the Admission of Members of the Armed Forces, the Internal Security Forces, the Iraqi National Intelligence Service and the National Security Agency to university studies	Instructions	Updated	2018
16	Regulation (3) of Scholarships, Fellowships and Financial Aid No. (3) of 2018	Regulation	Updated	2018

#	Legislation	Type	Amended or updated	Year of amendment/ update
17	Instructions No. (1) of 2019 implementing the provisions of the Employment Law for First Graduates No. (67) of 2017	Instructions	Updated	2019
18	Instructions No. (3) of 2019 facilitating the employment law for holders of higher degrees No. (59) of 2017	Instructions	Updated	2019
19	Law No. (20) of 2021 Principles of Equivalence of Arab and Foreign Academic Certificates and Degrees	Law	Updated	2020
20	Instructions No. (2) the first amendment to the instructions for implementing the Law No. (59) for the year 2017 for the employment of holders of higher degrees	Instructions	Amended	2021
21	Law No. (11) of 2021, the first amendment to the Law of the Internal Departments Support Fund in the Ministry of Higher Education	Law	Amended	2021

When evaluating what has been achieved from the National Strategy for Education and Higher Education in Iraq (2012-2022), the Ministry of Planning made some evaluative comments - on the higher education sector - about the level of project implementation:

1. The Ministry of Higher Education has made great achievements by establishing new university cities, colleges, institutes, and scientific departments, and expanding the capacity as of 2015.
2. There is a big gap in achieving the goal of establishing integrated technical institutes, as the plan was 30 integrated technical institutes, but the completion rate was 3.3%.
3. With regard to efficient education for all (i.e., enrollment, equality and efficiency), the enrollment rates still vary and a greater rate of progress has been achieved for females in higher education out of the total enrollment, whether in rural or urban areas.
4. The quality component – being one of the important components of the strategy - included 20 goals, 10 of which lacked in targets in the goals they have and quantitative indicators to measure them.

As for the rest of the goals, they were identified and measured, but despite the achievements that have been made in this component, there is a discrepancy in the achievement rate.

3. Challenges and Lessons Learned

Despite the serious work to implement the programs and projects of the National Strategy for Education and Higher Education 2012-2022, and the formation

of work teams specialized in M&E, many challenges emerged to constitute an obstacle to the integrated and efficient implementation of the strategy, the most prominent challenges are as follow:

1. The lack of sufficient financial allocations to fund projects in government budgets for many reasons, the most important of which is the decline in government revenues due to the sharp drop in oil prices.
2. The difficult security conditions due to ISIS terrorist occupation of a number of Iraqi provinces in 2014, which led to the suspension of most of the projects and directing the available resources to manage the security aspects and relief for the displaced.
3. Weak political and administrative stability, which led to a continuous change in the senior leadership of the two ministries, and then negatively affected developing government plans and priorities according to the emerging circumstances.
4. The great expansion of programs and projects, some of which were very ambitious and require stable economic and security conditions, and very large funding.
5. Weak coordination between the two ministries and some other government agencies, which led to the inability to implement some joint programs and projects.
6. The novelty of the experience, as it is the first comprehensive strategy at the national level in the field of education.
7. The administrative and legal constraints that formed a complex routine which prevented the proper implementation of some projects.

Chapter 2

Education sector and overall context in Iraq

The education system in Iraq was long viewed as a model of education in Arab countries. Today, Iraq faces numerous external and internal challenges in rebuilding its education system. This chapter outlines the context in which the education sector in Iraq exists.

2.1 Geographic Framework

The 19 governorates of Iraq and the autonomous region of Kurdistan constitute the Republic of Iraq. The Arab Gulf, Kuwait, and Saudi Arabia border Iraq in the south, Iran in the east, Turkey in the north, and Jordan and Syria in the west. Topographically, Iraq is divided into four regions: the alluvial plains of Mesopotamia and the Tigris and Euphrates rivers, the mountains in the north and east, and the desert in the south and west, which represents more than 40 percent of the country.

2.2 Constitutional Framework (Legislative Context)

After the transition of the political regime in 2003, the Iraqi Constitution of 2005 was issued. It placed education at the heart of development in Iraq and established it as a basic factor for the progress of society. Article 16 of this constitution stipulates that “equal opportunities are a guaranteed right for all Iraqis, and the State shall guarantee taking the necessary procedures to achieve this.” Article 29 confirms the right of children over their parents to upbringing, care, and education and the prohibition of all forms of violence and abuse in the family and school. Article 34 states that “Education is a basic factor for the progress of society and a right guaranteed by the State and shall be mandatory at the primary stage. The State shall guarantee combating illiteracy, and free education in its various stages shall be a right for all Iraqis. The State shall encourage scientific research for peaceful purposes in a manner that serves humanity, foster excellence, creativity, innovation and various aspects of genius, and private education shall be guaranteed and regulated by law.”

This constitutional framework establishes a strong foundation for the education and higher education sectors on which plans and strategies can be developed. Iraq’s Vision for Sustainable Development (2030) established that building a good-quality, comprehensive education system is a national priority for building human beings (Ministry of Planning 2019).

2.3 Economic Framework

Iraq’s economy is not well diversified, with oil providing about half of its GDP and more than 90 percent of government revenue. Iraq depends greatly on imports of basic commodities. The economy stagnated as a result of the war with ISIS in 2014, the increase in military spending, and the drop in oil prices, which limited job opportunities.

Less than half of Iraq’s population is economically active. The public sector, the largest employer, employs approximately 4 million people and accounts for approximately 43 percent of all job opportunities. The government controls mining, services, and banking and has a predominant role in construction and manufacturing and in the purchase, sale, and import of some commodities. Although the role of the private sector in the economy is underdeveloped and primarily informal, it is not considered to be an attractive employment option because it does not provide workers with privileges and protection. According to 2019 data, 57.8 percent of all employees in Iraq work in the service sector, compared with 19 percent in the agricultural sector. Figure 2.1 shows the estimated distribution of employment in Iraq according to economic sector from 2009 to 2019.

Unemployment is 8.9 percent overall and 16.6 percent among youth aged 15 to 24 (UNDP 2020). The construction, hospitality, and agricultural sectors employ the largest proportion of young people.

The Iraqi government approved the 2018-2022 five-year plan for national development with a view to promoting the economic reality, increasing the growth rate to 7 percent, achieving economic reform, providing an enabling environment for investment, and enhancing the role of the private sector to address the most prominent economic challenges facing Iraq. Which include deterioration of the investment climate and the imbalance in the production structure and balance of trade and the structure of the public budget, the increase in the public debt, the underdevelopment of the banking system, the expansion of the disorganized economic sector, and the limited role of the private sector. The plan depends on oil revenues as the most important financial resource for the national economy, accounting for 84.1 percent of total projected revenue.

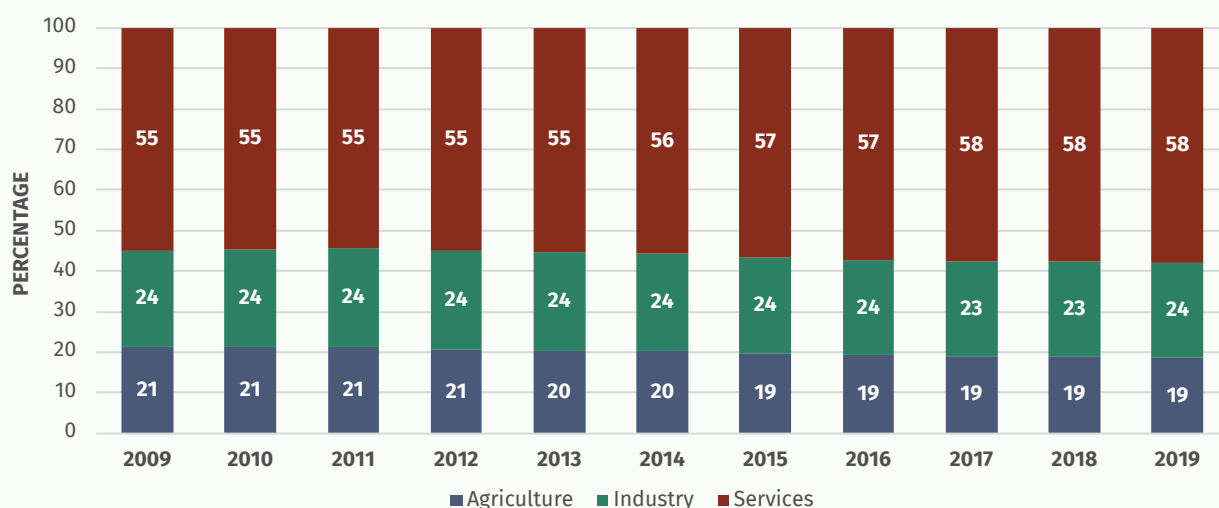


Figure 2.1. Share (percentage) of Total Employment per Sector, 2009-2019

Source: [Statista](#)

Between 2019 and 2020, GDP at current prices declined from 277.9 trillion dinars to 198.8 trillion dinars, and average per capita GDP at current prices for the same period declined from 5.7 million dinars to 5.0 million dinars as a result of the economic crisis, the spread of the COVID-19 pandemic, and the drop in global oil prices.

2.4 Population context

The last available population census for Iraq was in 1997. Population figures used in this document are based on projections that do not account for conflict-related deaths and population displacement inside and outside the country. The population of Iraq was estimated to be 40.2 million² in 2020. Approximately 69 percent of the population lives in urban areas. With annual population growth of 2.5 percent for 2015 to 2020, Iraq's population is young, with 38 percent younger than 15 (UN DESA 2020). The male : female ratio is 1.02 according to 2018 estimates. Population distribution rates vary between governorates, with the largest portion of the population living in Baghdad, the capital (21

percent), followed by Nineveh (10 percent) and Basra (7.6 percent). Muthanna Governorate has the lowest percentage of the population (2 percent).

The total population of Iraq is expected to reach more than 51 million in 2030. The percentage of the working population will rise, and the dependency ratio will decrease from 156 in 2020 to 134 in 2030. Moreover, the number of people aged 4 to 24 (general and university education age) will increase by 7 million (from about 18 million in 2017 to 25 million in 2030), accounting for 49 percent of the total population (UN DESA 2020). This constitutes a major challenge for planners and decision makers in the education, higher education, and scientific research sectors because of the great pressure that this many people will put on the educational system and the associated infrastructure and human and financial resources needed to provide educational services of the desired quality (Figure 1.2).

2.5 Social Context

Iraq ranks 120th in the Human Development Index (0.689), with 8.6 percent of the population living in multidimensional poverty (UNDP 2020). The literacy rate in Iraq is 43.7 percent. The average number of

² The population data shown are projections from the MoP. The United Nations Department of Economic and Social Affairs estimated 39.3 million for 2019 (<https://population.un.org/wpp/>).

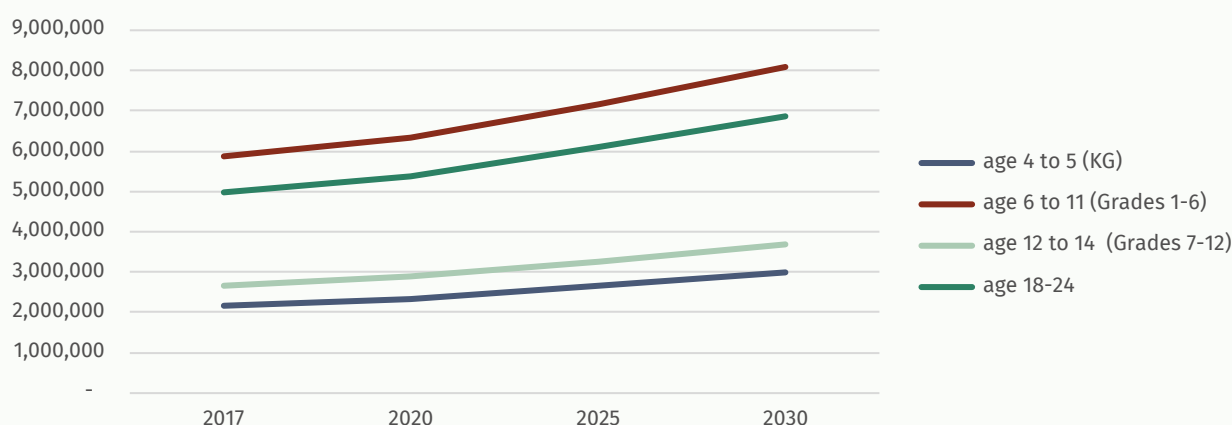


Figure 1.2. Current and Projected School-Aged Population According to Education Level, 2017-2030

Source: Ministry of Planning.

years of education is 7.3 (6 for girls, 8.6 for boys), lower than in other countries in the region (Figure 2.2).

Only 12.4 percent of women aged 15 and older are economically active (UNDP 2020), and a significant proportion of them work in public sector services such as education, social work, and health. The agricultural sector is also a major employer of women. Women's participation in work is expected to rise as a result of the increased support of youth for women's right to work.

Child labor in Iraq is considered informal work, is completely unregulated, and is often defined as occasional in war, displacement, and poverty,

especially since the ISIS insurgency in 2014. Child labor is a human rights problem across the country. The sixth version of the Multiple Indicator Cluster Survey in 2018 showed that child labor was increasing in Babylon, Erbil, Kirkuk, and Maysan and that it is high in boys and rural and out-of-school students. It is lower in wealthier families and those in which the mother has a higher education level (WFP 2019).

Cultural and social norms influence decisions regarding children's education according to views of the importance of education for girls and boys. Disability is stigmatized, with children with special educational needs having less access to education (UNICEF 2019).

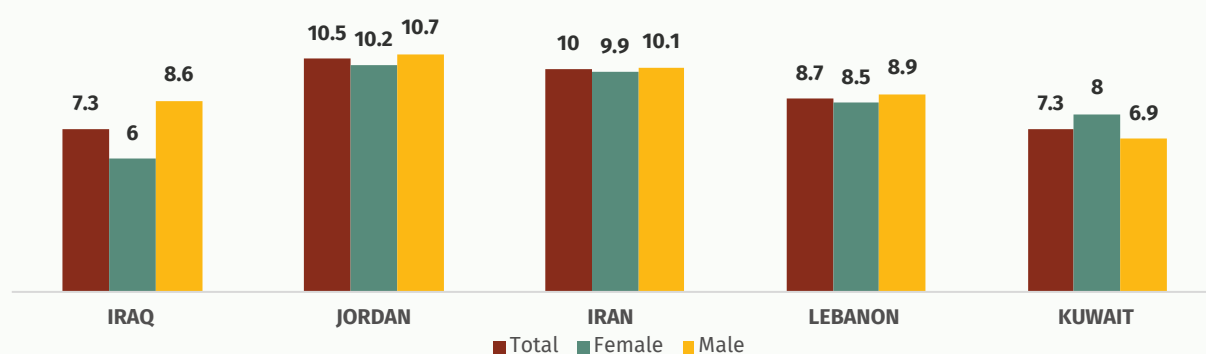


Figure 2.2. Average Years of Education in Comparator Countries, 2018 or Latest Year

Source: UNDP 2020

2.6 Security and Political Framework (Humanitarian Context)

Despite the great political transformation that Iraq experienced after 2003 and adoption of a democratic constitutional system, internal and international circumstances constitute a persistent threat to security and political stability. After the violence that occurred from 2006 to 2008, the situation stabilized, and security and political conditions improved, but ISIS terrorist groups initiated attacks in mid-2014, which removed several provinces from the Iraqi government's control and displaced millions of people, more than half of whom were school aged.

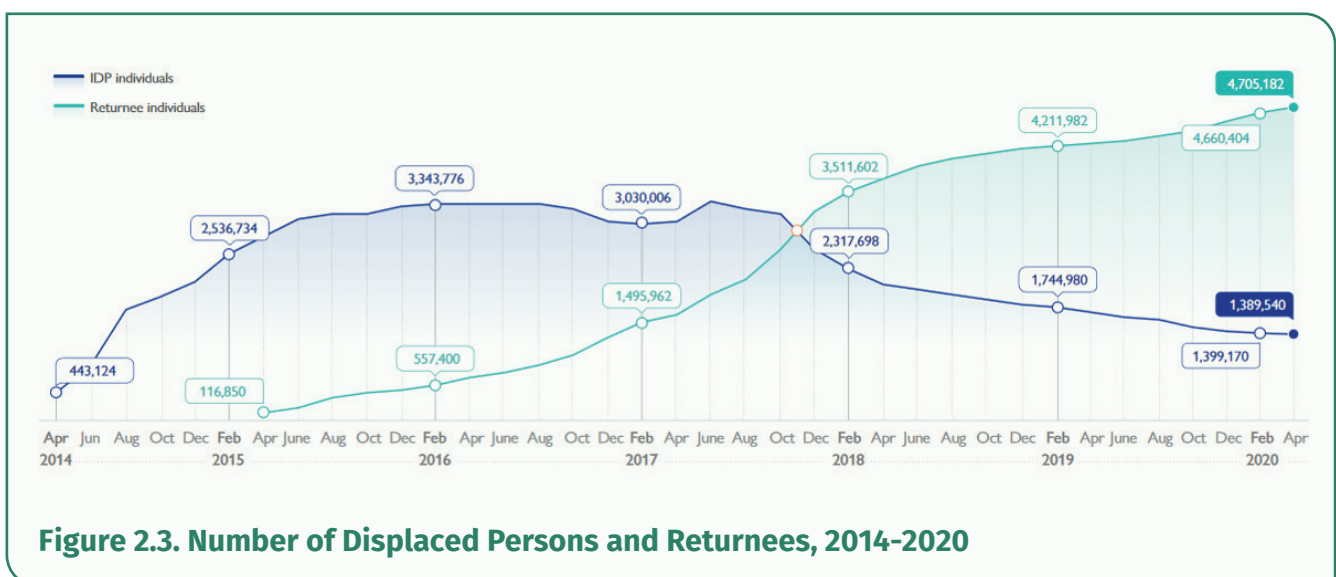
By 2017, the Iraqi government regained control over the governorates that ISIS had controlled, which led to noticeable improvement in security and political reality, but by mid-2019, Iraq once again faced civil unrest and peaceful popular protests calling for improvement in basic services; a reduction in youth unemployment; and reform of the electoral, constitutional, and security sector. Regionally, tensions escalated after the United States withdrew from the Iranian nuclear agreement (Joint Comprehensive Plan of Action) in May 2019 and a series of escalating actions occurred that led to insecurity and violence on Iraqi soil.

2.6.1 Displacement due to Insecurity

The unstable security conditions led to unprecedented displacements and migrations in Iraq, which put great pressure on the state's resources and the progress of the educational process. Data from the High Commission for Human Rights in Iraq, the Ministry of Migration and Displacement, and the UN Displacement Tracking Matrix indicate that there were approximately 7.5 million internally displaced persons in all governorates on July 1, 2018. According to the Displacement Tracking Matrix issued by the International Organization for Migration, there were 2 million internally displaced persons in Iraq in May 2018. The number of displaced persons fell to less than 1 million after the liberation of the areas under ISIS control and the return of most of the people to their homes. As of April 2020, about 1.4 million persons had been internally displaced, and 4.7 million had returned to their place of origin (Figure 2.3). Insecurity, lack of social cohesion and livelihoods, and destroyed or damaged houses have impeded people's ability to return home (UN Office for the Coordination of Humanitarian Affairs 2020).

2.6.2 Impact of Instability on Education

The 2014–17 conflict with ISIS and the subsequent large-scale population displacement have exacerbated the structural challenges facing the educational sector. Half of all school buildings were destroyed or damaged in the governorates that ISIS controlled, and



Source: IOM Displacement Tracking Matrix (March–April 2020).

unexploded ordinance in schools threatens students' and teachers' lives, killing six children in 2018. The war traumatized millions of people, with many students and teachers still managing the trauma of violence, homelessness, and loss. Furthermore, ISIS policies in schools and universities targeted education, national identity, and affiliation to society and the state, which damaged society, especially the concept of citizenship. The Iraq Education Cluster estimates that 775,000 Iraqi children are living inside and outside camps and about 150,000 in host communities and that there are 98,000 Syrian refugee children. Eight governorates host the most children in need of educational services: Anbar, Diyala, Dohuk, Erbil, Kirkuk, Nineveh, Saladdin, and Sulaymaniya (Iraq Education Cluster 2019). All these displaced persons are vulnerable and need educational assistance. Rehabilitation of schools has been one of the priorities of the liberated areas stability restoration program, which the government adopted with the assistance of the United Nations and the support of the international community to restore stability and enable the safe, sustainable return of displaced persons to their cities.

ISIS attacked universities, colleges, and educational institutions, which displaced many students—101,980 from 84 public colleges and educational institutions and three private colleges. The multiple humanitarian crises, which lasted for several years in Iraq, are now changing their direction as the post-conflict environment redirects the humanitarian context into long-term planning, with a focus on the need for permanent solutions.

2.7 Risks and Natural and Health Crises

Iraq faces a variety of risks and natural hazards because of its geographic location, diverse climate, and unpredictable harsh weather conditions, such as heat and irregular rainfall, which can lead to drought and floods. Iraq is particularly susceptible to the effects of climate change, with drought and floods affecting 2.4 percent of the population. Earthquakes are also a problem, especially in the border region between Iran and Iraq. The last earthquake, in 2019, caused considerable damage.

Climate change models show that floods will affect people living 5 meters below sea-level (CIESIN 2013). Basra may be under water by 2050 (New York Times 2019).

2.7.1 COVID-19

The spread of COVID-19 led to a complete lockdown of the country. In April 2020, about 1,500 cases of COVID-19 were reported in the country, although limited testing capacity may have led to an underestimation. Schools, universities, and training institutes closed in March 2020.

The pandemic and its social and economic impact have exposed the population in Iraq to new risks, in addition to existing vulnerabilities, and placed an additional burden on the public sector, which was already vulnerable. Because oil accounts for 92 percent of the country's exports, the drop in oil prices plunged the country into a financial and economic crisis. As the Iraqi government struggled to pay the salaries of 6 million public sector employees, millions of workers in the private and informal sector lost their jobs and livelihoods.

Millions of Iraqis will fall below the poverty line as the epidemic continues. The pandemic has affected 2.3 million internally displaced persons and 4.8 million returnees (REACH 2021), as well as students, whose education and training prolonged school closures have greatly affected. Chapter 3 provides more information on the effects of COVID-19-related measures on education.

2.8 Technical Framework

Rapid population growth, the explosion of knowledge, and the effects of COVID-19 highlighted the need to develop education; use information, communication, and Internet technology in training and education; and provide equipment and train teachers to enhance the role of information technology in providing quality education to contribute to providing the outputs of education and higher education with the skills and capabilities required for the twenty-first century and achieving sustainable development. Scientific research must also be supported by building capacity and increasing expenditures. The rate of expenditure on research and development in Iraq did not exceed 0.04 percent of GDP in 2017, which is lower than average global expenditures (2.23) and less than the rate of expenditure on scientific research in countries such as Japan (3.14), Saudi Arabia (0.82), South Korea (4.23), Turkey (0.88), the United Arab Emirates (0.96), and the United States (2.74).

Chapter 3

Analysis of the Education Sector

This chapter presents key aspects of the education system in Iraq, including its structure and sector performance in pre-primary, primary, secondary, TVET, and higher education. This includes indicators related to schooling, quality and management of education and training, inclusion, and internal and external efficiency. It is a synopsis of the full education situation analysis that was conducted from March to July 2020.

Education data used in this chapter stem from the annual school census prepared by the Central Bureau of Statistics, Directorate of Educational and Social Statistics; the Multiple Cluster Indicator Survey for Iraq 2018; and various humanitarian and development partner reports. The 2017/18 annual statistical abstract is the first full dataset assembled since the ISIS crisis, covering all 19 governorates. Data that were made available after finalization of the education situation analysis are considered in this chapter to the extent possible. Analysis provided in this chapter does not cover the province of Kurdistan, namely the governorates of Erbil, Dohuk, and Sulaymaniyah.

Table 3.1 provides a snapshot of key information on schools, including number of schools, number of teachers, number of students per education level, and school age population projection (aged 4–24) for 2031.

Table 3.1.

Number of Schools, Teachers, and Students Enrolled in the Public Education System, 2017/18, and School-Aged Population Projection (Aged 4–24) for 2031

	Pre-primary (aged 4–5)	Primary (aged 6–11)	Secondary (aged 12–17)	University (aged 18–24)
Schools	719	14,901	6,630	35
Teachers	5,952	271,597	153,506	42,907
Students	202,937	6,197,870	2,910,816	743,825
School-aged population (2031)	2.7 million	7.2 million	6.2 million	3.6 million

Source: CSO 2018.

Iraq is facing a “human capital crisis fueled by a learning crisis” (World Bank 2021). A child in Iraq can expect to complete only 6.9 years of schooling, amounting to 4.0 learning-adjusted years of schooling, by age 18. Prolonged school closures due to the COVID-19 pandemic are likely to reduce learning-adjusted years of schooling even further (World Bank 2021). Education service delivery and education quality are highly inefficient, as explained in this chapter.

3.1 Structure and Key Features of the Education System

The MoE and the MoHESR manage the education system in Iraq. The MoE supervises pre-primary schools (kindergarten but not preschool, which the Ministry of Labor and Social Affairs (MoLSA) supervises), primary schools, secondary academic schools, secondary vocational schools (grades 10–12), fine arts institutes, institutes for the blind and deaf, and the Open Education College, which awards bachelor’s degrees to teachers. The MoE provides alternative and informal education in three programs: the Accelerated Learning program, the Literacy Center, and Youth Schools. Learners at risk of or already out of school benefit from these programs.

Pass rates at the end of the preparatory level determine students’ post-secondary pathway. Graduates of the literature stream can qualify for advanced studies in education (at university or the Open Education College), literature, and other social sciences. Students who opt for biology at the preparatory secondary level qualify for medical specialties, pharmacy, science, and veterinary studies, and graduates of the applied sciences track qualify for engineering colleges and other science colleges. Vocational school graduates can progress to a technical college or a university of technology, although because of capped numbers, a very small percentage of vocational school graduates go on to a polytechnic or technical university. Education at all these levels is free, but only primary education is compulsory in Iraq.

The MoHESR supervises education at universities and colleges, except for the Open Educational College. Higher education technical programs are offered at technical institutes (2 years, Diploma) and colleges (4 years, Bachelor of Arts). Higher education provided at public universities is free in all disciplines, including subsidized residence in university dorms for students from other governorates. This does not apply to students enrolled in parallel and evening programs, for which students must make a partial contribution. Figure 3.1 shows the structure of the education system in Iraq.

Figure 3.1. Structure of Education System

Iraq tends to have lower GERs than other countries in the region except for primary education, in which Iraq has the highest GER. Enrollment in pre-primary education in Iraq is the lowest of five comparable countries (Table 3.2).

	Iraq العراق	Kuwait الكويت	Jordan الأردن	Saudi Arabia السعودية	Egypt مصر
Pre-primary	11	62	27	21	29
Primary	121	92	81	100	106
Secondary	67	98	63	118	88
Higher education (undergraduate)	22	55	33 ^a	71 ^a	39

32

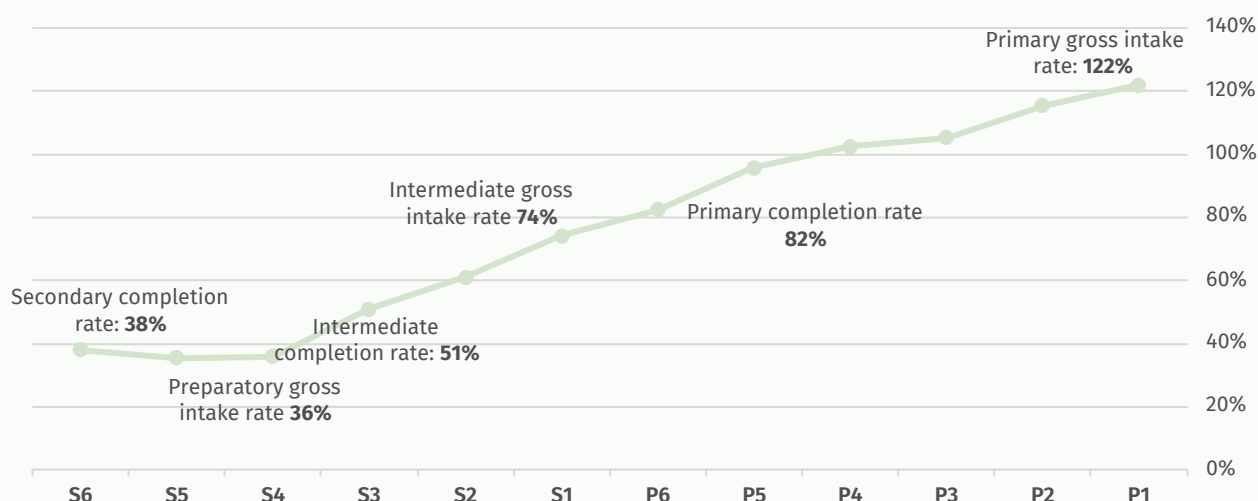


Figure 3.2. Transversal Primary and Secondary Schooling Profile, 2017/18 (Gross Access Rate)

Source: CSO 2018.

3.2.1 Pre-Primary Education

Between 2009 and 2017, schooling indicators steadily progressed, with the most significant growth for pre-primary education, with a GER of 11 percent in 2017/18. The high level of growth can be attributed to a catch-up phenomenon, with extremely low enrollment (7 percent) in 2009, and particular interest by the private sector covering 15 percent of the total kindergarten enrollment. Despite this growth, enrollment in kindergarten remains low, possibly because of high levels of economic inactivity among women³ and lack of awareness among parents of the importance of registering their children in early childhood development programs. Supply-related challenges include inadequate infrastructure and lack of teachers.

3.2.2 Primary Education

Enrollment at the primary level also increased from 105 percent 2009 to 121 percent in 2017. Despite the primary gross intake rate at grade 1 of 122 percent (3.2), not all children access primary education.

High GERs result from out-of-age children returning to school to catch up on schooling opportunities missed during the war against ISIS and significant numbers of under- and over-age students. Even though primary education is mandatory according to the Iraqi Constitution, only 82 percent of pupils complete the primary cycle (primary completion rate). Therefore, universal primary completion has not been achieved in Iraq.

3.2.3 Secondary Education

Enrollment at the secondary level also grew considerably. Lower secondary enrollment was 86 percent in 2017/18, although enrollment in upper secondary was 46 percent. Completion rates at the end of upper secondary more than tripled, from 12.8 percent in 2009 to 38 percent in 2017 (ESA 2020). Greater schooling coverage can be attributed to greater supply as a result of government investment in post-primary education, public awareness of the importance of secondary education, and parents' interest in better employment opportunities for their children (table 3.3).

³ The Multiple Indicator Cluster Survey 6 (CSO, KSO, Ministry of Health, UNICEF, 2018) indicates that close to 87 percent of women in Iraq are economically inactive (not working or not looking for work) and 78 percent are housewives.

Table 3.3.
Schooling Coverage Indicators According to Subsector, 2009/10 and 2017/18

	2009/10	2017/18	Growth 2009-2017
		%	
Pre-primary	7	11	57
Primary	105	121	15
Intermediate	66	86	30
Preparatory (total, including vocational schools)	33	46	39
Preparatory vocational schools	--	2	--
Higher education	--	93 ^a (2020)	--

Source: CSO 2018, authors' computations.

Note: Calculations are based on population estimates provided by the Ministry of Planning, which applies a constant growth rate of 2.6 percent to 2.7 percent. When using UN population data, schooling coverage indicators improve slightly as lower population growth rates are applied.

a. 2020.

Access rates were 74 percent for intermediate education and 36 percent for preparatory education. Only 38 percent of one cohort of students reached the end of the preparatory education cycle in 2017/18 (Figure 3.3).

There are two key concerns about the education system in terms of enrollment and completion.

First, **repetition levels have increased significantly since 2014/15**. As of 2017/18, the share of repeaters

was 13.9 percent for primary, 27 percent for lower secondary, and 20.6 percent for upper secondary. All values are far from the national target of 3 percent (National Development Plan, 2018-2022). Figure 3.3 shows that repetition levels peaked in grade 5, with 25 percent of all pupils repeating 1 year before the primary education certificate examination,⁴ then again in the last year of lower secondary (S3), with 41 percent of students repeating, and in the final year of upper secondary (S6), with a repetition rate of 37 percent.

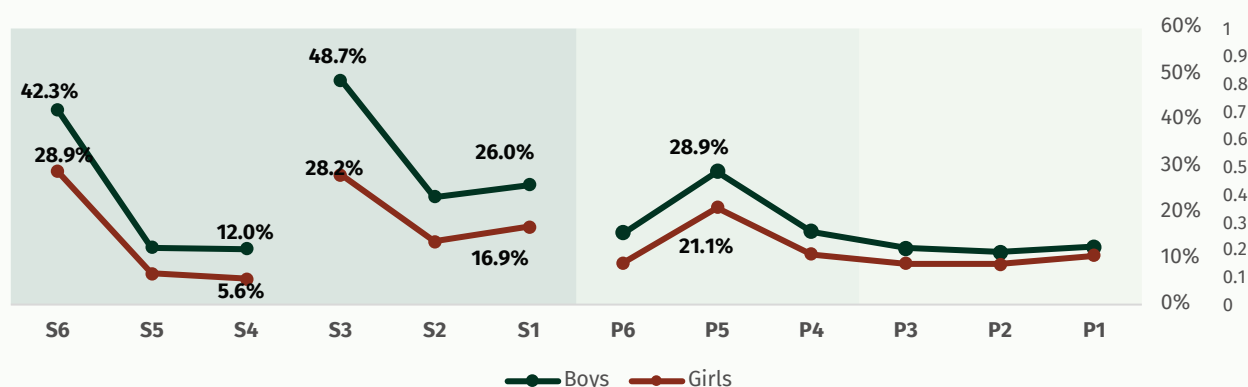


Figure 3.3. Share of Repeaters According to Education Level and Gender, 2017/18

Source: CSO 2018.

⁴ MoE staff indicated that high repetition rates in grade 5 are mostly because this is the first grade in which written examinations are taken, and anecdotal evidence suggests that teachers encourage underachieving learners to repeat a year before taking the primary leaving examination to increase their chances of success.

Second, **despite the decline in the share of out-of-school children since 2011, 2,039,436 out-of-school children aged 6 to 17 is significant.** Girls account for one-third of out-of-school children (Multiple Indicator Cluster Survey of 2018). As of 2018, more than 400,000 primary school-aged children and almost 500,000 of lower secondary school-aged children were out of school. More than 1 million upper secondary school-aged children are out of school, although the data do not include TVET.

3.2.4 Informal Education

Non-formal education is provided in all governorates but remains insufficient with regard to demand.

The Accelerated Learning program provides a fast-track second-chance opportunity for learners aged 12 to 18. Schools condense the 6-year primary curriculum into a 3-year course of study. Upon completion, students receive a formal certificate and (if they are still school aged) can enroll in secondary school. As of 2017/18, 33,149 students were enrolled in this program across 351 schools.

The Adolescents Schools or Schools for Youth for learners aged 10 to 15 enroll over-age learners who did not start primary education before age 10. As of 2019, 10,507 learners attended 80 adolescent schools. The youth program squeezes the 6-year basic education cycle into 4 years. The number of Adolescent Schools has decreased considerably since 2009/10 because of the increase in the number of accelerated learning programs.

Literacy and adult education centers welcome learners aged 18 and older and are the largest provider of informal education, with 72,242 students and 983 centers across the country. The program is divided into two levels: basic and complementary. Each level runs for 7 months and provides a certificate that is equivalent to the fourth-grade certificate in formal education. Textbooks cover math, Arabic language, and general education (MoE, UNESCO, 2011).

3.2.5 Vocational Education

Post-primary vocational education does not meet the quantitative needs of the substantial youth population or the qualitative requirements of the labor market.

As of 2017/18, only approximately 50,000 students were enrolled in Iraq's 314 vocational schools. The low GER of approximately 2.1 percent is due to lack

of acceptance of TVET among the population and limited supply. In 2020/21, enrollment in vocational schools doubled, reaching more than 109,000 students and resulting in a GER of 4.3 percent. Ministry staff indicate that regulatory changes that allow students in vocational schools to continue their post-secondary education explain the increase. The MoE responded to the sudden increase in enrollment by allowing schools to offer afternoon and evening sessions and use volunteer teachers, but because the 2020/21 academic year took place mostly online because of the COVID-19 pandemic, learning outcomes and continued trends in vocational school enrollment are unclear.

3.2.6 University Education

University education does not meet the quantitative needs of Iraq's youth despite a remarkable increase in the number of universities and in enrollment over the past decade. As of 2017/18, there were 35 public higher educational institutions and nine private universities.

More than 640,000 students were enrolled in 2020. Ninety-three percent of students completing grade 12 enter a public or private higher education institution. Higher education accommodates approximately 18,500 undergraduate students per 100,000 inhabitants aged 18 to 20, of whom there are approximately 3,470,000.

3.2.7 Gender Disparities

In 2017/18, gender disparities were nonexistent in kindergarten and relatively low at the beginning of primary school, with 95 girls for every 100 boys, but widen significantly at the intermediate level, with 78 girls for every 100 boys (Table 3.4). At the preparatory level, the gender gap falls slightly to 94 girls for every 100 boys. Although girls start dropping out after primary school at a higher rate than boys, enrollment of boys decreases significantly at the preparatory level and almost equals that of girls. Gender disparities disappear at the end of the preparatory level, with the same completion rate for girls and boys. Economic activities may prevent boys from completing preparatory education. Social practices such as early marriage and the unstable security situation have prevented many girls from accessing post-primary education.

Table 3.4.

Gross Enrollment Rates of School-Aged Population According to Gender and Level and Gender Parity Index, 2017/18

Level	Male	Female	Total	Gender parity index
	%			
Pre-primary	11	11	11	1.00
Primary	118	125	121	0.95
Secondary	60	73	67	0.83
Intermediate	75	96	86	0.78
Preparatory	45	47	46	0.94

Source: CSO 2018.

Table 3.5.

Number of Students Enrolled in Vocational Schools and Undergraduate Studies According to Gender, 2017/18

Level	Male	Female	Total	Percentage of students who are female
	Number			
Ministry of Education vocational schools	38,219	11,820	50,039	24 (30 in 2021)
Private and public undergraduate studies	425,615	377,136	141,549	47

Source: CSO 2018.

Girls are underrepresented in vocational schools (table 3.5). Of 50,039 students pursuing agricultural, commercial, applied arts, information technology, or industrial studies in 2017/18, 11,830 were female. Low female participation rates in vocational schools and great disparities between governorates (9 percent female participation rate in Al Anbar versus 37 percent in Thi Qar) are linked to social and cultural perceptions of the role of women in technical and vocational occupations. Trainings that the MoLSA offers are perceived as better suited for female students, leading to occupations primarily held by women, such as sewing.

The gender imbalance in technical universities that the MoHESR runs could be decreasing, with 43 percent of students in 2017/18 being female, compared with 29 percent in previous years. Low enrollment of female students indicates prevailing cultural norms that lead to men occupying technical jobs.

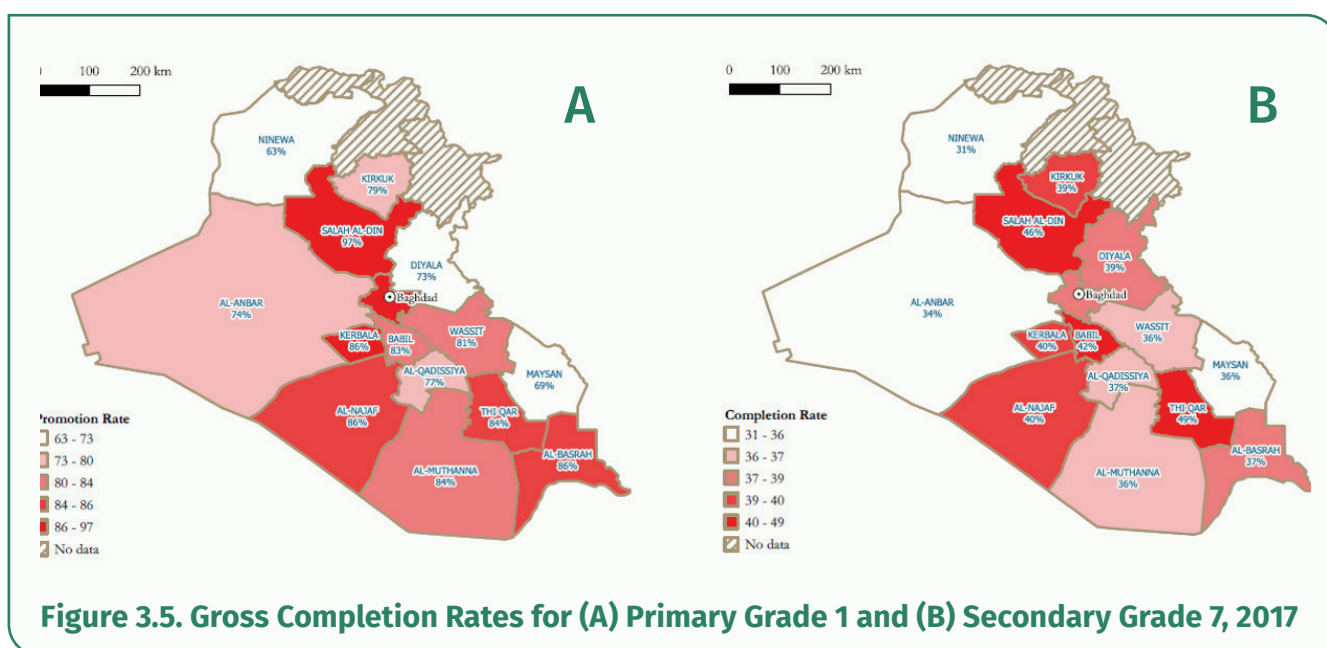
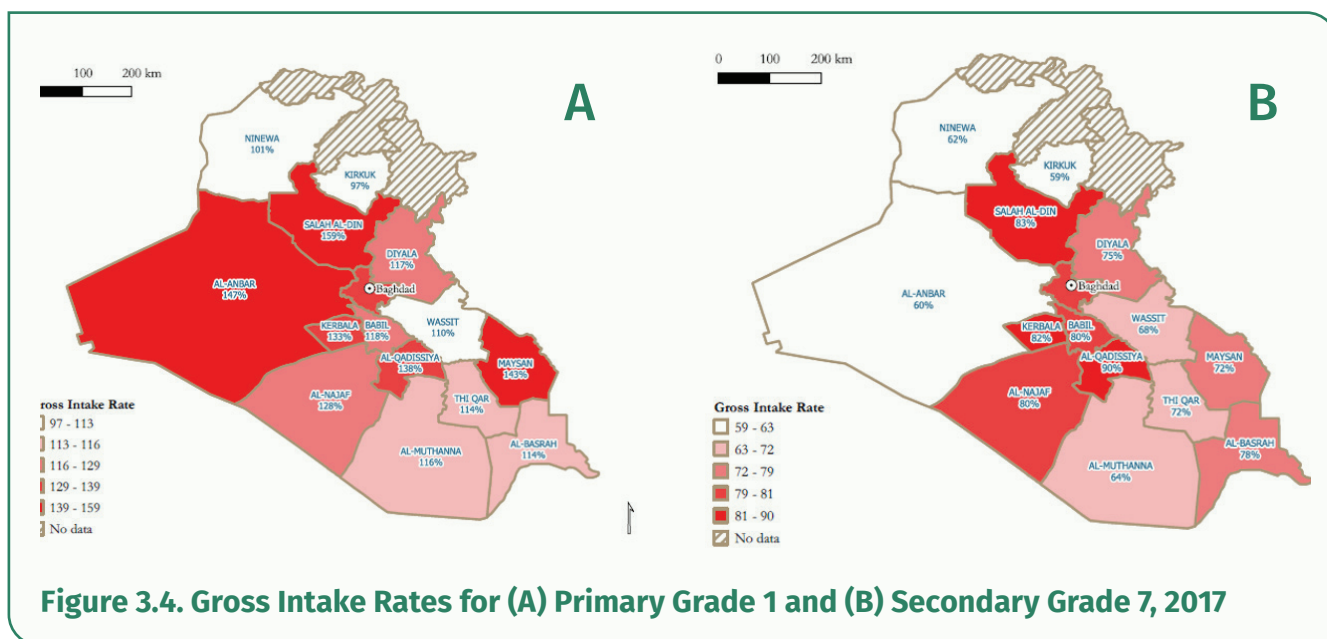
Female participation rates are much higher in academic universities and institutes, with almost half of students being female.

3.2.8 Governorate-Level Disparities

National averages hide important disparities between governorates. Analyses conducted at the governorate level show the effects of conflict and insecurity on schooling coverage. In general, governorates experiencing conflict and insecurity have lower coverage rates, as seen in Ninewa and Al Anbar. This is a consequence of disruptions in service delivery. High levels of population movement, including displacement across governorates, and returns make it difficult to provide an accurate picture of schooling coverage.

There are large regional discrepancies in pre-primary education, with highest coverage rates in oil-rich governorates where the private sector shows great interest in investing in pre-primary education. Access to primary education varies from 159 percent in Salah al-Din to 97 percent in Kirkuk (Map 1A). Disparities prevail throughout the cycle. The lowest completion rates for primary education were recorded for Ninewa (63 percent) and the highest for Salah al

Din (97 percent) (Figure 3.4). At the secondary level, access rates vary between 59 percent (Kirkuk) and 97 percent (Al Qadissiyah) (Map 1B). Completion rates are approximately 35 percent to 40 percent for most governorates (Figure 3.4). Repetition rates not homogenous across governorates either. The most pronounced disparities were recorded at the preparatory level in 2017/18, ranging from 3.4 percent in Salah al Din to 30.2 percent in Al Muthanna.



Source: CSO 2018.

There are high levels of under- and over-age students, although some governorates are particularly affected. Governorates affected by conflict and insecurity may have larger shares of over-age learners, who catch up after years of limited access to education. Over-age learners are at higher risk of dropping out, adding to existing inefficiencies. Current education opportunities to catch up or re-join educational offers, specifically for displaced students, are not sufficiently targeted to their needs.

Governate disparities also affect access to vocational training. Industrial and commercial vocational schools are unevenly distributed.

3.2.9 Barriers to Access and Completion in Pre-University Education

Several key factors influence the likelihood of attending and succeeding in school: low levels of mothers' education, poverty, gender, and special needs. Conflict and insecurity also limit access to education and training. On the supply side, lack of schools poses the most significant challenge for education access and completion.

One of the most significant factors affecting whether a child will receive an education is their mother's level of education. In 2018, primary school-aged children of mothers with pre-primary or less education were almost 6 times as likely not to be in school as children of mothers with upper secondary education or above (Multiple Indicator Cluster Survey of 2018). The education level of mothers of lower secondary school-aged children plays an even more significant role. Poverty also affects access to education. In 2018, children from the richest households were 3.5 times as likely to go to school as their peers from the poorest households. Poor children, especially poor urban children, are disproportionately unlikely not to be in school. Forty-six percent of primary school-aged out-of-school children are from the poorest quintile. Gender plays a varying role in education access and success. Although boys are twice as likely as girls to go to school, they are more likely to repeat.

The government of Iraq has committed to ensuring education for all no matter a person's individual

abilities. Educational care for children with disabilities is provided through institutes specializing in mental and physical disabilities and through establishment of integrated classes for special education in mainstream schools, although only 2 percent of mainstream schools provide classes for students with special needs.⁵ As of 2017/18, 1,325 schools were hosting 15,683 learners with special needs. Despite this small progress, there is a significant shortage of educational and school facilities for special needs learners. Lack of effective regulation, lack of suitable transport, limited number of teachers with special education needs qualifications, limited number of diagnostic centers, overcrowded classrooms, limited teaching and learning materials, stigma and embarrassment of parents about having a child with disabilities, poverty, and insecurity are barriers to education for learners with special needs.

Trauma, lack of resources, and poverty are barriers for many displaced, returnee, and refugee children. When families are unable to return to their place of origin because of suspected Islamic State of Iraq and the Levant affiliation, their children face obstacles to entering school because their access to civilian documentation required for schooling is limited (UNAMIS/OHCHR 2020). When displaced students manage to find a school, they face extreme overcrowding and limited school places (UNAMIS/OHCHR 2020). Barriers to educational participation disproportionately affect adolescent girls because of social practices.

3.2.10 Lack of Schools

Another limit on access is lack of school infrastructure. A combination of population growth, slowed infrastructure investment, and infrastructure damage since 2013⁶ has led to a shortage of 8,147 school buildings as of 2017/18 (MoE). According to the MoE's needs assessment, terrorist operations have destroyed or damaged 15.5 percent of all school buildings. In 2020, internally displaced persons were occupying almost 400 schools across six governorates, further limiting access to education of displaced learners. Lack of a school in the neighborhood is the main reason of 30 percent of learners aged 6 to 11 for not enrolling in school (World Bank 2018). This applies particularly to girls and children in rural

5 Following the 10-year National Framework for Inclusive Education, the MoE works toward ensuring inclusion of children with learning difficulties for grades 1 to 4, with additional support in form of resource rooms rather than special classes.

6 The World Bank Damage and Needs Assessment of seven directly affected governorates (2018) concluded that 18 percent of education infrastructure was completely destroyed, with a total cost of conflict damages estimated at 2.8 trillion dinars (USD 2.4 billion), with total reconstruction and recovery needs estimated at 5.4 trillion dinars (USD 4.6 billion).

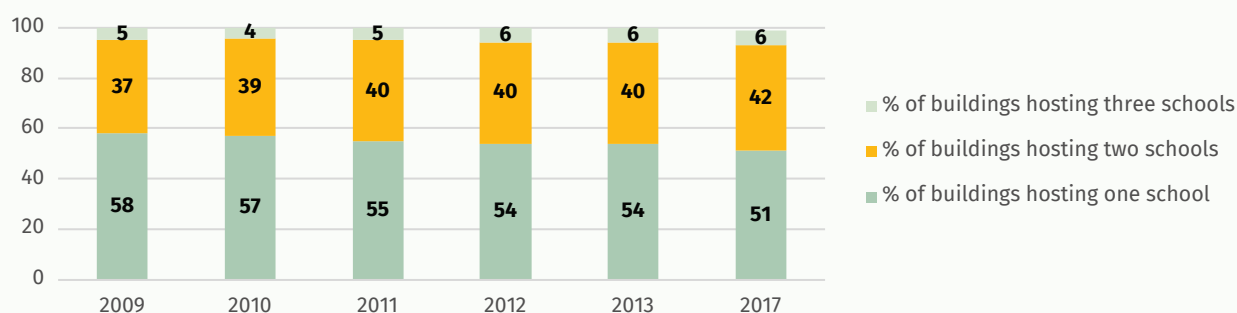


Figure 3.6. Share of School Buildings According to Occupancy, 2009/10-2017/18

Source: CSO 2017/18.

Note: Data for 2014/15 and 2016/17 are incomplete because of Islamic State of Syria occupation and are not shown in this graph.

areas. The MoE addresses lack of school buildings using a multiple shift system. Almost half of Iraq's primary and secondary schools share their building with one or two other schools (3.6).⁷

3.3 Educational Quality

Quality in education is based on four interrelated factors: input-related resources such as classrooms and facilities; learning materials; and competent,

motivated teachers. Various process-related factors are critical for learning, including pedagogy, strategies for teaching and learning, curriculum implementation, feedback mechanisms, support and supervision of pupils and teachers, school leadership, and professional development for teachers. Contextual factors also determine quality of education (Figure 3.7). In Iraq, insecurity; displacement; inadequate funding for the educational sector; and effects of economic sanctions on attracting and ensuring a qualified, motivated teacher workforce greatly influence this context.

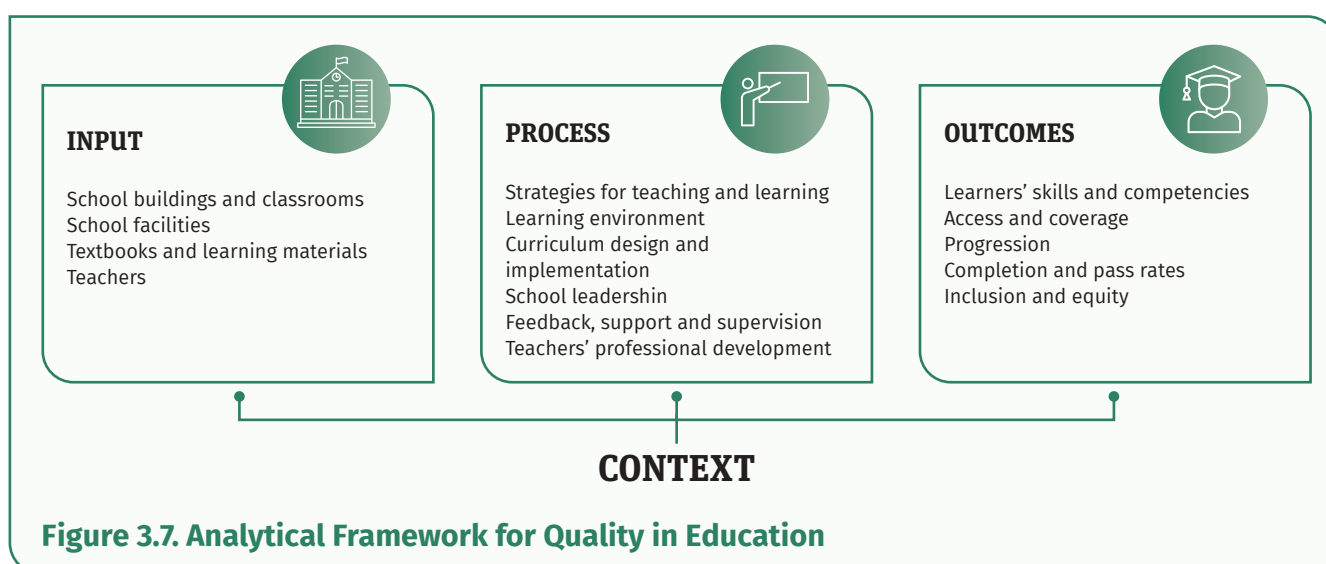


Figure 3.7. Analytical Framework for Quality in Education

Source: IIEP Learning Portal

⁷ The number of schools with multiple shifts has increased in recent years, particularly for secondary education, growing annually by 15.7 percent between 2013/14 and 2017/18, allowing the MoE to increase access to secondary education. The increase in double- or triple-shift primary schools is slightly lower, yet significant, with an annual growth rate of 10.4 percent.

3.3.1 Schools, Classrooms, and Facilities

In Iraq, the MoE responds to lack of school infrastructure by resorting to double and triple occupancy of school buildings. When multiple schools share buildings, school management and administration teams are different for each school. When the same level of education shares school buildings (e.g., two primary schools sharing one building), lessons are run in shifts. This means that the school to which the building is assigned, popularly called the “owner,” starts lessons in the morning, and after lessons are finished, the second school starts lessons. In school buildings where more than one level of education is accommodated (e.g., pre-primary and primary), lessons can run concurrently.

As of 2017/18, only 51 percent of primary and secondary schools in Iraq had their own school building. The share of buildings hosting two or three schools has increased 2.5 percent and 3.8 percent, respectively, since 2009, and single occupancy has decreased by 0.6 percent. With an average school size of slightly fewer than 400 students and with more than half of schools sharing facilities in multiple shifts, “a shortage of educational infrastructure profoundly affects the ability to deliver quality education services” (World Bank 2021).

In 2017/18, terrorist operations had destroyed or damaged almost 16 percent of all schools, and 45 percent of Iraq’s schools needed maintenance. Most of these school buildings are in Baghdad, Diyala, Najaf, Nineveh, Qadisiyah, Salah al-Din, and Thi Qar. In 2017/18, 6 percent of primary schools and 9 percent of secondary schools were temporary structures that did not meet national minimum construction standards. This includes mud, tent, and caravan schools. Many schools lack basic facilities and equipment, with 19 percent of primary and secondary schools lacking drinkable water, 27 percent lacking sanitation, and 17 percent lacking adequate health facilities. Although 47 percent of primary schools have libraries, only 8 percent have computer labs. Although 51 percent of secondary schools have libraries, only 39 percent have science labs and 39 percent have computer labs.

Areas affected most by insecurity and resulting effects such as displacement have the lowest share of primary and secondary school facilities. In 2019, the MoE introduced minimum standards for school infrastructure, with the goal of improving teaching

and learning conditions, including in view of climate change-related weather conditions.

3.3.2 Food Insecurity and School Meals

Iraq faces declining levels of food production because of drought and armed conflict, including displacement of farmers and damage to irrigation systems. Approximately 60 percent of Iraq’s rural population faces food insecurity, with several governorates⁸ at even greater risk (WFP 2019). Areas with high levels of food insecurity also have high levels of student absenteeism and dropout. Iraq’s poverty alleviation programs that reach the most vulnerable families through food rations fail to support education (WFP 2019). The National School Feeding Program that the MoE, Ministry of Health, and World Food Program designed addresses this. In 2019/20, 330,000 primary students (approximately 18 percent of primary students) received a fresh, healthy breakfast at the beginning of the school day. In 2020/21, 255,000 learners in 872 primary schools were reached. The MoP’s poverty and food security indicators are used to select beneficiary schools. COVID-19-related school closures and economic crisis are likely to have a negative impact on children’s nutrition.

3.3.3 Class Size

Lack of school buildings and teacher shortages have increased class sizes. Classes in urban areas and in areas hosting displaced persons are especially overcrowded. The average student:section ratio is 48:1 for kindergarten, 37:1 for primary, and 42:1 for secondary, although differences between governorates are significant, with ratios as high as 82:1 in Basrah at the pre-primary level. At the primary level, several governorates have student:section ratios of more than the average 40:1 (Baghdad, Basra, Ninewa), and Basrah (43:1), Baghdad (43:1), and Ninewa (44:1) have similar high ratios at the secondary level.

3.3.4 Safety in Schools

Prolonged conflict, protracted displacement, economic crises, and subsequent social and political tensions continue to expose learners, teachers

⁸ More than 75 percent of people are food insecure in the governorates of Dhi Qar, Dohuk, Karbala, Maysan, Muthanna, Najaf, Qadisiyah, and Sulaymaniyah. There are no data on food security in the governorates of Anbar and Nineveh (WFP 2019).

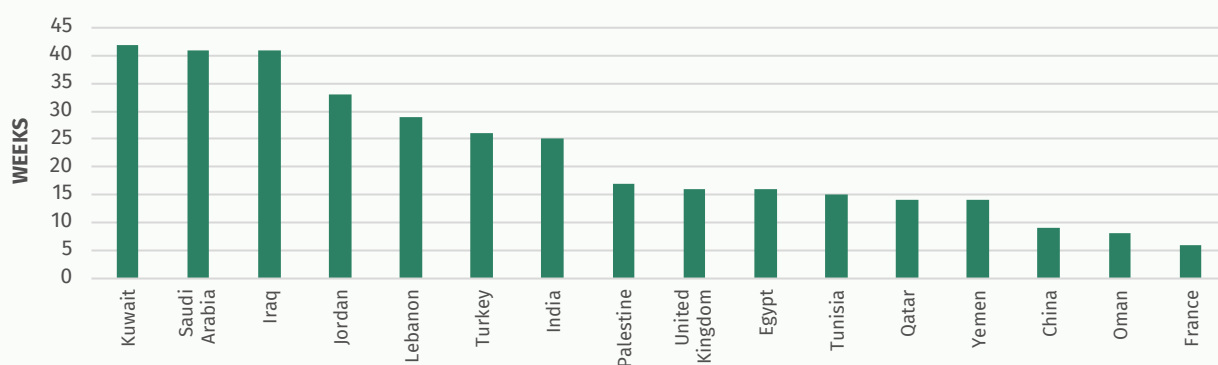


Figure 3.8. Duration (in weeks) of Full School Closures in Selected Countries, March 2020–March 2021

Source: UNESCO 2021.

and non-teaching staff to violence and hardship, damaging their emotional, mental, and physical well-being and development. Violence against children in schools and at home is high (UNICEF 2015). Ten percent of schools lack fencing, an important factor for student and teacher security. Governorate disparities are significant.

3.3.5 School Calendar and Instructional Time

With 137 days for primary and 151 days for secondary education, the school year in Iraq is the shortest in the region. The theoretical length of a school day is 4.5 hours, but schools that host two or three shifts reduce instruction time to 4 hours and 3.5 hours, respectively. Instructional time is on average 3.6 hours a day—too little to teach Iraq’s comprehensive curriculum (USAID 2012). Reasons for too little instructional time are multifold. In the case of insecurity or imminent emergency, schools close to protect learners and staff. Student absenteeism at the primary level (age 6–14) is particularly high in areas affected by insecurity, such as districts in Anbar province, which have an absenteeism rate of 15 percent (WFP 2018). Girls are more likely to be absent than boys. Teacher strikes and natural disasters add to missed instruction time. Between February 2020 and May 2021, schools were closed for 303 days because of the COVID-19 pandemic, one of the longest school closures globally (Figure 3.8).

In response to the pandemic the MoE combined remote learning with in-person instruction, the impact of which on immediate and long-term learning

losses, learning inequality, and intergenerational earning losses remains to be seen.

3.3.6 Teacher Qualifications

Based on the MoE teacher qualification law, as of 2017/18, approximately 28 percent of teachers did not meet the minimum qualification requirements at the pre-primary level. At the primary level, almost 17 percent were not qualified, whereas all teachers at the secondary level met the required qualification. At vocational schools, 47 percent of teachers were qualified below the bachelor’s level, and 75 percent of teachers had a technical qualification, meaning that most had no long-term formal teaching qualification. Although information on the quality of teaching in vocational schools is not available, the low qualification level of teachers has implications for the quality of pedagogical practice in schools, although the experienced teacher population balances the low level of academic qualifications, with 72 percent of teachers aged 40 and older. The fact that teachers are unwilling to leave secure jobs explains this.

The current teacher training system lacks a strategic vision and appropriate financial resources to meet pre- and in-service teacher training needs. The Institute for Training and Educational Development, which is represented in the center and has 20 branches across all 14 governorates, provides teacher in-service training. The Open Education College of the MoE, based in Baghdad, provides continuous training for teachers to complete their bachelor’s degree. The budget for teacher training

within the general education budget is not set and is insufficient and consequently dependent on external funds. No official body is responsible for accrediting training programs, and trainings lack follow-up and evaluation to ensure relevance and quality. Teacher training institutes are poorly equipped in terms of human, financial, and material resources. Teacher promotions are based purely on number of years in service and not on performance, which reduces teachers' motivation and interest in improving teaching skills. Teacher training must be decentralized to reach all teachers, including those in areas that are insecure. Moreover, the MOE does not necessarily hire faculty of education graduates to work in schools, which has led to an uncertain job market for these graduates, discouraging enrollment in bachelor-level teacher education programs.

In higher education, almost 17 percent of instructors held a PhD as of 2017/18, and 92 percent of instructors teaching at public higher education institutions held a master's-degree. Academic promotion of instructors depends on a point-based system based on academic criteria, such as number of published research papers, academic position, research activities, academic activities, and social activities. There is no requirement for faculty members to hold a doctorate. Promotion to higher management levels is based on years of experience and teaching performance, so there is no incentive for teachers to invest in research excellence and academic promotion.

3.3.7 Curriculum, Teaching Practices, and Textbooks

The revised national curriculum framework (2012) addresses critical topics, including civic education, social cohesion, safety, disaster risk reduction, and environmental change, although it remains knowledge based rather than competency based. Many modifications were made to the school curriculum recently, with varying degree of quality, and have not been completely implemented. Teacher training has not caught up with the revisions made to the curriculum, which therefore remains only partially implemented.

Instruction is primarily teacher centered and based on students' rote memorization despite the curriculum framework's emphasis on student-centered learning, active student participation, and promotion of analytical and problem-solving skills. This is mainly because of lack of teacher training in student-centered approaches and use of learning

activities, as well as a very short school day and year, which forces teachers to focus on covering the curriculum.

Sufficient numbers of textbooks are not provided because of the significant reduction in the MoE's budget for textbook printing in recent years. In 2016, 50 percent fewer textbooks were provided, prompting families to buy costly textbooks at bookstores, adding a financial burden to families' tight budgets after the economic crisis.

With only 14 percent of vocational schools having an Internet connection, systematic online study or the use of Internet resources in class is challenging. Almost all vocational schools have sanitation and drinking water, but only 54 percent have libraries, and these libraries are poorly resourced, with only 313 books each on average.

In general, vocational schools are poorly resourced, with a shortage of training materials, and those that are available have not been updated for many years. Existing programs are not competency based, and in many cases, parts of the curriculum can be delivered only in theory because of lack of facilities, equipment, and materials needed for any practical application. There is also little linkage with the private sector and a credibility gap with the skills, and therefore employability, of graduates. The government has been addressing these challenges through its TVET reform programs. During Phase 1, hundreds of teachers were trained in competency-based teaching methods. Curricula and teaching materials for 99 modules at levels 1 to 3 (suitable for vocational schools and vocational training centers) were developed, and some of these have been piloted in MoE vocational schools and MoLSA vocational training centers.

In higher education, curriculum is one of eight standards of the accreditation of national institution standards project for higher education institutions in Iraq. As such, they have been set and ratified, and the Department of Quality Assurance supervises their implementation. They are systematically assessed based on Iraqi national standards. For each academic specialization, a council is in charge of assessing institutional and academic performance in accordance with international quality indicators. The council monitors the latest developments in academic curricula in the university stage and reviews the academic programs against labor market needs. Academic committees in each academic department are responsible for determining the number of subjects, hours, and requirements for obtaining academic degrees.

3.3.8 School Inspection and Educational Supervision

Iraq has an established system with national guidelines for school inspection⁹ and educational supervision. School principals and educational supervisors provide classroom-based observation and supervision.¹⁰ There are performance assessment criteria for teachers, head teachers, and educational supervisors (British Council 2020). Newly recruited teachers and poorly performing teachers receive four visits per year, average-performing teachers three visit per year, high-performing teachers two visits per year, and excellent teachers one visit per year. In recent years, the frequency of supervision visits has increased, with 95 percent of planned visits executed; 75 percent of teachers are visited once every 2 to 3 months. External evaluation committees provide follow-up. Challenges to ensuring frequent, good-quality supervision include lack of access to certain regions because of lack of security and military operations, lack of specialization among supervisors, and failure to disburse transportation allowances for travel to remote schools. Despite an increase in supervisors (4,696 as of 2017/18), there are too few, and they lack training and support.

3.3.9 Learning Outcomes and Transferrable Skills for the Labor Market

In the past decade, Iraq has not participated in internationally comparable systems of student learning assessments and relies therefore on alternative sources of education data.

According to the Early Grade Reading Assessment and Early Grade Mathematics Assessment undertaken in 2011/12 for grades 2 and 3¹¹ (USAID 2012), students did not master the basics of literacy and numeracy. Assessments administered in a sample of primary schools showed overall poor results for most students at the end of the second and third grade. Approximately one-quarter of second and third graders tested could not read with sufficient fluency to be able to understand the text. Their limited mastery of letters contributed to very low scores in word decoding and oral reading fluency, with 48

percent having a score of 0; 26 percent did not have proficient reading skills, and 18 percent did not have sufficient reading comprehension skills. In contrast, students' listening comprehension was somewhat stronger, with 89 percent of students succeeding.

National leaving examinations occur at the end of primary (grade 6), intermediate secondary (grade 9), and preparatory secondary (grade 12) school. There was a significant decline in pass rates from 2009 to 2017. Data on educational achievement are limited. The 2012 Early Grade Reading Assessment showed that 53 percent of grade 2 students and 29 percent of grade 3 students, after reading an age-appropriate text, could not answer a single reading comprehension question. Reading comprehension was the weakest in Iraq, followed by Egypt and Jordan. "Early learning gaps exist for nearly all students, not just for those who performed poorly. These struggles to build key foundational skills early in life linger and contribute to high rates of illiteracy (44 percent) and unemployment (17 percent) among Iraqi youth" (World Bank 2021).

Only 48 percent of 14-year-old displaced, refugee, and returnee children surveyed mastered basic literacy and numeracy skills, only 5 percent of 8- to 9-year-olds were able to read and solve math problems at the appropriate grade level, and 45 percent of 8-year-olds were unable to identify at least four Arabic letters of 10 required (UNICEF 2019).

Prolonged school closures during the COVID-19 pandemic exacerbated the learning crisis. The World Bank (2021) estimated that, because of school dropout and quality shortcomings of distance learning solutions during the pandemic, there would be an average loss of 0.9 learning-adjusted years of schooling from a baseline of 4.0. Learning loss is likely to be more significant among the 7.4 million marginalized students at the primary, secondary, and tertiary levels who did not have the means to access remote learning that the MoE provided through its two online platforms 'newton' and 'e-parwarda' (OCHA 2020).

Despite the variety of vocational courses offered in preparatory education, they do not meet labor market needs. Despite a high demand for individuals literate in information technology in the job market,

9 The general education directorates in the governorates select and appoint school inspectors.

10 The General Directorate of Educational Supervision in the MoE selects educational supervisors in a competitive process who then receive specialized training.

11 The study sample consisted of 54 schools with a total of 1,153 students in the governorates of Anbar, Baghdad, Karbala, Maysan, Najaf, and Wasit.

Table 3.6.
Teachers According to Level of Education, 2017/18 and 2019/20

	Kindergarten	Primary	Intermediate	Preparatory	Vocational	University and college
2017/18						
Number of teachers (public)	5,952	271,597	95,040	58,466	11,245	42,925
Rural (%)	2.9	35.4	25		25.1	
Female (%)	100	65.9	42.5	46.4	56	45.3
2019/20						
Number of teachers (public) in	6,190	268,312	68,321	34,362	10,741	50,701
Female (%)	100	56.7	56.8	47.8	46.5	

there were only 10 computer and information technology vocational schools in Iraq in 2017/18, teaching approximately 2,000 students. Enrollment in information technology programs skyrocketed during the pandemic, reaching 18,000 students in 2021. A labor market survey conducted in 2017 (UNESCO 2017) revealed that courses offered in vocational schools do not equip enough students with the skills required in the construction labor market.

The MoHESR provides career guidance and counseling to university graduates. Thirty-five career development centers have been established in Iraqi universities to develop seniors' and graduates' skills to meet labor market needs, but male students tend to monopolize job opportunities, and opportunities for female students are limited.

3.4 Management of the Educational System

The education workforce accounts for approximately 20 percent of public employees, of whom 12 percent are part of the pre-university workforce (World Bank 2021). Education is no exception to mal practices in

the public sector, which is plagued by unchecked recruitment, indiscriminate hiring, and fraudulent payroll practices. The World Bank Public Expenditure Review 2021 notes that, "between 2015 and 2019, the number of employees in the (...) ministries of education and higher education increased from 250 thousand to 772 thousand (187 percent)." In 2019/20, there were approximately 440,000 teachers in public educational institutions (Table 3.6).

Women fill half of the teaching positions from primary to higher education.

3.4.1 Teacher Management

There are 485,207 teachers and trainers in general education. The teacher: class ratio at the primary level was 1.7:1 in 2018/19, which means that, for every 10 teachers, there are seven not allocated to a class. This might be because there are many ghost teachers¹² and because human resource management practices do not exclude staff on leave (e.g., for maternity or study leave) from the Department of Education's payroll. Data from the Central Statistics Organization show that some primary school teachers were hired for specific subjects, which creates additional inefficiencies. At the time of writing, there was no exact information on the number of ghost teachers

12 Although data regarding ghost workers in the education sector are inexact, the Iraqi Parliament's Finance Committee detected 150,000 people who earned dual salaries in the public sector in 2020 (Al-Masalah 2020). The committee previously estimated the number of ghost workers to be 200,000 to 300,000.

in primary education. The teacher: class ratio in secondary schools is 2.2:1. High teacher: class ratios indicate waste of public resources.

At vocational schools, the average student: section ratio was 19, although ratios differ according to specialization. Agricultural vocational schools have the lowest student: section ratio, with 16 students per section on average, whereas section classes are slightly bigger, with 20 students per section, at arts vocational schools and vocational schools offering computer and information technology courses.

In general, class sizes, and student: staff and student: class ratios have remained constant over the last 5 years, despite the increase in the number of schools over the same period. Large class sizes can make it difficult to use certain teaching methods and to manage the classroom and therefore limit student learning, although the number is only marginally greater and within the ability for a skilled teacher to keep motivated. The number of students per school is decreasing—from 294 in 2014/15 to 215 in 2018/19—which makes it easier for school administrations to supervise students. The situation with regard to libraries and workshops has also improved as extra facilities have been added over the 5-year period, bringing down the ratios of students to resources to 507 from 613 for libraries and to 80 from 96 for workshops.

The ratio of students to teaching staff members in higher education has decreased in public universities; student: teacher ratios were slightly higher in 2017/18 in humanities departments (13:1) than science departments (9:1). The average for Organization for Economic Cooperation and Development (OECD) countries is 18:1. In private universities, the ratio has increased in recent years in scientific and humanities departments. Student: teacher ratios were significantly higher in 2017/18 in humanities departments (49:1) than science departments (19:1).

Despite the large teacher workforce, Iraq faces teacher shortages in rural and marginalized communities and increasingly relies on untrained volunteers to keep schools open, especially in camps and returnee areas. The teacher shortage is a result of multiple factors, including inability to implement the policy that requires teachers to serve a minimum number of years in rural or disadvantaged communities. Displacement of teachers because

of insecurity has hampered effective teacher allocation; allocation of female teachers for mixed schools in the intermediate and preparatory stages is challenging because of gender restrictions at the secondary level; and there are too many graduates in specializations that the MoE does not need and too few in specializations that are needed.

In addition to the described inefficiencies, a recently announced hiring freeze and change in retirement age from 63 to 60 has led the MoE to resort to hiring contract teachers from among graduates of faculties of education and other teacher colleges that have a mandatory teaching period as part of the fourth year of training to address teacher deficits.

3.4.2 Non-Teaching Staff

Twenty-eight percent of education staff are non-teaching staff working in schools, departments of education, or the MoE. The share of non-teaching staff varies across governorates, with the highest share in Ninewa (52 percent), followed by Missan (43 percent) and Al Anbar (37 percent).

In higher education, half of approximately 45,000 staff are non-teaching staff. Large shares of non-teaching staff place a significant financial burden on the education sector and contribute to internal inefficiencies.

3.4.3 Student:Teacher Ratio

Inefficient allocation of teachers has increased the student: teacher ratio in recent years, with the most significant increase at the primary level (from a national average of 18:1 in 2009/10 to 26:1 in 2017/18). Student: teacher ratios were 29:1 in kindergarten and 18:1 at the secondary level in 2017/18. In higher education, student: teacher ratios vary from 5:1 (agriculture) to 27:1 (technical institutes).

3.4.4 Education Management and Leadership at Sub-Central Levels

Within the decentralization reform that started in 2005, education authorities at sub-central levels and schools perform selected administrative, financial,

educational, and legal functions.¹³ Challenges at the organizational level include lack of a decentralized financial system to secure basic operating expenses and scarcity of material resources and other equipment, including computers, office equipment, transportation, and infrastructure. Roles and responsibilities of education managers at governorate and district education authorities lack clarity, including in form of job descriptions and performance-based career progression structures. Clear guidance and coordination and communication mechanisms for educational planning, budgeting, and management, including feedback loops to correct and improve planning and management processes, are missing. Lack of transparency and accountability are a challenge.

Following a recent pilot, the MoE has begun to upscale the school-based management approach to strengthen school leadership and increase community participation in school management. As of 2021, more than 1,500 schools were trained in application of the school-based management approach, which uses a series of practical approaches to improve learning outcomes and ensure a safe, healthy learning environment by including principals, deputy principals, teachers, community representatives, and parents. To measure school performance, schools assess themselves against 25 defined standards.

13 This includes contracting and renewing contracts for temporary appointed staff; preparing training plans; approving transfer of employees; managing school canteens (administrative functions); preparing operational and investment budgets; transferring funds within annual budget item lines; disbursing student grants (financial functions); preparing plans; supervising school construction and maintenance; implementing the compulsory education law; managing investments in pre-primary education; strengthening the role of parent-teacher associations; managing education information and data; school mapping; coordinating and cooperating with nongovernmental organizations; opening, closing, merging, separating, and naming schools (educational functions); and accepting and managing contributions and donations, including of land for educational purposes (legal functions).

Chapter 4

Vision, Mission, and Strategic Directions

4.1 Vision and Mission

The INES 2022-2031 forms part of a government-wide effort to improve the quality and efficiency of public services. This is reflected in the national development goal for Iraq's future as stated in the National Development Plan 2018-2022:

Establish the foundations for an effective development state with social responsibility.

To contribute to this transformation, an ambitious vision that has been formulated considering the overall socioeconomic development vision of the country; the education laws, including provision of compulsory, fee-free basic education; and Iraq's international commitments in education according to SDG 4 guide the education system. It emphasizes equitable access to high-quality education that provides young people with the skills required in a 21st century knowledge society to enable them to contribute to Iraq's sustainable economic and social development.

This vision can be summarized as:

To aspire to an education system that is committed to values; provides sustainable, innovative, high-quality learning opportunities for all; promotes building a knowledge society; and conforms to global standards.

To this end, the mission of the INES is to:

Strengthen educational institutions for community partnership and build systems that promote scientific and creative thinking, ensure outputs that meet the needs of the labor market, and help build a diversified knowledge economy in line with the requirements of sustainable development.

4.2 Core Values

The INES is built on 12 core values, highlighting the education system's role in Iraq's future development:

1. Belonging and citizenship, leading to global citizenship
2. Respect for diversity and acceptance of others

3. Justice
4. Transparency
5. Integrity and impartiality
6. Community partnership
7. Compatibility with the needs of the Iraqi society and sustainable economic development
8. Excellence and high-quality service delivery
9. Teamwork
10. Academic freedom
11. Competitiveness
12. Lifelong learning and continuous development of those in charge of education services

4.3 Strategic Directions

To achieve the above-mentioned vision, the education sector, led by the MoE and MoHESR, in close collaboration with line ministries, communities, and external partners, strives to work toward strategic goals specific to three sub-sectors: general education, TVET, higher education.

Each sub-sector has a strategic goal:

- General education: Ensure provision of high-quality education for all at all levels
- TVET: Ensure provision of high-quality education that meets labor market needs
- Higher education: Develop a high-quality educational system that encourages broad participation through multiple effective, efficient educational paths for excellence in teaching, learning, scientific research, and community services that will achieve a comprehensive renaissance in Iraq

Key themes across the three sub-sectors are:

- ensuring equitable access to education and training
- improving the quality and relevance of education and training and strengthening linkages with the labor market
- improving the governance of education, including human resource development, educational planning, and management
- improving human resource and financial management for better service delivery and greater value for money

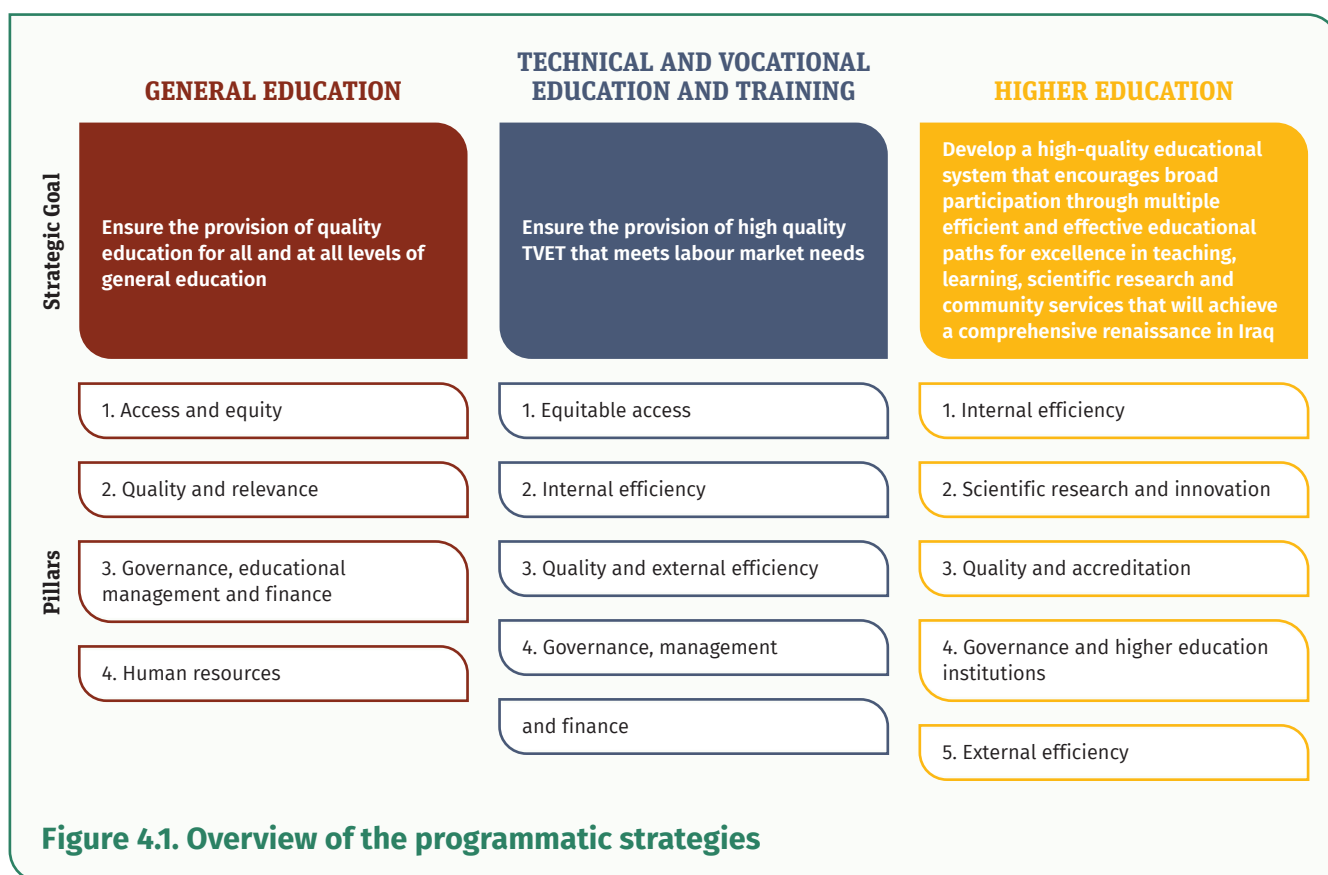


Figure 4.1 provides an overview of these broad sub-sector strategies.

The strategies build upon and are consistent with recent policy, planning, and programming decisions that the MoE and MoHESR have made. Within the overarching sub-sector strategic goals, the INES 2022-2031 emphasizes various cross-cutting factors, including gender, rural-urban divides, children and youth affected by conflict and insecurity, and poverty. Proposed programs address the vulnerabilities of children and youth in Iraq. Key performance indicators (KPIs) are disaggregated according to

gender and other categories to help the MoE and MoHESR identify and address sources of vulnerability and disadvantage.

The INES also proposes programs to address the effects of past and ongoing crises, including prolonged school closures due to COVID-19, displacement, destruction of education infrastructure, and psychosocial needs of learners. The strategy also identifies concrete avenues for increasing the resilience of the education system by institutionalizing crisis-sensitive planning and response.

Chapter 5

Policy Priorities, Strategic Objectives, and Priority Programs

This chapter outlines the foundation of the INES 2022-2031, describing policy priorities, strategies, and related priority programs for general education, TVET, and higher education. Each of the priority programs consists of several sub-programs. A “priority program” can be understood as a set of sub-programs designed to achieve the same program objective. Some priority programs are more complex than others because the challenge that they address is more significant or has a more diverse range of causes.

Identification of policy priorities and development of the strategies and priority programs was based on a comprehensive analysis of education sector performance conducted in 2020 that identified priority challenges. This was followed by the development of sub-sector-specific problem trees, which provided an overview of all known causes and effects related to these priority challenges. Subsequently, each sub-sector team prepared solution and objective trees by reversing the negative statements that form the problem tree into positive statements. Next, the solution and objective trees were analyzed to identify and strengthen linkages between general education, TVET, and higher education.

This chapter describes the results of this work, providing a condensed version of the solution and objective trees after highlighting key challenges each sub-sector faced as of 2018/19. In 2021, the COVID pandemic had not changed the nature of these challenges but had probably deepened them. Although the effects of COVID-19 containment measures (e.g., prolonged school closures) on education access, quality, and governance are not yet fully visible, the government expects learning loss and potentially higher drop-out rates, especially among the most marginalized learners.

Successful implementation of the INES depends on several factors, including the soundness of the proposed strategies and programs and:

- availability and effective provision of required financial resources and effectiveness of and transparency in their use
- availability of committed, skilled human resources at central and decentralized levels of the educational administration and in schools
- capacity of personnel to participate in the reforms that achievement of INES demands, which will require that many of them change ingrained working habits
- effective coordination and alignment of efforts between the MoE, the MoHESR, and other ministries, including the MoLSA, the Ministry of Finance, the MoP; external partners; and the private sector
- sustainability of political commitment to the vision of INES and its long-term objectives
- regular M&E and the willingness of decision makers and implementers to learn from M&E exercises

5.1 General Education

General education encompasses 2 years before primary school (kindergarten) through preparatory education (academic stream). Despite a considerable increase in access to general education over the past decade, completion levels remain low, and unequal access to education remains a key challenge. The most recent crises, including the COVID-19 pandemic, not only jeopardized access to education, but also limited learning achievements, especially for the most marginalized children and youth.

The list below summarizes key challenges in general education.

Despite considerable increase in access to basic education over the past decade, completion levels remain low.

- Despite significant growth in pre-primary enrollment, gross enrollment remains low, at approximately 10 percent for kindergarten in 2018/19.
- Despite a primary gross intake ratio of 122 percent in grade 1 in 2018/19, not all children access primary education. High GERs result from out-of-age children returning to school to catch up on schooling opportunities missed during the war against ISIS and significant numbers of under- and over-age students.
- Even though primary education is mandatory according to the Iraqi Constitution, only three-quarters of pupils completed the primary cycle in 2018/19.
- GER at the intermediate level is 85 percent. Only half of students completing intermediate education entered preparatory education in 2018/19.
- Completion rates at the end of preparatory education remain low even though they more than tripled from approximately 13 percent in 2009/10 to 40 percent in 2018/19.

Despite progress, unequal access to education remains a challenge.

- Significant disparities in access disproportionately affect girls, rural learners, and children from poor families.
- Progress was made in gender equality, especially in primary education, with girls accessing almost at the same rate as boys. The gender gap widens significantly at the intermediate level and then narrows at the preparatory level because of higher dropout rates for boys.
- The large number of out-of-school children and youth remains a key challenge. More than 2 million children and youth aged 6 to 17 were out of school in 2018.
- Informal education programs designed to accelerate learning and (re)integrate out-of-school children and dropouts are insufficient to absorb the number of out-of-school children, which is expected to grow given challenges in terms of progress and completion.

The most recent crises further limited learning achievements, especially for the most marginalized children and youth.

- On average, children in Iraq receive only 4 years of actual learning (learning-adjusted years of

schooling) by age 18 because effective years of schooling are 40 percent less than actual years of schooling, which is on average 6.9 years (World Bank 2018).

- Prolonged school closures during the COVID-19 pandemic are likely to lead to more learning loss, projected to amount to 0.9 learning-adjusted years.¹⁴

Public resources are wasted because of internal inefficiencies.

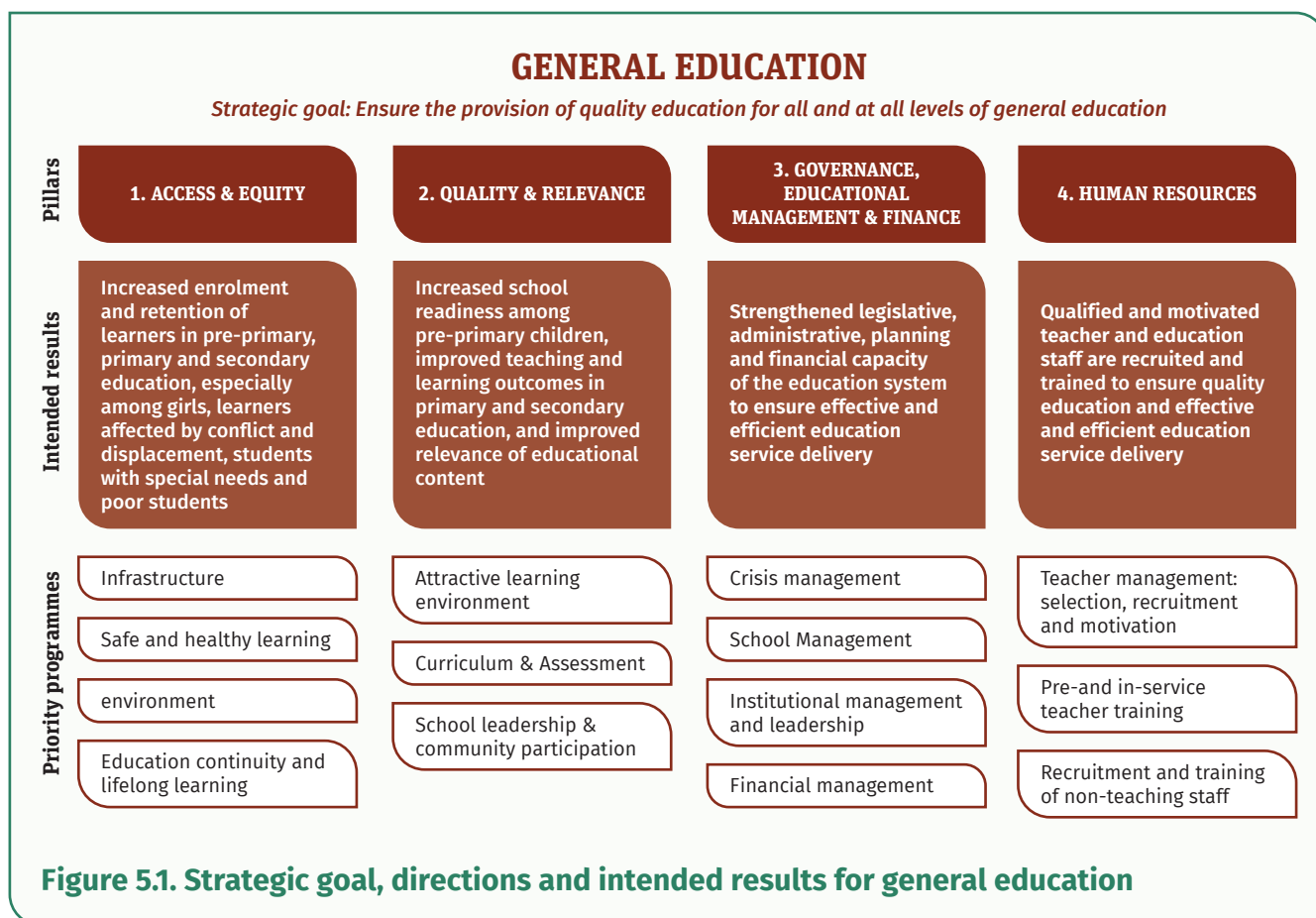
- A key concern for general education is high repetition rates, which peak for grade 5, with one in four students repeating 1 year before the primary education certificate examination,¹⁵ and then again in the last year of intermediate (grade 9; 41 percent) and the final year of preparatory (grade 12; 37 percent). Girls repeat at lower rates than boys.

- In 2018/19, teacher: class ratios were 1.7:1 for primary and 2.2:1 for secondary education. These ratios indicate that the teaching force is ineffectively used, because the additional teachers do not add value. This places a financial burden on the education system.

Figure 5.1 describes the strategic goal for general education, its four proposed strategic directions (pillars), and their respective intended results and associated priority programs.

The goal of general education is to meet the following policy goals:

- Increase school readiness of young children by increasing enrollment, including by increasing the share of public provision of kindergarten, although that will remain minor
- Ensure universal completion of primary education



¹⁴ Sedmik et al. 2020. "Iraq: Can COVID Teach Us How to Build a Better Education System for All Children?" World Bank blogs (November 6, 2020). <https://blogs.worldbank.org/arabvoices/iraq-can-covid-teach-us-how-build-better-education-system-all-children>.

¹⁵ MoE staff indicated that high repetition rates in grade 5 are mostly because this is the first grade in which written examinations are taken. MoE staff also mentioned that anecdotal evidence suggests that teachers encourage underachieving learners to repeat a year before taking the primary leaving examination, to increase their chances of success.

- Address internal inefficiencies by reducing repetition and dropout rates
- Improve teaching and supervision conditions by reducing the average number of students per teacher
- Provide better support for children with disabilities as part of the education sector's inclusion agenda
- Reduce waste of resources by reducing teacher: class ratios and address teacher gaps by improving teacher allocations
- Increase private sector engagement in education to meet increasing needs
- Improve the education process and the quality of education through school-based management
- Improve governance in management and oversight at all levels of the education administration for better value for money and sector performance

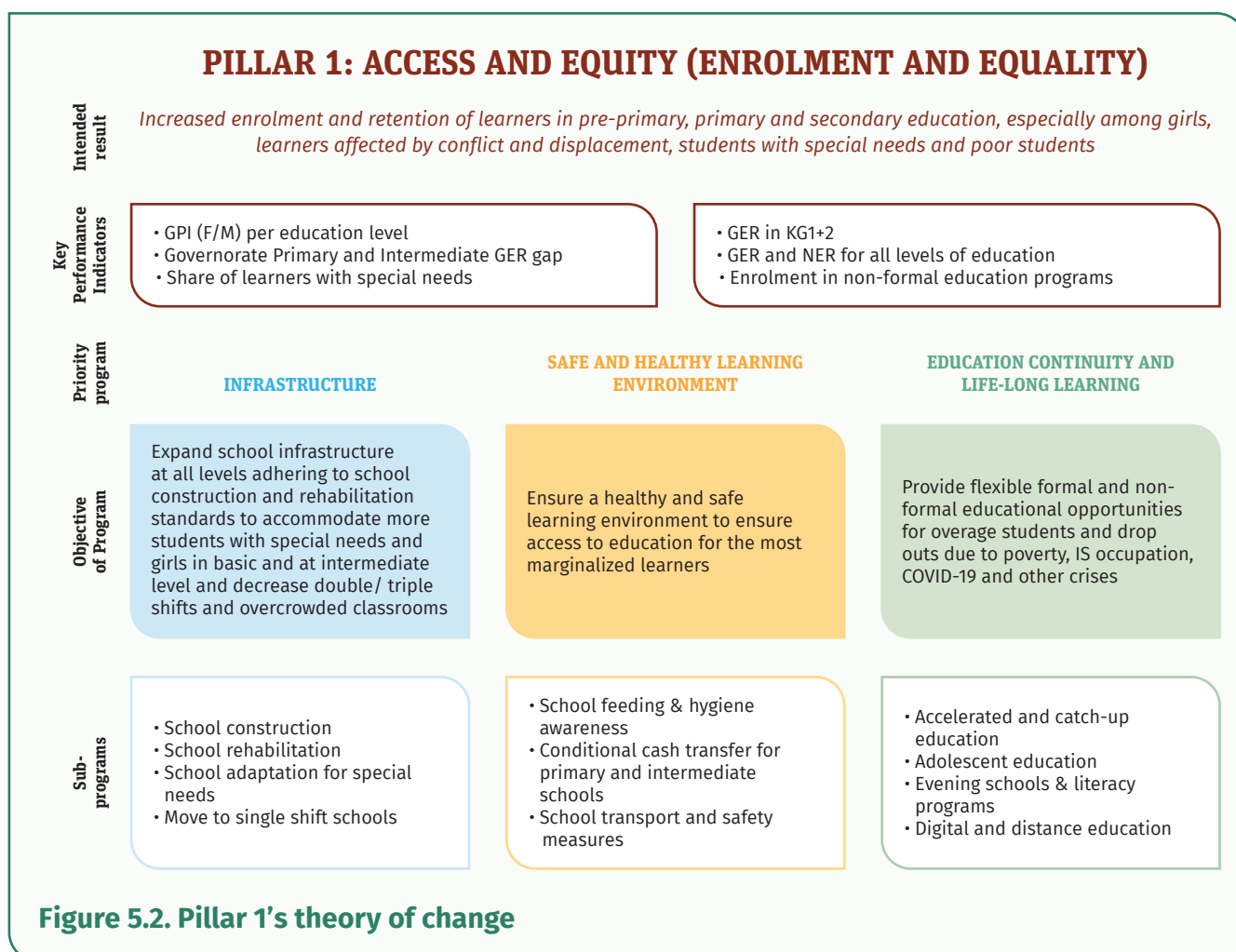
The following sections explore each of the four pillars in detail, providing first a theory of change detailing the objectives of the respective priority programs and related sub-programs. Progress toward pillar objectives is measured according to KPIs (outcome

level indicators; Figures) and selected output level indicators for the priority programs. (See the results matrix in chapter 6 for a full list of indicators per sub-sector.)

5.1.1 Pillar 1: Access and Equity

Pillar 1 contains three priority programs that contribute to the intended result based on the following theory of change:

If school infrastructure expands to accommodate an increasing number of students and decrease multiple shifts, if children and youth have access to a healthy and safe learning environment, and if children and youth have access to flexible learning opportunities, then enrollment and retention of learners in pre-primary, primary, secondary, and informal education programs will increase, especially among girls, learners affected by conflict and displacement, students with special needs, and poor students (Figure 5.2).



INFRASTRUCTURE

Significant investment in infrastructure (construction and rehabilitation) is needed to absorb increasing numbers of students, including in pre-primary education, and to reduce double and triple shift classes because multiple-shift classrooms are a key determinant of low-quality education because they reduce contact time between teachers and students and contribute to scarcity of resources (e.g., libraries). To promote access to education of special needs learners, school facilities will be equipped with disability-friendly infrastructure.

Operational plans derived from the INES 2022-2031 must examine creative ways to upscale construction and rehabilitation projects, such as providing low-cost schools to meet immediate needs.

Further information on the scale of infrastructure investment is provided in chapters 6 (cost and financing) and 7 (results matrix).

SAFE, HEALTHY LEARNING ENVIRONMENT

To reduce disparities between learners, pillar 1 provides a series of sub-programs to encourage

(re-)enrollment and retention, especially of girls and learners affected by conflict and displacement, disability, and poverty. These programs include expanding the national school meal program to reach all pupils in public primary schools and the most vulnerable children attending kindergarten by 2031. To increase enrollment in pre-primary education, the MoE aims to increase awareness of the positive role of early childhood development. School transport and safety measures are designed to ensure that learners in rural areas and areas affected by insecurity and natural hazards have access to a safe learning environment. The MoE also plans to explore the affordability and practicality of providing conditional cash transfers for marginalized learners, potentially linked to social welfare packages that the MoLSA provides.

This series of interventions forms part of the MoE's overall strategy to address widespread internal inefficiencies that hamper students' completion of the primary and secondary education cycle, including high levels of repetition, lower enrollment of girls and rural and poor children, and overall large

numbers of out-of-school children. Other strategies to reduce high repetition rates include introducing new regulatory policies limiting repetition, increasing the number of places available in the next education level, and increasing the number of students selecting into vocational preparatory education, as outlined in section 5.2.

EDUCATION CONTINUITY AND LIFE-LONG LEARNING

The MoE aims to provide flexible learning opportunities for children and youth who are out of school or at risk of dropping out by strengthening established informal education programs. The MoE works closely with external partners to fill gaps in the provision of informal education programs, which often take place in the afternoon or evening, including the Schools for Youth for learners aged 10 to 14, the accelerated learning programs for learners aged 12 to 18 who catch up on missed education through a compressed 3-year curriculum, and adult literacy centers for learners aged 18 and older who lack basic literacy and numeracy skills. Further coordination is needed with the MoLSA, which offers training courses for out-of-school youth to ensure that they receive not only literacy and numeracy skills, but also vocational skills that enable them to enter the labor market. More work is also needed to ensure that displaced learners without civil or legal documentation can transition to the formal education system.

The MoE plans to continue investing in distance and digital education to expose students to modern learning techniques and to ensure education continuity during school closures, such as COVID-19 related school closures. The MoE recognizes the need to explore strategies to reduce disparities in accessing distance and digital education between governorates, rural and urban areas, and female and male students due to limited access to devices and connectivity. The MoE also explores strategies to support students who have fallen behind their peers during school closures due to COVID-19 through remedial classes. To this end, it would be important to develop a 2-year operational plan that details implementation of the INES 2022-2031 provides an opportunity to adjust existing and develop new approaches to helping students catch up on their education.

PILLAR 2: QUALITY AND RELEVANCE

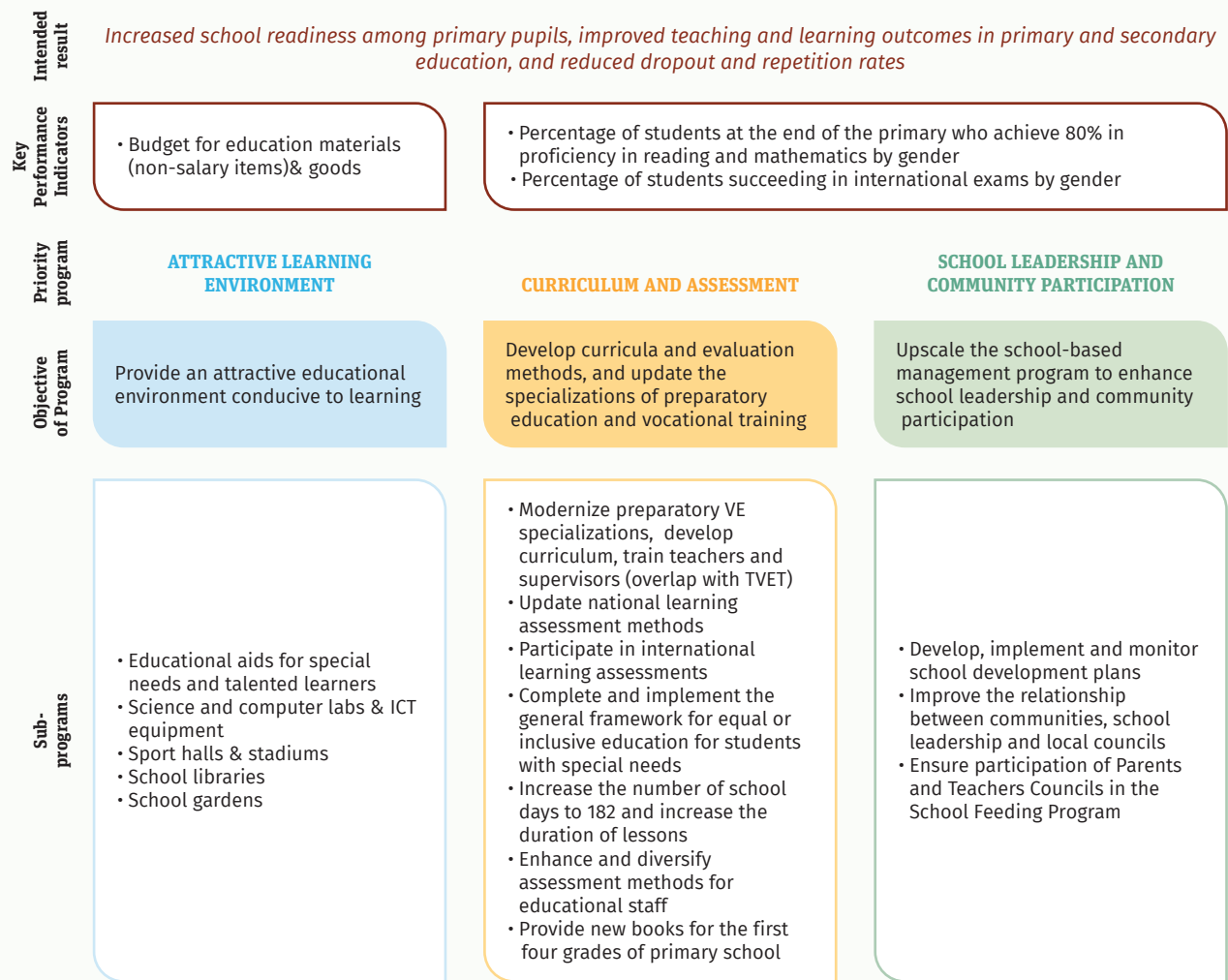


Figure 5.3. Pillar 2's theory of change

5.1.2 Pillar 2: Quality and Relevance

Pillar 2 contains three priority programs contributing to the intended result based on the following theory of change:

If schools provide an attractive education environment conducive to learning, *if* curricula and evaluation methods are updated, and *if* school leadership and community participation are enhanced through the school-based management program, *then* pupils at the primary level will be better prepared, teaching and learning outcomes in primary and secondary education will improve, and high repetition and drop-out levels will decrease (Figure 5.3).

ATTRACTIVE LEARNING ENVIRONMENT

Improving the quality of education requires a range of educational tools, including learning aids for special needs learners, school gardens and libraries, sports infrastructure, science and computer labs, and information and communications technology equipment. Doing so motivates school staff, in particular teachers, and students. Moreover, availability of these tools enables a wider variety of teaching methods, helping change in teaching habits toward more student-centered, active learning.

CURRICULUM AND ASSESSMENT

Curriculum review and development is a long-term, resource-intensive process, with implications for almost all educational staff, including teacher trainers, supervisors, and teachers. Two main considerations guide curriculum reviews:

- The need to strengthen the link with national development, including labor market needs
- Results from student assessments and examinations, which highlight where curriculum improvement is most urgently needed

With regard to this second consideration, Iraq's planned participation in international learning assessments such as the Trends in International Mathematics and Science Study 2023 will allow the MoE to use learning results as a basis for comprehensive curriculum review and development.

An update of national learning assessment methods will accompany curriculum reform. Close to finalization, the general framework for equal and inclusive education for students with special needs will provide further guidance.

SCHOOL LEADERSHIP AND COMMUNITY PARTICIPATION

By 2031, the MoE aims to expand the school-based management program to all primary and secondary schools in Iraq. The school-based management program empowers schools, parents, and communities to make school-related decisions collectively to improve the school environment and children's learning. The approach is rooted in Iraq's decentralization efforts, using participatory structures and tools such as school management committees, school and learning improvement plans, and training programs to build capacities of departments of education, school leaders, supervisors, teachers, and communities; foster community engagement; and monitor implementation and results of school improvement plans. Within the school-based management program, the MoE aims to provide school grants to primary and secondary public

schools that school management committees will manage. The committee is also tasked with assessing schools and developing school improvement plans identifying priority areas to improve the school environment and children's learning outcomes. By transferring financial and administrative functions to schools and increasing accountability at the school level, the MoE, with support from key partners including UNICEF and the World Bank, hopes to increase the effectiveness and quality of education based on school-specific needs and capacities. This includes enhancing student achievements in general education and increasing demand for and supply of pre-primary classes. The country-wide introduction of the school-based management program requires an institutionalized approach to school-based management, including strengthening feedback loops between central MoE departments, departments of education, and school leaders, which will enable decision makers to base policy and planning decisions on school-level experiences.

5.1.3 Pillar 3: Governance and Educational and Financial Management

Pillar 3 contains five priority programs contributing to the intended result based on the following theory of change:

If educational staff capacities for planning, M&E, coordination, and administration are strengthened; *if* financial management capacities at the institutional, organizational, and individual level are strengthened; *if* schools have access to operational funds to implement their school improvement plans and raise funds through school canteens to address school-specific needs; *if* central and decentralized educational authorities have the tools to plan for and respond to crises; and *if* partnerships between education stakeholders and the private sector are strengthened, *then* the legislative, administrative, planning, and financial capacity of the educational system will be strengthened to ensure effective, efficient education service delivery (Figure 5.4).

PILLAR 3: GOVERNANCE, EDUCATIONAL AND FINANCIAL MANAGEMENT

Intended result

Strengthened legislative, administrative, planning and financial capacity of the education system to ensure effective and efficient education service delivery

Key Performance Indicators

- Share of schools with operational budget available to implement school improvements plans (School-based management approach)
- Annual joint sector reviews and mid-term review of INES

- EMIS active at central and governorate level and used for planning, M&E processes
- Share of DoEs that have specific plans and budgets to ensure education continuity during crises, with a total of 20 plans (19 governorates and educational districts in Baghdad + 1 centre)

Priority program

INSTITUTIONAL MANAGEMENT AND LEADERSHIP

FINANCIAL MANAGEMENT

SCHOOL MANAGEMENT

CRISIS MANAGEMENT

PARTNERSHIP WITH PRIVATE SECTOR

Objective of Program

Strengthen staff capacities at the Ministry of Education's office and general directorates in aspects of planning, coordination and administrative skills to ensure service delivery, monitoring and evaluation

Strengthen financial management capacities for effective and efficient education service delivery

Provide schools with operational funds to implement their school improvement plans and implement the school canteen strategy to allow schools to address school-specific needs

Institutionalize crisis-sensitive planning and strengthen response capacities at central and decentralized levels to ensure education continuity during crisis,

Strengthen partnerships with the private sector to improve the quality of education

Sub-programs

- Create and amend educational laws to meet quality standards
- Strengthen inter-ministerial coordination to ensure implementation and monitoring of progress of INES
- Strengthen coordination with international partners to ensure alignment, implementation and monitoring of progress of the National Education Strategy
- Strengthen the supervision function through review of tasks and development of supervision reform linked to decentralization efforts
- Support the roll out of administrative and electronic educational systems, incl. EMIS and Kobo toolbox
- Determine roles and responsibilities between the Ministry's office and the general directorates of education to match the requirements of the educational process

- Develop the Financial Management Information system
- Build the capacities and motivation of employees in the financial department to plan for disbursement of funds for program/project implementation
- Develop diversity of education financing (Investment)
- Provide accurate financial data and information with a high level of transparency

- Provide schools with block grants to implement school improvement plans
- Amend the school canteen system to be dedicated to the school's operational budget
- Develop the school administration selection system
- Promote Human Rights Awards in Schools
- Expand the school-based management program to all schools

- Establish a crisis cell in the Ministry of Education and its offices in the governorates
- Train staff of the Crisis Cell in the Ministry of Education and its offices in the governorates to respond to crises and to ensure the continuity of education during crises

- Amend the private school system
- Amend legislation for the purpose of encouraging private sector investment in education

Figure 5.4. Pillar 3's theory of change

INSTITUTIONAL MANAGEMENT AND LEADERSHIP

The MoE plans to undertake a series of interventions to strengthen the legislative framework for education, such as by amending educational laws to better meet quality standards, planning and management capacities of education staff at central and decentralized levels. Several key areas require special attention, including:

- piloting and implementation of the educational management information system, the geographic information system, and other data collection tools such as the Kobo Toolbox for rapid needs assessments
- M&E progress and achievements made toward objectives such as through annual sector reviews, as well as mid-term and end-of cycle reviews, which allow education stakeholders to take stock of challenges and opportunities faced during implementation of the INES
- inter-ministerial coordination to ensure implementation and monitoring of progress of the INES
- coordination with international partners to ensure alignment, implementation, and monitoring of progress of the INES, such as with the Education Sector Coordination Group
- revision of the supervision function through review of tasks and development of a supervision reform linked to decentralization efforts and greater school autonomy
- division of roles and responsibilities between the MoE and the general directorates of education to match the requirements of the educational process, such as through development of job and department descriptions

FINANCIAL MANAGEMENT

Financial management capacities must be increased to ensure effective, efficient service delivery, including through:

- development of the financial management information system to provide accurate financial data and information with a high level of transparency to allow for efficient disbursement of funds to ensure program and project implementation
- preparation and effective disbursement of education budgets
- diversification of education financing, including through the private sector

SCHOOL MANAGEMENT

In line with the school-based management program listed under Pillar 2 (Quality), Pillar 3 ensures provision of block grants to implement school improvement plans. Furthermore, the school canteen system will be reviewed to allow schools to increase their operational budgets.

CRISIS MANAGEMENT

Building on the MoE's experience with managing crises and in close collaboration with national and international partners, the MoE aims to institutionalize its crisis response by establishing a crisis cell in the MoE and its offices in the governorates to coordinate and plan better for crisis response.

PARTNERSHIP WITH PRIVATE SECTOR

A joint approach between public and private actors is needed to meet demand for education and training over the coming decade. The INES 2022-2031 aims to increase private sector engagement in general education significantly, relieving pressure on the public education system. To this end, the MoE aims to amend how it regulates and supports the private school system, including by issuing legislation and guidance to encourage private sector investment in education.

5.1.4 Pillar 4: Human Resources

Pillar 4 contains three priority programs contributing to the intended result based on the following theory of change:

If the teaching environment attracts and retains qualified, motivated teachers, if teacher qualifications are improved through pre- and in-service training, and if non-teaching staff are recruited and trained based on needs, then qualified and motivated teacher and education staff will ensure quality education and delivery of effective, efficient educational services (Figure 5.5).

PILLAR 4: HUMAN RESOURCES

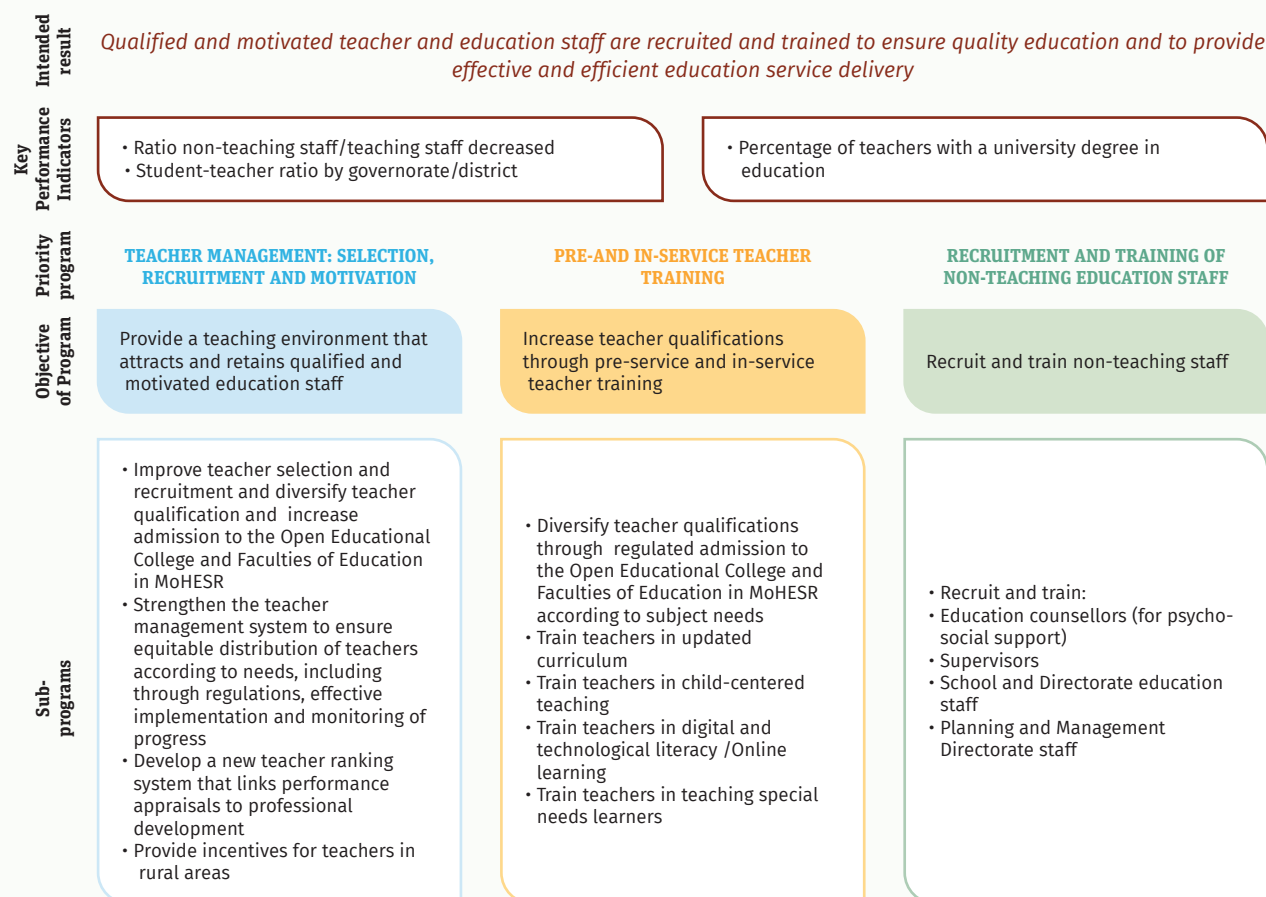


Figure 5.5. Pillar 4's theory of change

TEACHER MANAGEMENT

The teaching workforce is the most valuable resource in any education system; teachers have a major impact on the quality and effectiveness of the education offered and on learning. Teacher salaries are also the largest budget item. Despite high teacher: class ratios, many schools, especially in disputed areas, have teacher gaps, with the result that volunteer teachers with few qualifications teach afternoon shifts. The high wage bill leaves little room to invest in anything else. Strategies to reduce the wage bill must include identifying and removing ghost staff and staff on leave from the MoE wage bill, regulating the size of employment by increasing the number of teaching hours, further reducing the teacher: class ratio and reducing teacher allowances. A comprehensive review of current teacher management practices is required in the first years of the INES 2022-2031 to identify the nature and scope of these strategies. Teacher selection and recruitment regulations need immediate improvement to ensure that teacher

allocations are based on needs. To fill teacher gaps in rural areas, the MoE must renew incentives for teaching in rural areas. The MoE must also pursue development of a career progression pathway that links performance appraisals to professional development and promotion

PRE- AND IN-SERVICE TEACHER TRAINING

Reforms are needed to improve teacher pre- and in-service training to achieve more-effective teaching and learning approaches, which boost literacy and numeracy, digital literacy, child-centered learning, and inclusive education. It is essential to link those who provide pre- and in-service training with those who work on development and renewal of the curriculum to ensure that teacher training adapts, when needed, to the requirements of the new curriculum. Teacher training should also focus on helping teachers use the more attractive learning environment to change their teaching styles and approaches to student-centered teaching.

RECRUITMENT AND TRAINING OF NON-TEACHING STAFF

Non-teaching staff account for a considerable portion of the overall wage bill. The INES 2022-2031 proposes a reduction of the overall non-teaching staff: teacher ratio for all levels of education, with the most considerable reductions in kindergarten and higher education. The MoE plans to provide training for non-teaching staff in key areas, including for implementation of the school-based management program, to strengthen counseling, including for provision of psychosocial support in schools; improve supervision and mentoring; and provide educational planning and management.

5.2 Technical and Vocational Education and Training

The INES 2022-2031 pays special attention to vocational education (TVET). Because TVET is offered at the preparatory and tertiary education levels, it falls under the responsibility of the MoE and the MoHESR. Its organization also requires close coordination and alignment with other ministries offering informal vocational training courses, such as the MoLSA. The vocational stream of preparatory education marks the beginning of vocational education. Successful vocational school graduates can progress to a technical college or a university of technology.

The list below summarizes key challenges in the TVET system:

Low demand for vocation and technical education and training

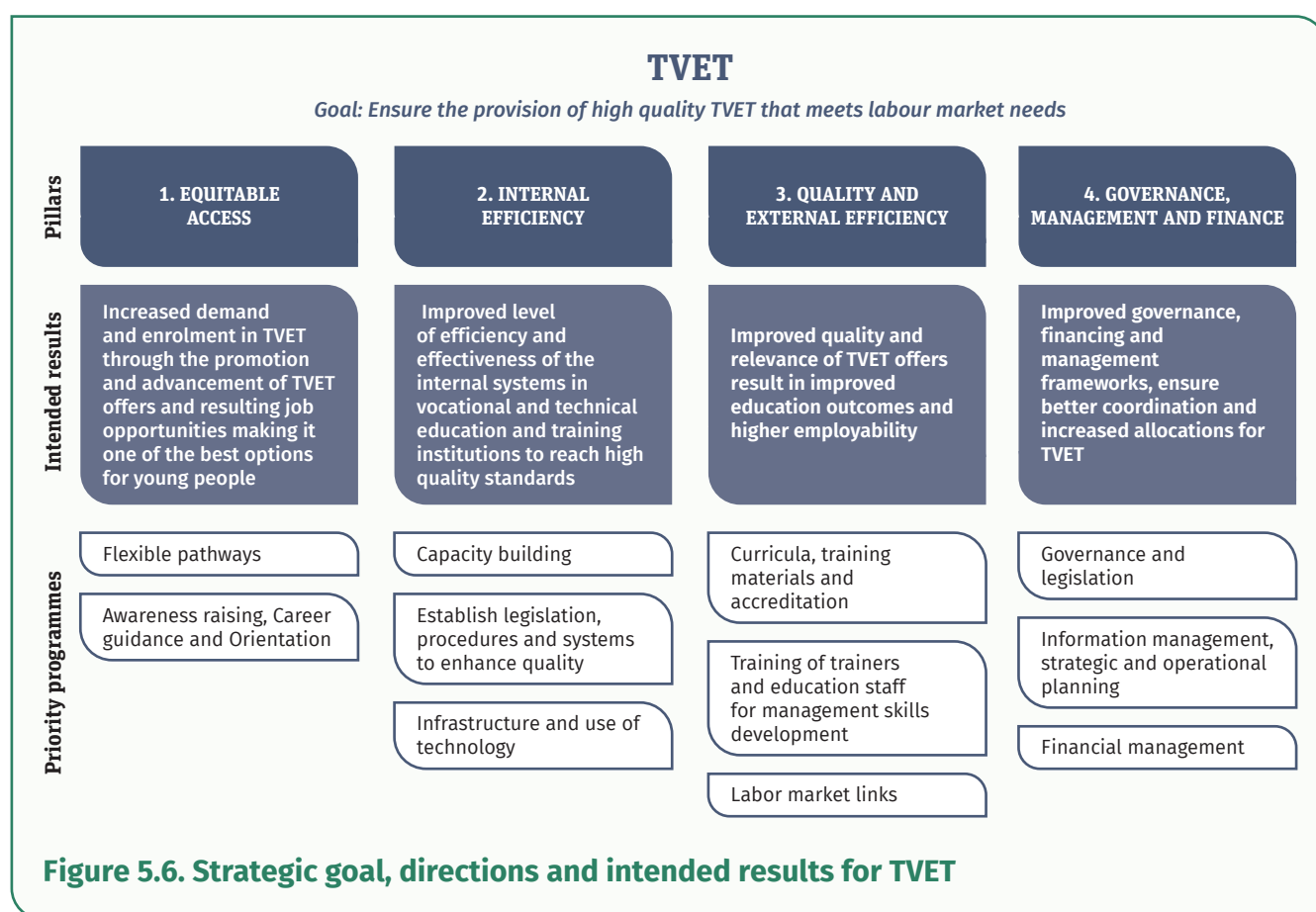
- The capacity of vocational education at the preparatory education level is limited because of lack of demand and supply, with a GER of 2 percent in 2018/19.
- A sudden increase in demand occurred in 2020/21, increasing the GER to almost 7 percent, because of policy changes that allow vocational school students to enter universities after grade 12. This may seem like a positive development, but it poses short- and medium-term challenges in terms of the capacity of TVET institutions (at the preparatory and post-secondary levels) to

absorb this sudden increase in enrollment in terms of teaching staff and infrastructure.

- Because the number of institutions is capped, a very small percentage of vocational school graduates enter tertiary technical education in polytechnic universities and technical universities.
- Negative perceptions associated with vocational work and the lower social status of technical professions compared to academic professions, limit demand for and enrollment in vocational education. Social factors also limit the enrollment of women.
- Absence of an institutional framework to ensure quality of TVET outputs, the unimplemented Technical and Vocational Qualifications Framework (TVQF), outdated curricula and training materials, insufficient training and qualification opportunities for teachers and trainers, and the need to upgrade schools and training centers contribute to inefficiencies and low quality of TVET offers, which in turn lead to high repetition and dropout rates.
- TVET institutions are unable to meet the needs of the Iraqi labor market and the socioeconomic development needs of the country, in part as a result of the lack of a strong working relationship between business owners and TVET institutions, as well as the lack of a comprehensive labor market information system that can guide decision makers, students, and graduates.
- Lack of coordination; insufficient institutional capacity to ensure quality assurance; and limited funding, with an almost unique reliance on government allocations, limit the MoE's capacity to manage and supervise reform of the TVET system effectively.

These challenges exist against a difficult economic background. Iraq's unemployment rate was 12 percent in 2018, and the youth unemployment rate was 34 percent. The private sector creates too few job opportunities, and as a result, the public sector is the first and last resort for job seekers, with 40 percent of all job opportunities, but the public sector cannot continue to expand. Lack of employment opportunities and the ensuing despair about the future can feed into disaffection and violence.

Figure 5.6 shows the strategic goal for TVET, its four proposed strategic directions (pillars), their respective intended results, and associated priority programs. Several documents and work processes were critical for development of the TVET section of the INES, including the TVET strategy 2014-2023 and the TVQF.



The government of Iraq is committed to developing and implementing a more flexible, responsive, demand driven TVET system to address Iraq's most pressing challenges, including high unemployment rates, especially among young people. To this end, TVET has the following policy goals:

- Expand vocational education at the preparatory level to help substantially more students enter the labor market after completing grade 12 or access further technical education in higher education institutions, considering the needs of the national economy
- Offer an attractive alternative to academic specialization, allowing students to enter the labor market as soon as they complete grade 12 or continue their studies in technical and academic universities
- Coordinate with ministries and institutions offering informal vocational training to offer dropouts, graduates, jobseekers, and in-service workers life-long vocational training to enhance their employment opportunities.

The following sections explore each of the four pillars in detail, providing first a theory of change detailing the objectives of the respective priority programs and

related sub-programs. Progress toward set objectives is measured through a series of KPIs (outcome-level indicators) and selected output level indicators at the level of priority programs. (The results matrix in chapter 6 provides a full list of indicators according to sub-sector.)

5.2.1 Pillar 1: Equitable Access to TVET

Pillar 1 consists of two priority programs contributing to the intended result based on the following theory of change:

If career guidance and orientation based on labor market needs is provided to students; *if* communities are increasingly aware of the benefit of TVET; and *if* students, especially girls, out-of-school youth, students with special needs, and students at risk of dropping out, have access to flexible pathways that facilitate entry into the labor market or lead to better employment opportunities, *then* demand and enrollment in TVET programs will be increased through the promotion and advancement of TVET offers and resulting job opportunities, making it one of the best options for young people (Figure 5.7).

PILLAR 1: EQUITABLE ACCESS TO VOCATIONAL AND TECHNICAL EDUCATION AND TRAINING

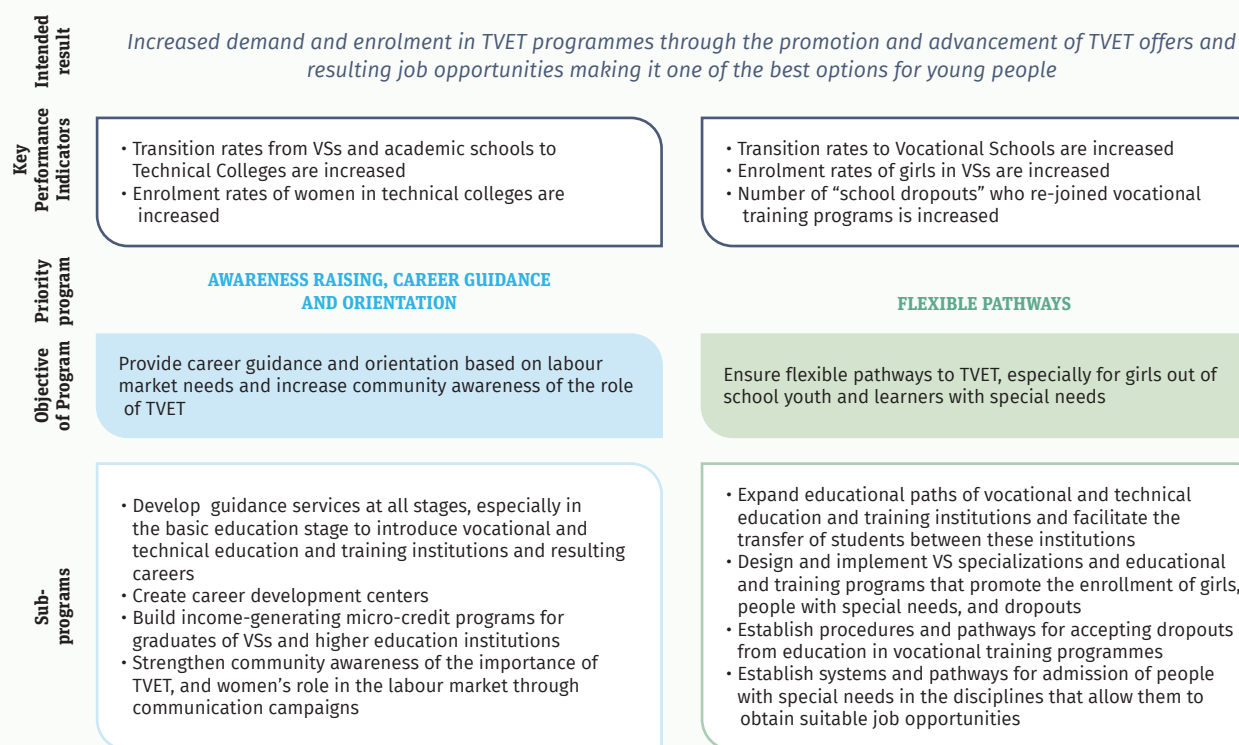


Figure 5.7. Pillar 1’s theory of change

AWARENESS RAISING, CAREER GUIDANCE AND ORIENTATION

A key impediment to upscaling TVET is limited demand for TVET, which reflects and reinforces lack of interest in professions based on vocational and technical skills. Pillar 1 includes interventions designed to increase awareness of the importance of TVET. The supply of high-quality, relevant TVET programs will be critical for changing perceptions. To this end, the INES 2022-2031 aims to create career development centers and provide guidance and counseling services at all stages of students’ learning trajectories, outlining TVET offers and resulting careers. Income-generating programs such as micro-credits for vocational school, technical college, and university graduates can minimize risks and motivate young entrepreneurs to start their own businesses.

FLEXIBLE PATHWAYS

To increase demand for TVET, educational paths of TVET institutions will need expansion and flexibility. This includes making it easier to transfer between

institutions, allowing students to move easily and readily between different training sectors to increase their skills and qualifications. Greater recognition of qualifications of graduates will also enhance mobility—nationally and internationally. The TVQF provides the framework in this regard. It is hoped that flexible pathways to TVET will provide an opportunity for the many young men and women who have missed out on education and training in recent years to develop skills to allow them to earn a living and enrich their communities. To make vocational education offers more attractive, the MoE plans to design and implement additional specializations in response to labor market needs to attract women, students with special needs, and young people who have dropped out of school.

5.2.2 Pillar 2: Internal Efficiency

Pillar 2 consists of three priority programs contributing to the intended result based on the following theory of change:

PILLAR 2: INTERNAL EFFICIENCY

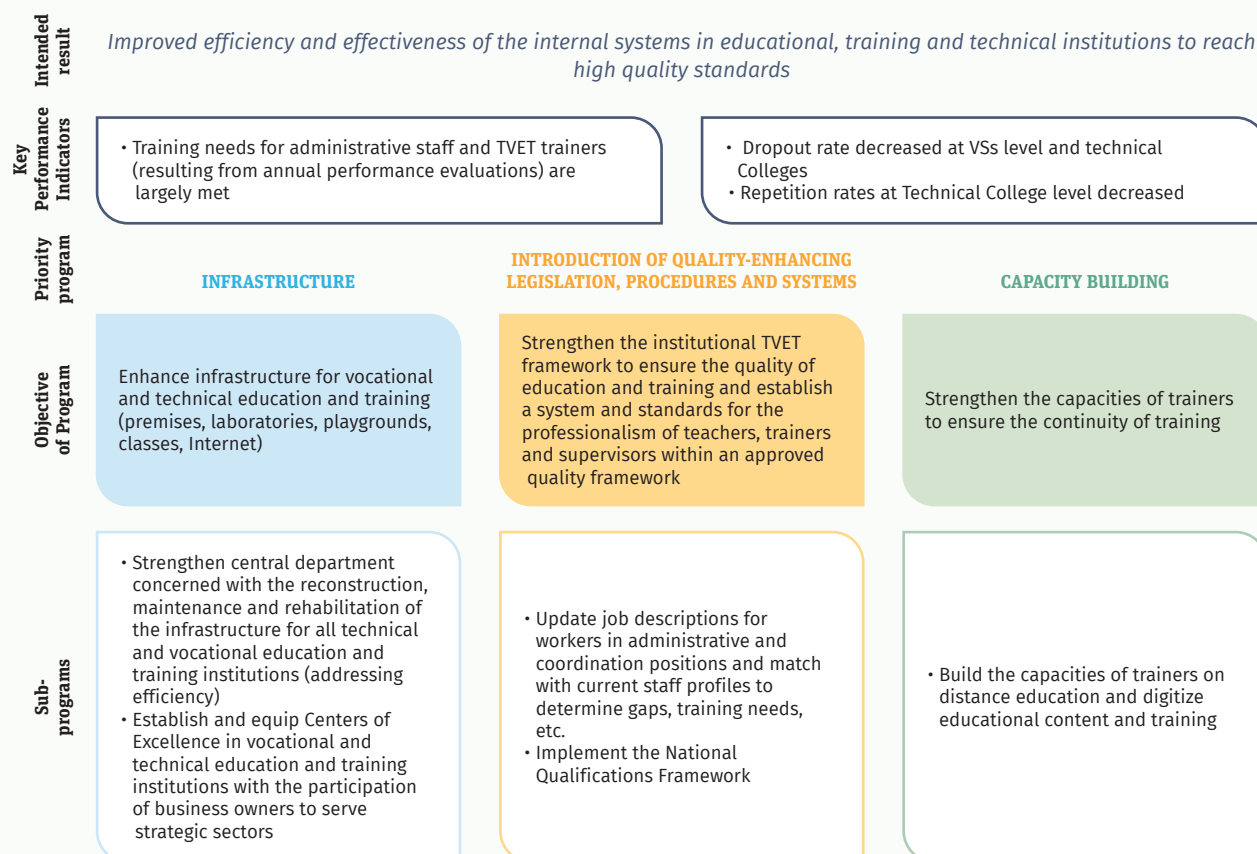


Figure 5.8. Pillar 2's theory of change

If the infrastructure for TVET is enhanced; if the institutional framework of TVET is strengthened such as through standards for professionalization of teachers, trainers, and supervisors within an approved quality framework; and if trainers increase their capacities through in-service professional development, then the efficiency and effectiveness in educational, training, and technical institutions will increase, and repetition and dropout rates will decrease (Figure 5.8).

INFRASTRUCTURE

To meet the envisioned increasing demand for TVET, Pillar 1 indicates the need to strengthen the central department concerned with reconstruction, maintenance, and rehabilitation of TVET institutions and address inefficiencies, including related to the location of TVET institutions.

Sectoral centers of excellence that emphasize employers' engagement in providing TVET, addressing sectoral needs, will present a TVET model that eliminates the prevailing mismatch between supply and demand with a sectoral focus.

QUALITY-ENHANCING LEGISLATION, PROCEDURES, AND SYSTEMS

The TVQF¹⁶ has been established as a normative reference framework covering TVET and professional education that permits recognition of existing in-country qualifications. The Iraqi Inter-Ministerial Working Group, the European Union, and UNESCO officially endorsed the TVQF in 2018. The qualifications will come into full effect once the draft law on the national TVET council is approved. Until then, the framework supports piloting new curricula and competency-based training and assessment. Once activated, the TVQF functions as a structure

¹⁶ UNESCO supported development of the TVQF with funding from the European Union.

for development of all new regulated qualifications and serves as an instrument for the comparison and alignment of international qualifications. The TVQF includes a framework for governing (see Pillar 4) and ensuring the quality of the framework.

Enhancing management of TVET institutions requires that clear roles and responsibilities be defined. Pillar 2 is designed to update job descriptions of staff in key positions and determine existing gaps in skills and tasks, which in turn will help identify training needs.

CAPACITY BUILDING

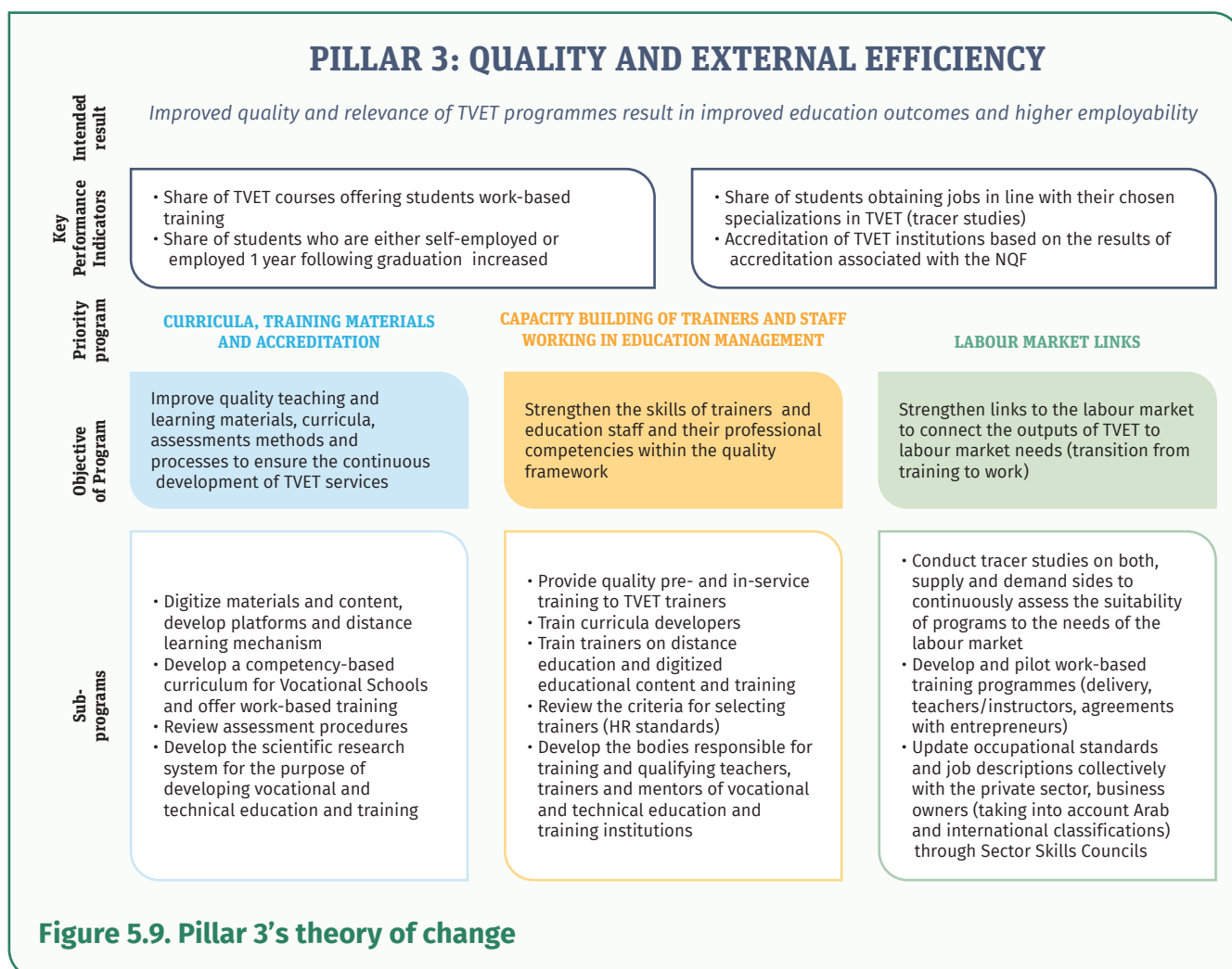
Training of teachers and trainers is a key part of the INES 2022-2031. Under Pillar 2, the focus is on increasing the capacities of trainers on distance, digital, and blended learning and the need to digitize educational and training content. The aim is to reach more students through this approach., . Although digital training has the potential to help teachers and parents support learning away from home and

to reach more learners, strategies are needed to address increasing disparities in access, as evidenced in the digital divide discourse.

5.2.3 Pillar 3: Quality and External Efficiency

Pillar 3 includes three priority programs contributing to the intended result based on the following theory of change:

If the quality of teaching and learning materials, curricula, assessment methods, and scientific research that inform TVET development is improved; *if* trainers and education staff have better skills and greater competencies according to quality standards; and *if* links between TVET and the labor market are strengthened by connecting the outputs of TVET to labor market needs, *then* the quality and relevance of TVET programs will improve, strengthening education outcomes and increasing the employability of TVET graduates (Figure 5.9).



CURRICULA, TRAINING MATERIALS, AND ACCREDITATION

By improving teaching and learning materials, including by updating curricula, the INES 2022-2031 aims to ensure that TVET offers better meet labor market needs. The inclusion of work-based training will allow students to acquire practical skills in partnership with employers. As mentioned in Pillar 2, digitization of training requires an institutionalized approach with well-designed platforms and mechanisms.

CAPACITY BUILDING OF EDUCATION MANAGEMENT TRAINERS AND STAFF

A series of interventions are planned under this priority program targeting curriculum developers and trainers (through pre- and in-service training). Selection criteria for trainers require a review to ensure the best fit in terms of teaching and professional skills needed. This program also involves restructuring and developing existing institutions and departments responsible for training teachers and trainers in the ministries responsible for TVET.

LABOR MARKET LINKS

“The relationships between the TVET institutions, the employment sector and labor market institutions are considered a reciprocal relationship and cannot be separated” (TVET strategy 2014-2023). Considerable efforts are needed to strengthen this relationship. Pillar 3 outlines several opportunities to ensure that the output of TVET institutions—concretely skilled women and men—are matched to the needs of the labor market, allowing for a smooth transition from school or training to work. Expansion of work-based training programs will be critical for this transition. Occupational standards and job descriptions will be updated in coordination with the private sector and business owners through sector skills councils, considering Iraqi, Arab, and international classifications. Tracer studies that follow graduates will allow decision makers to assess the relevance and effectiveness of programs based on labor market needs.

5.2.4 Pillar 4: Governance, Management, and Finance

Pillar 4 includes three priority programs contributing to the intended result based on the following theory of change:

If the institutional framework for delivery of quality TVET programs is improved; if strategic and operational planning capacities, including information management, are strengthened; and if financial management is made more effective, allowing for timely, needs-based financial allocations based on strategic and operational plans with a diversified funding portfolio, then governance, financing, and management frameworks will be improved and can ensure better coordination and greater funding for TVET (Figure 5.10).

GOVERNANCE AND LEGISLATION

The law on the national TVET council, which is awaiting approval from the Council of Ministers and the Parliament, can be the starting point for new institutional arrangements. This also includes the above-mentioned Iraq TVQF, which will come into full effect once the draft law on the national TVET council is approved. Additional legislation is needed to encourage private sector participation in TVET, such as through sectoral skills councils.

STRATEGIC AND OPERATIONAL PLANNING

The national TVET council that will be established once the draft law on the national TVET council is approved (see process outlined above) will be tasked with managing, coordinating, and ensuring the quality of the development of a TVET system that responds to labor market needs. Coordination and collaboration between the respective ministries tasked with providing TVET is a pre-condition for the success of Iraq’s ambitious agenda to transform TVET into a driver of Iraq’s economic growth and employment and consequently contributing to social cohesion. A vocational qualifications department responsible for implementation of the TVQF will be established under the auspices of the national TVET council.

With joint establishment of the labor market information observatory, ministries and private sector partners will be able to identify and analyze market trends and challenges while the Iraqi TVET sector and the labor market are undergoing profound transformations in their structures, mechanisms, and relations. Key thematic areas for analysis will include the future of work and skills, private sector labor needs and vacancies, occupational outlooks, youth unemployment, the effects of digitization and greening of the economy on employment and skills, and labor mobility.

PILLAR 4: GOVERNANCE, MANAGEMENT AND FINANCE

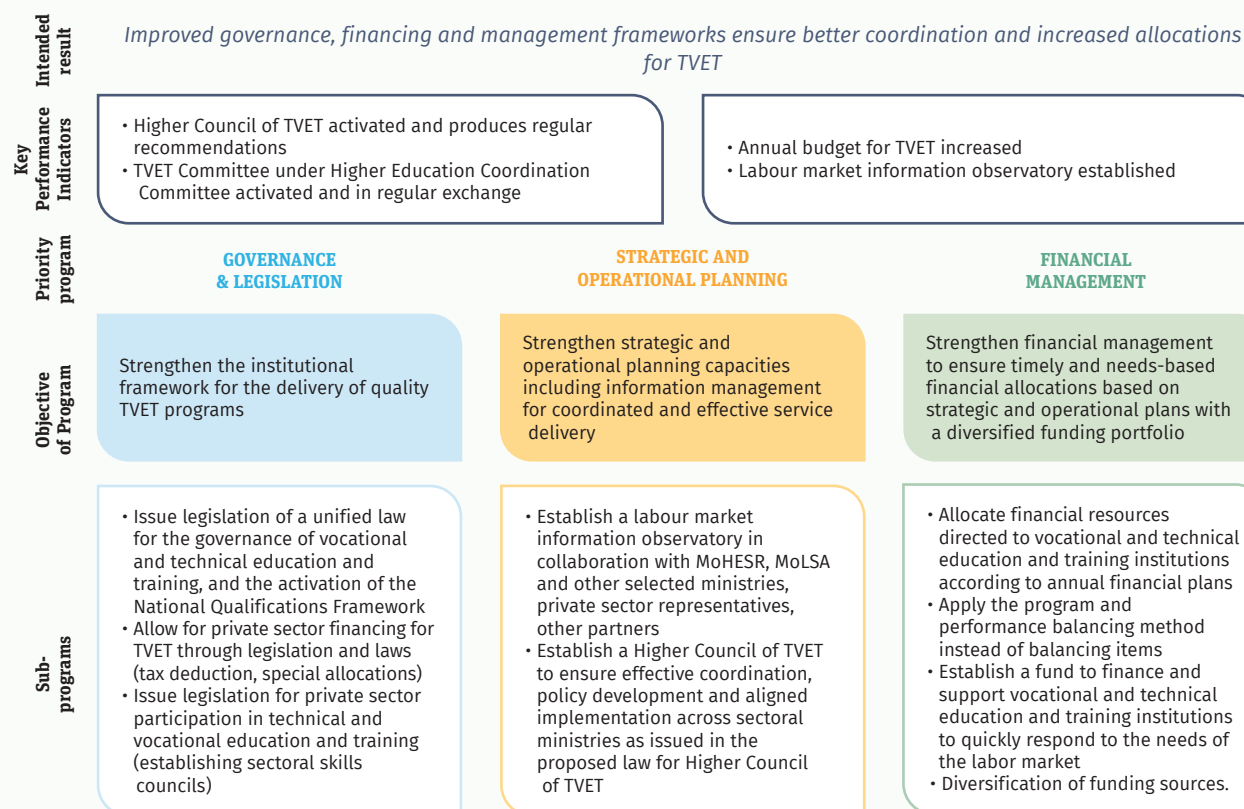


Figure 5.10. Pillar 4's theory of change

Given the need for extensive sources to establish and maintain an effective observatory, Pillar 4 foresees close collaboration between key stakeholders. This collaboration and coordination is also needed when evidence resulting from work the observatory informs policy changes.

FINANCIAL MANAGEMENT

The government source of financing for TVET institutions is within the central investment and operational budget of the federal state because TVET institutions are not part of governorate authorities. On a much smaller scale, some institutions can self-finance some of their offers through production activities. Pillar 4 proposes several actions to improve the financial management of TVET services to ensure timely, needs-based disbursement of budgets and to diversify the current funding portfolio. This includes creating a training fund (or levy) that will allow employers to contribute with a percentage of their payroll bill for the TVET sector to respond to the direct needs of these employers in a sustainable manner. It is hoped that a clear strategy for expanding TVET will

generate external funds by amplifying the government's intentions and readiness to strengthen TVET.

5.3 Higher Education and Scientific Research

The INES 2022-2031 recognizes the importance of higher education to achieve its sector goals. Its organization requires close coordination and alignment with the MoE, which is responsible for pre-university education and universities in Iraq. Despite key improvements in the provision of higher education, technical and academic university education is insufficiently oriented toward the socioeconomic development of the country. As mentioned in section 5.4, Iraq's unemployment rate was 12 percent in 2018, and youth unemployment was 34 percent. Lack of employment opportunities leaves university graduates with a bleak future.

HIGHER EDUCATION

Strategic goal: To develop a high-quality educational system that encourages broad participation through multiple efficient and effective educational paths for excellence in teaching, learning, scientific research and community services that will achieve a comprehensive development in Iraq

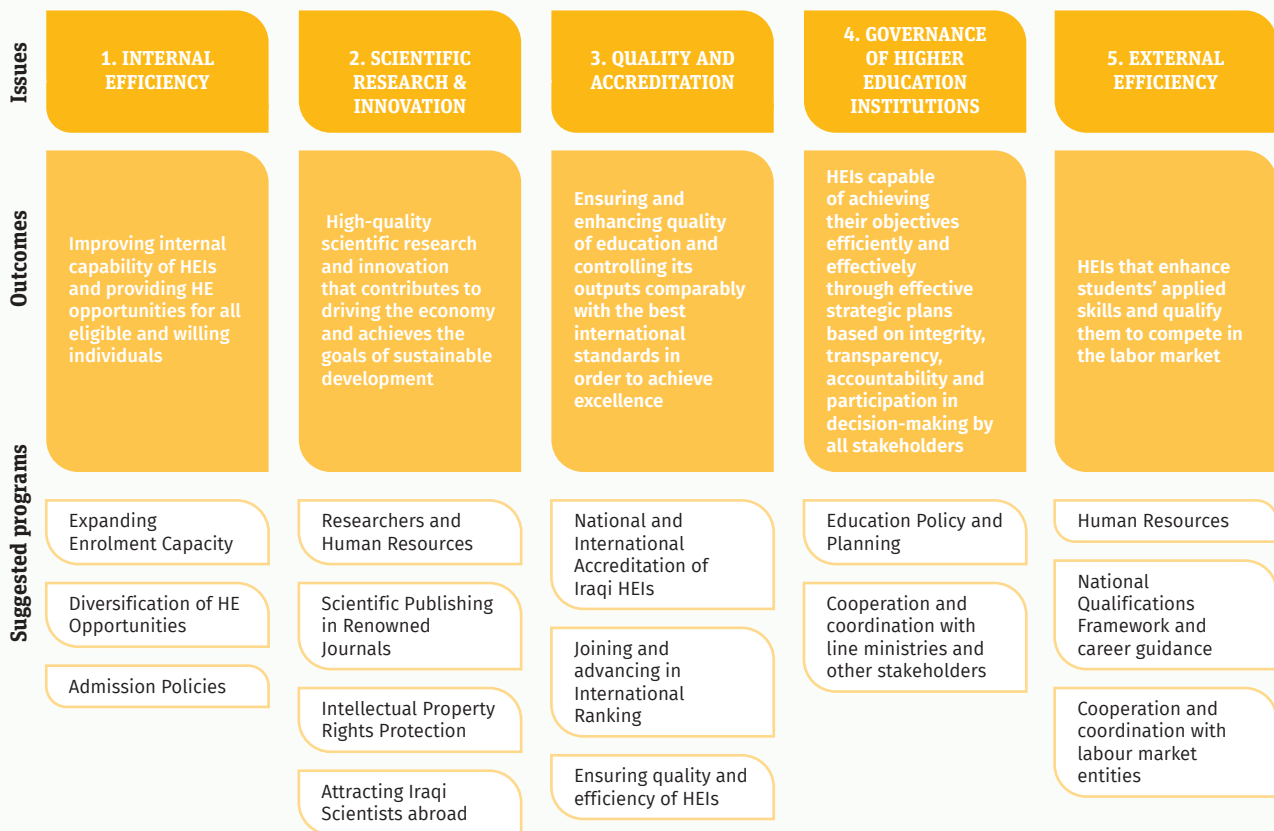


Figure 5.11. Strategic goal, directions and intended results for higher education

Figure 5.11 shows the strategic goal for higher education, its five proposed strategic directions (issues), their respective intended results, and associated priority programs.

The government is committed to developing and implementing a flexible, responsive, demand-driven higher education system to address Iraq's needs for economic development. To this end, the goal is to **develop a high-quality educational system that encourages broad participation through multiple efficient, effective educational paths for excellence in teaching, learning, scientific research, and community service that will enable a comprehensive renaissance in Iraq.**

More concretely, the INES 2022-2031 aims to:

- Provide higher education opportunities based on demand and labor market requirements

- Upgrade the quality of higher education system components so that high-quality graduates are produced
- Provide graduates with applied skills relevant to local and regional labor market needs
- Ensure a sustainable funding model for Iraqi universities
- Ensure a flexible, accountable, transparent, well-integrated higher education system
- Ensure high-quality scientific research

The following sections explore each of the five issues in detail, providing a theory of change detailing the objectives of the priority programs and related sub-programs. Progress toward objectives is measured using a series of KPIs (outcome-level indicators) that are shown in the figures, as well as selected output-level indicators at the level of priority programs. (See the results matrix in chapter 6 for a full list of indicators according to sub-sector.)

5.3.1 Issue 1: Internal Efficiency

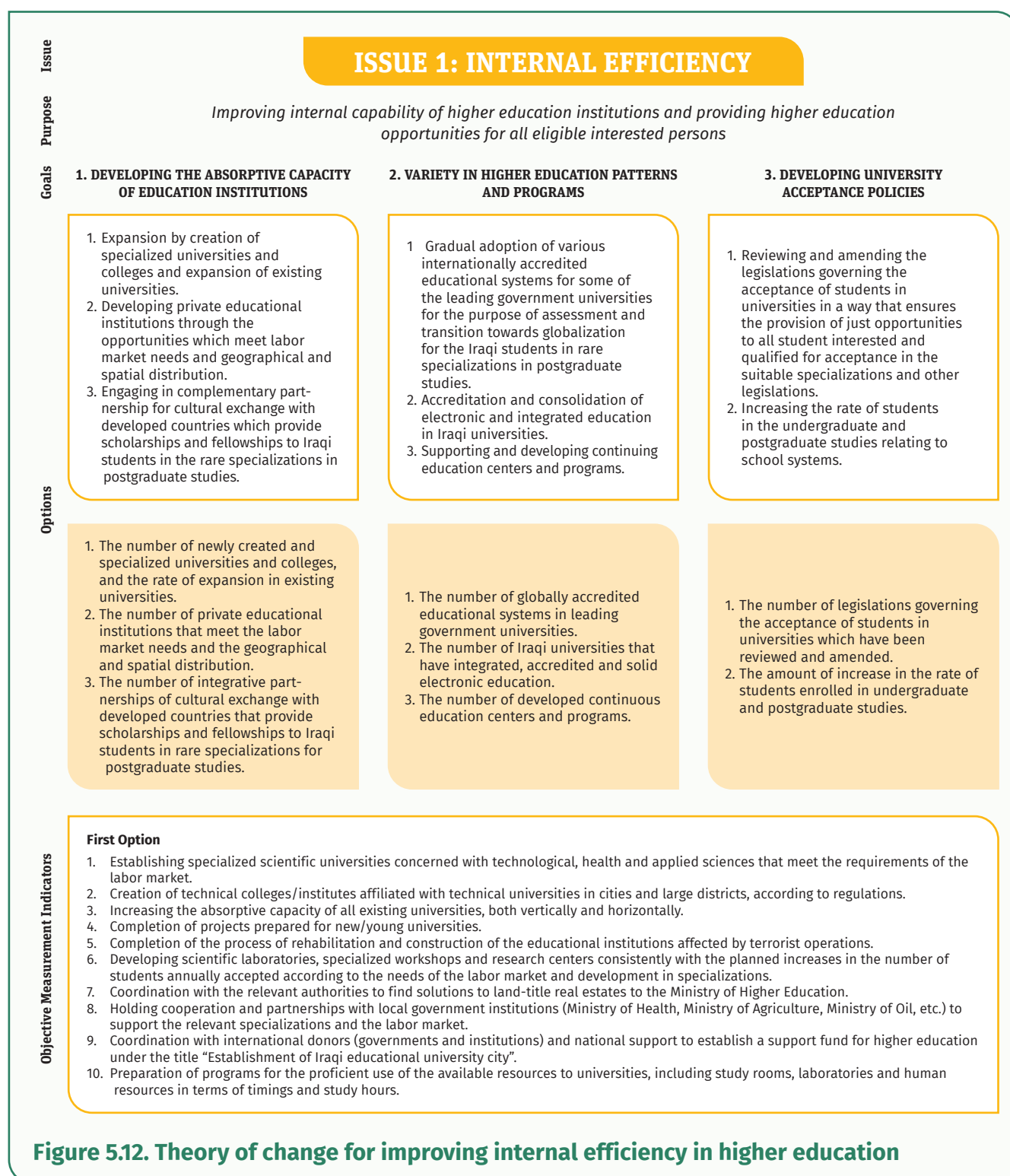
Issue 1 includes three priority programs contributing to the intended result based on the following theory of change:

If more higher education institutions are established with diverse types and programs, and *if* admission policies are reformed, *then* greater access and internal efficiency will be achieved, with greater

enrollment capacity for all those willing and well qualified (Figure 5.12).

INCREASING ENROLLMENT CAPACITY OF HIGHER EDUCATION INSTITUTIONS

A key driver for enhancing the capacity of higher education institutions is to establish new universities and higher education institutions of different types



in various geographic locations and new programs of study and fields of specialization that respond to the need for development of Iraq. This includes expansion of public and private higher education institutions, with special attention paid to areas with destroyed or damaged infrastructure. Development partners must work together with the government to establish a fund that supports the development of the higher education sector, including establishment of an education and innovation city and similar new models of higher education institutions. These strategies are aligned with governance and finance, with coordination needed with relevant ministries and local authorities to ensure allocation of land and properties.

DIVERSIFYING TYPES AND PROGRAMS OF HIGHER EDUCATION INSTITUTIONS

New modes of learning and specializations based on sound international practices are required with robust impact studies to ensure effective implementation and relevance according to objectives established in national development plans. A scholarship plan is included to support creation of the human capital needed to run programs in new fields (e.g., artificial intelligence, mechatronics, nanotechnology) because there is a significant shortage in these fields at Iraqi universities. Lifelong learning, continuing education, eLearning, and distance education are other strategic options for increasing access to higher education and flexibility of admission and enrollment, so the strategy capitalizes on formalizing and recognizing these modes of learning to be well integrated into the higher education system in Iraq.

REFORMING UNIVERSITY-LEVEL ADMISSION POLICIES

Admission policies for higher education must be updated and reformed to consider admission factors other than grades obtained at the preparatory level. A university admission examination should be introduced to assess students' competencies that allow access to required specializations. This requires coordination with the MoE to ensure integration in the science and humanities tracks as well as opening pathways for graduates of vocational schools. To provide equitable access to higher education institutions, the MoHESR also aims to admit or re-admit candidates with training and work experience after secondary education.

5.3.2 Issue 2: Scientific Research and Innovation

Issue 2 aims to achieve a quality scientific research and innovation system that contributes to sustainable development, including as outlined in the SDGs. Issue 2 consists of four priority programs contributing to the intended result based on the following theory of change:

If Iraqi researchers' competencies and skills are upgraded to increase their research productivity, *if* funding for research is provided for the purpose of publishing in renowned academic journals while protecting intellectual property rights of universities at Iraq, and *if* Iraqi scientists find an attractive research context that makes *them* return to their country, then high-quality research is to be produced, leading to innovation and economic development (Figure 5.13).

UPGRADING IRAQI RESEARCHERS' SKILLS AND INCREASING QUALITY SCIENTIFIC PRODUCTIVITY

Research skills are the main prerequisite to producing high-quality scientific research, in addition to provision of basic means of scientific research, including improvement in research infrastructure, and provision of quality research facilities and equipment. Quality scientific research can then respond to the needs of industry. This requires more funding for research in collaboration with industry.

FUNDING OF RESEARCH FOR PUBLISHING IN RENOWNED ACADEMIC JOURNALS

Increasing funding for scientific research generally increases research productivity by increasing access of researchers to international knowledge resources and databases. This in turn enables researchers to publish in internationally peer reviewed journals, recognizing quality research aspired by Iraqi universities. Policy reforms will be introduced to ensure adequate funding for research in collaboration with industry.

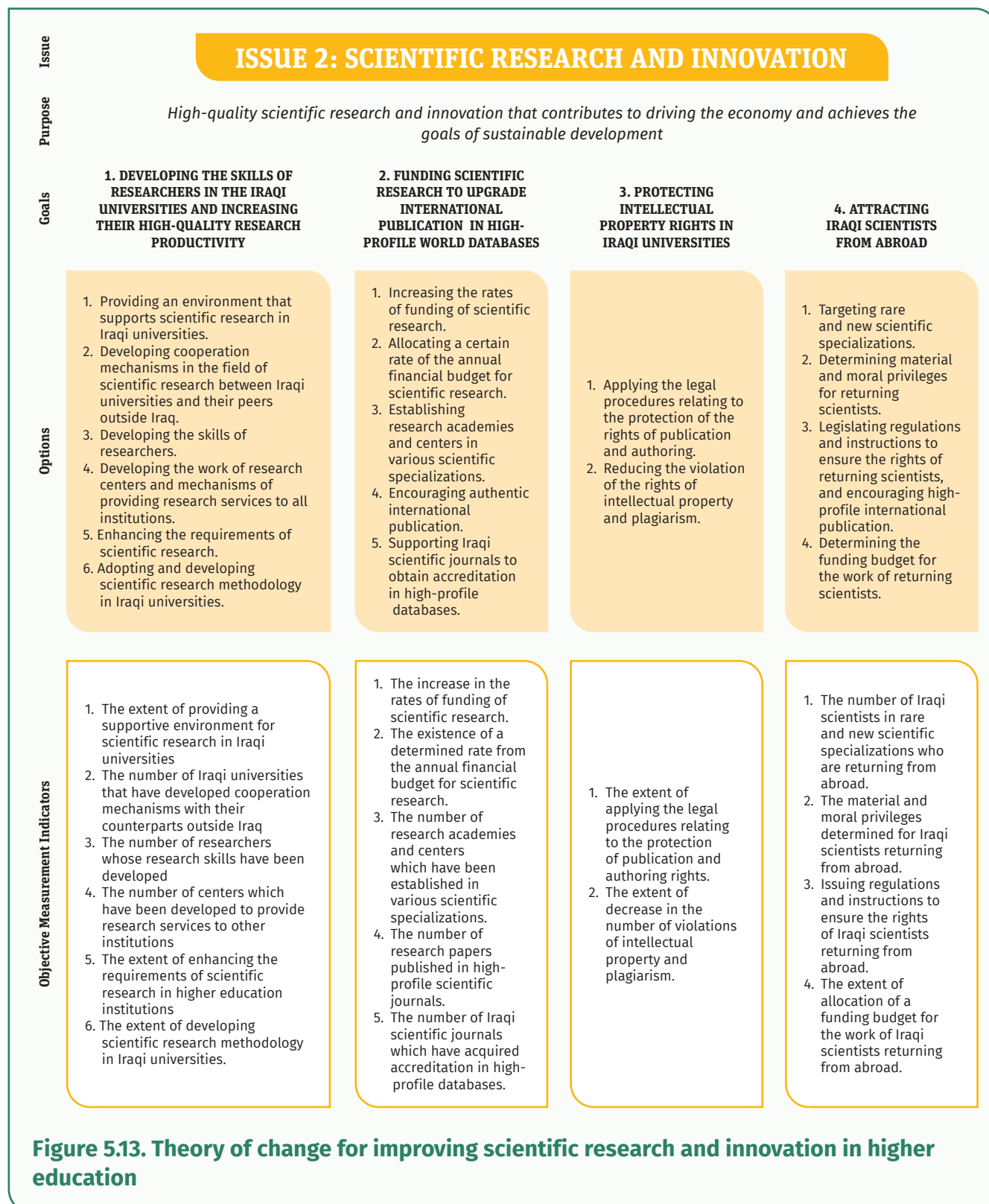
INTELLECTUAL PROPERTY PROTECTION FOR UNIVERSITIES

Protecting intellectual property rights of universities is another critical requirement for enhancing research productivity and return on investment. Enforcement of intellectual property protection and reform of intellectual property registration procedures are critical.

ENCOURAGING EXPATRIATE IRAQI SCIENTISTS TO RETURN TO IRAQ

The war against terrorism resulted in considerable brain drain, with hundreds of Iraqi scientists leaving the country. It is hoped that improving the research

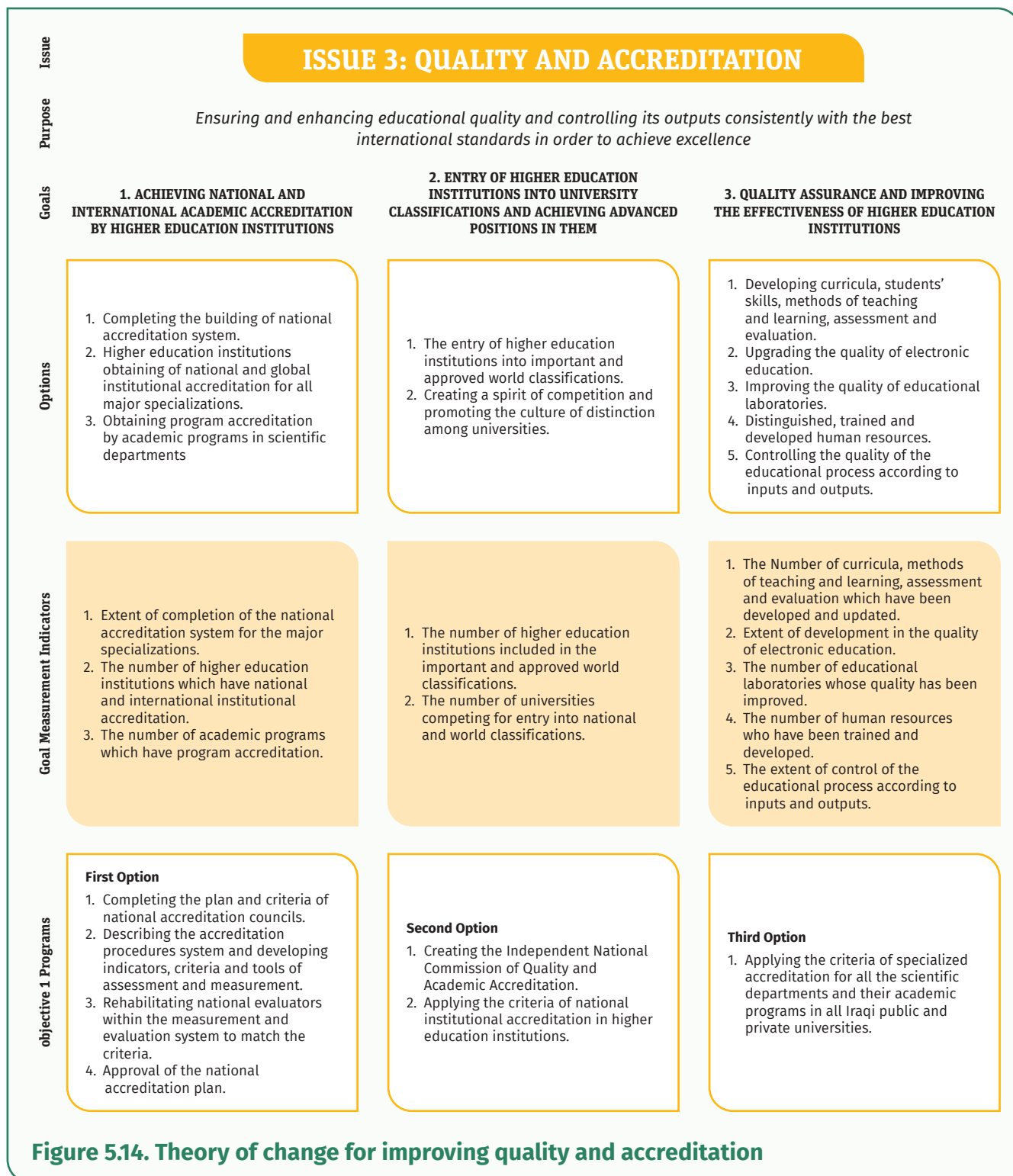
environment at Iraqi universities will encourage Iraqi scientists to return, contributing to high-quality research. Special focus is to be placed on rare and modern fields of specialization required for successful implementation of Iraq's vision, outlined in the Iraq National Strategy 2030.



5.3.3 Issue 3: Quality and Accreditation

Issue 3 aims to ensure and enhance quality of education and control over educational outcomes in accordance with the best international quality standards. Issue 3 includes three priority programs contributing to the intended result based on the following theory of change:

If more higher education institutions can obtain national and international academic accreditation to ensure quality and increase the effectiveness of their services, and if Iraqi universities can engage in international university ranking systems and league tables and consequently achieve an advanced position, then higher education will produce better learning outcomes (Figure 5.14).



NATIONAL AND INTERNATIONAL ACADEMIC ACCREDITATION

Options adopted in this goal focus on reestablishing a national accreditation system. Higher education institutions will be able to apply for national and international accreditation in all academic disciplines and for specific academic programs. National academic reference standards for each academic discipline will be developed, and a comprehensive data system including standards and indicators must be developed to guide decisions about quality assurance and accreditation.

INTERNATIONAL UNIVERSITY RANKING SYSTEMS AND LEAGUE TABLES

Iraqi universities should participate in international university ranking systems, which would help create a competitive higher education ecosystem that drives excellence and continuous improvement through national and international benchmarking. Issue 3 will help universities increase their readiness to participate in international university ranking systems, including through financial support for such things as subscription fees. The MoHESR aims to improve the ranking of Iraqi universities in international league tables within the strategy period.

QUALITY ASSURANCE AND INCREASING THE EFFECTIVENESS OF HIGHER EDUCATION INSTITUTIONS

Issue 3 plans for development of curricula and improvement of new and existing university programs and academic disciplines, including modes of delivery that integrate e-learning and blended learning. Capacities of teaching and administrative staff and physical resources in the form of infrastructure, laboratories, and learning resources will be increased while prioritizing safety and security standards and measures. The academic supervision division at the MoHESR or a quality assurance agency to be established will ensure the quality of inputs and outputs.

5.3.4 Issue 4: Governance of Higher Education Institutions

Issue 4 aims to enable higher education institutions to achieve their goals efficiently and effectively by improving their performance through strategic plans that have transparent, participatory, accountable goals. Issue 4 includes two priority programs contributing to the intended result based on the following theory of change:

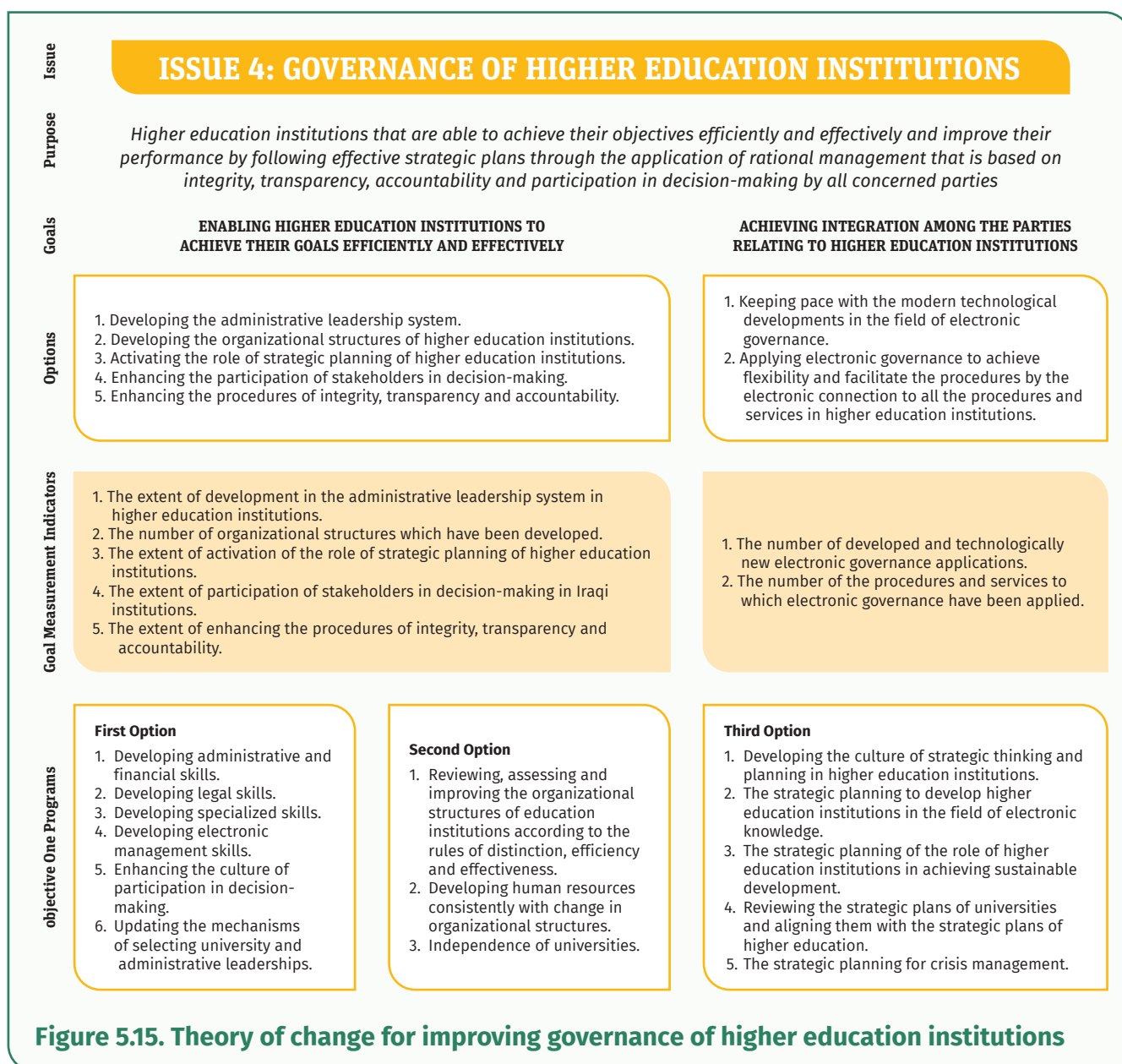
If the governance and finance system of higher education institutions is improved, and if Iraqi universities can engage all stakeholders, then higher education institutions will be empowered to achieve their goals in an effective, efficient way (Figure 5.15).

LEADERSHIP SUPPORT

Enhancement and support of university leadership structures will empower Iraqi higher education institutions while engaging stakeholders in the decision-making process through a well-structured university strategic plan that provides transparency, accountability, and integrity. The strategy will use professional development of university staff in leadership and management skills, establishment of national and institutional electronic management information systems, and reform of leaders' recruitment and selection systems for this purpose.

INTEGRATION

Periodic mapping of internal and external stakeholders; their involvement on boards of academic departments, faculties, and universities; and sharing of outcomes of their involvement are the primary areas this option focuses on. Outcome indicators will be adopted to ensure active, meaningful stakeholder participation in governance of Iraqi higher education institutions.



5.3.5 Issue 5: External Efficiency

Issue 5 aims to enhance students' applied skills to enable them to compete in the labor market. Issue 5 includes three priority programs contributing to the intended result based on the following theory of change:

If higher education institutions can establish solid partnerships with industry, *if* university professors can integrate global advancements in their teachings, and *if* graduates can better respond to labor market and society needs, *then* higher education graduates will be more employable (Figure 5.16).

SUSTAINABLE PARTNERSHIPS WITH LABOR MARKET AND INDUSTRY

Sustainable partnerships with the labor market and industry are critical for increasing the external efficiency and relevance of higher education. This requires comprehensive labor market information to inform higher education policies, planning, and teaching, as well as strong scientific and applied links between higher education institutions and business entities.

PROFESSIONAL DEVELOPMENT

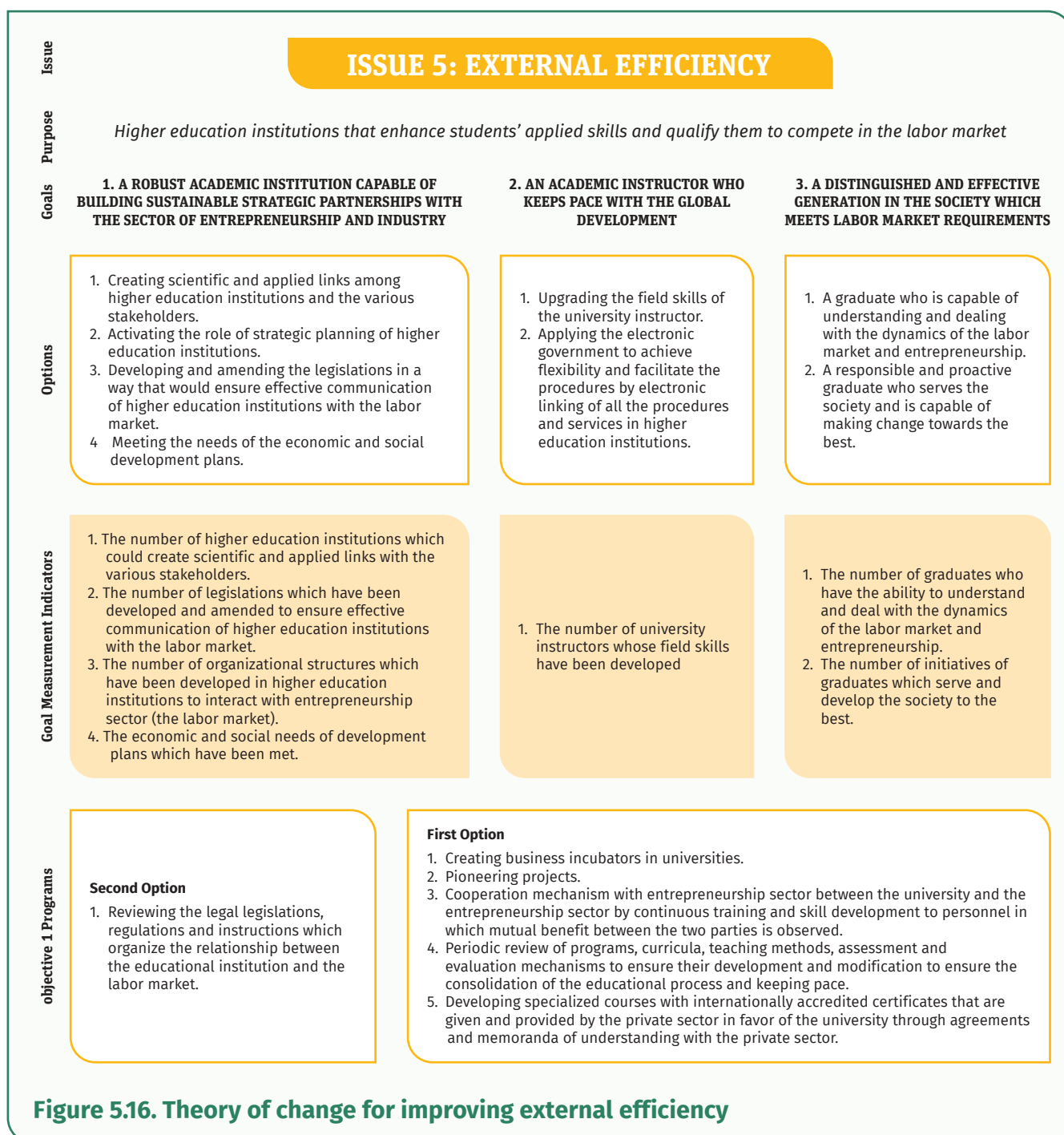
Continuous professional development on global advancements will enable university teaching staff

to ensure that academic courses are linked to real-world problems, teaching students' relevant knowledge and skills applicable in the labor market, but also contributing to the development of industry and hence creating win-win partnerships.

HIGHER EDUCATION OUTPUTS RESPOND TO LABOR MARKET NEEDS

This option enables higher education institutions to address emerging skills gaps and compete in a rapidly changing world by strengthening students' skills

and creating stronger links with business entities. An integral part of this option is development of a national qualifications framework that supports integration of professional experience into learners' credentials. Establishment of the labor market observatory in cooperation with other line ministries (see also section 4 TVET in this chapter) is a mechanism to continuously observe and evaluate labor market needs and job requirements. Close collaboration and coordination with general education and TVET stakeholders is a success indicator for this option.



Chapter 6

Cost and Financing

This chapter outlines the costs and resources required to implement the INES 2022-2031 and achieve its policy targets. In preparation of the INES, three quantitative scenarios were developed and discussed¹⁷ that anticipate different levels of educational achievements and then project the human and physical resources required to accommodate expected enrollment at the general education, TVET, informal education, and higher education levels. The projected costs of the third scenario amplify ambitious, yet realistic and holistic development of the education system over the next 10 years through strategic investments aimed at increasing access to and quality of education and addressing inefficiencies in the education system

The chapter begins with introducing the Iraq Simulation Model and then outlines the criteria used for the simulation and the macro-economic framework, provides a detailed description of the cost implications of agreed-upon policy targets, and discusses financing gaps.

17 The first scenario reflects no major changes to the education system development and can therefore be considered as “business as usual.” The second scenario works on the assumption that strategic policies and sub-sector strategies will be implemented, specifically regarding increasing access, internal efficiency, and quality of education services from kindergarten to higher education. A policy note was prepared to present scenarios 1 and 2 as a basis of discussion for development of scenario 3. Scenario 3, which is the scenario adopted for the INES 2021-2031, results from trade-off discussions based on scenario 2 between the MoE, MoHESR, and MoP and ensures alignment between the three sub-sectors and careful consideration of technical, financial, and human capacities to implement the identified priority programs.

6.1 The Iraq Education Simulation Model

The Iraq Education Simulation Model is a spreadsheet that has been used to create quantitative scenarios for development of education and assess the financial and human resource consequences of the education objectives in the INES 2022-2031.

The model includes:

- projections of enrollment based on demographic projections and specific access and efficiency objectives (e.g., gross intake ratios, repetition rates)
- projections of numbers of teaching and non-teaching staff, classrooms, learning materials, and other resources needed to accommodate the projected number of students
- projections of key operational¹⁸ costs of the national school meal program, school grants for all public kindergartens and primary and secondary schools, provision of scholarships in preparatory and higher education, and inclusion of learners with special needs
- projection of financial resources required

Education data within the model are official data from the Central Statistical Office of the MoP. Financial data were taken from the Ministry of

Finance Open Budget Survey website and the World Bank Public Expenditure Review conducted in 2020. Additional macroeconomic data (GDP, GDP growth rates) were obtained from the International Monetary Fund (IMF) online database. The MoP provided information on macroeconomic data, although that was limited to 2019 and was similar to IMF projections. References made in this chapter, as throughout the strategy document, to “Iraq” exclude the governorates of Dohuk, Erbil, and Sulaymaniyah. Tables and graphs presented in this chapter are directly derived from the Iraq Education Simulation model. The official enrollment projections from the Central Statistical Office, which apply a constant annual growth rate of 2.6 percent to 2.7 percent, have been used as the demographic framework for the INES 2022-2031.¹⁹ According to these projections, the overall school-aged population is expected to increase by approximately 27 percent from 2021 to 2031, amounting to almost 20 million children and youth aged 4 to 22. Table 6.1 shows the population projections according to age group.

The INES 2022-2031 projects a steady increase in enrollment over the 10-year period of the strategy (Figure 6.1). The number of enrolled learners in pre-university education is projected to increase from 11.4 million at the start of the INES (2022) to 16.2 million in 2031. It is projected that enrollment in higher education will grow from 729,000 students in 2022 to 1.3 million in 2031.

Table 6.1.
Population Projections According to Age Group, 2021-2031

Age (school level)	2021	2031 (end of INES)
4-5 (kindergarten)	2,119	2,695
6-11 (primary)	5,661	7,200
12-14 (intermediate)	2,527	3,214
15-17 (preparatory)	2,346	2,984
18-22 (higher education)	2,847	3,621
Total	15,500	19,713

Source: Central Statistical Office.

¹⁸ According to terminology used in Iraq, “operational” costs are equivalent to “recurring” costs.

¹⁹ The last full population census was conducted in 1997.

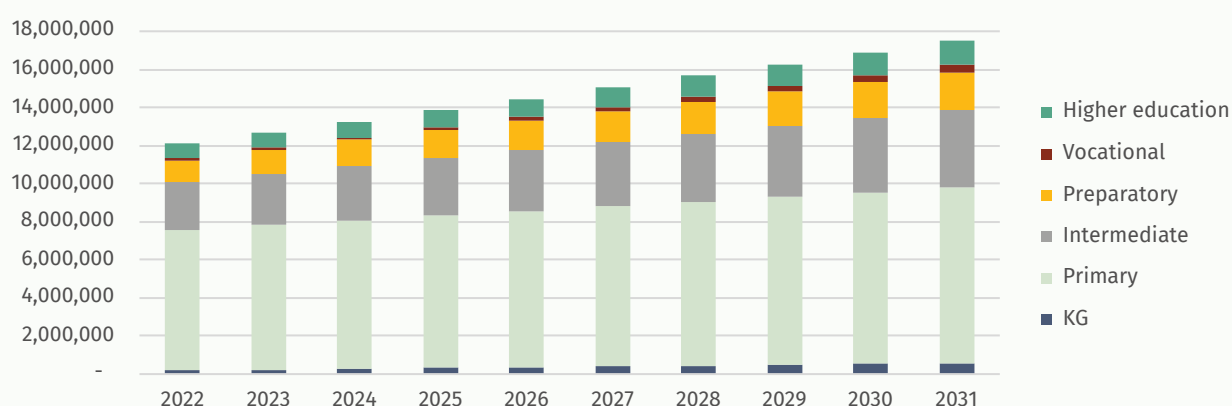


Figure 6.1. Enrollment Projections According to Educational Level, 2022-2031

Source: Iraq Simulation Model 2021.

Key reforms within the INES 2022-2031 are projected to start in 2022, with gradual improvement in key education parameters in the first 5 years of the strategy. The assumption is that many regulatory and policy frameworks must be put in place before effective change such as in enrollment, repetition, and dropout will be observed. Key regulations concerned with teacher recruitment and allocation and infrastructure should be addressed in the initial phase of the strategy (2022-2025). Annual salary increases are projected to begin in 2025. Non-salary operational expenditures, including in-service teacher training, provision of teaching and learning materials, school meals, and school grants, are projected from the beginning of the strategy.

6.2 Targets and Hypotheses Used for the Simulation

The cost of INES 2022-2031 depends primarily on its policy objectives, which are translated into specific intake and internal efficiency targets as part of the government's overall vision for sustainable development. The INES 2022-2031 defines several key targets:

- Increase school readiness of pre-primary-aged children by increasing enrollment from 10.7 percent of the gross enrollment rate (GER) in 2018/19 to 30 percent by 2031, including by increasing the share of private provision of kindergarten to 25 percent
- Increase completion of primary education to 100 percent by 2025 from approximately 83 percent in 2018/19
- Increase completion of intermediate education to 100 percent by 2028 from approximately 62 percent in 2018/19
- Increase GER for preparatory education (academic) from 47 percent in 2018/19 to 68 percent by 2031 for boys and from 46 percent to 66 percent for girls
- Expand vocational education at the preparatory level to so that more students complete grade 12 or access technical training (GER from 7 percent/3 percent in 2018/19 to 14 percent/13 percent by 2031) so that they are prepared to enter the labor market, considering the needs of the national economy
- Ensure controlled expansion of higher education with 82 percent of academic preparatory graduates (compared with 100 percent in 2018/19) and 50 percent of vocational preparatory graduates projected to enter universities and colleges by 2031 while keeping current levels of private provision at 25 percent
- Improve teaching and supervision conditions by decreasing kindergarten and primary education class sizes (from 64 to 30 for kindergarten and 40 to 35 for primary), keeping class sizes at 35 at the secondary level, and increasing class sizes in vocational schools to 30
- Increase the number of students participating in informal education programs from approximately 70,000 to 300,000
- Increase internal efficiency by regulating repetition, reducing the teacher: student ratio, applying new teacher allocation regulations, and increasing the number of teaching hours

Funding is also provided to establish several programs to reduce disparities in access to education in terms of gender, disability, displacement, location, and poverty and improving the quality of education:

- Gradual increase in provision of school meals to all public primary school students by 2030 and to the most vulnerable children attending pre-primary education (25 percent of all children attending kindergarten by 2031)
- Provision of school grants as part of the school-based development program to all public kindergartens and primary and secondary schools from 2022 on²⁰
- Provision of textbooks in the primary (three per student for grades 1-3, 7 for grades 4-6), intermediate (five per student), and preparatory (five per student) levels and a set of learning materials for vocational education
- Provision of scholarships to 2 percent of all students in preparatory and 5 percent of all students in higher education

The pillars and priority programs were described according to sub-sector in chapter 5, highlighting how these objectives will be achieved. Related policy targets are described in the results matrix in Chapter 7.

6.3 The Macroeconomic Framework for the INES 2022-2031

Over the past decade, education spending as a share of total government expenditure and GDP has declined steadily. In 2019, education in Iraq center received only 9.7 percent of total government spending, down from 13.6 percent in 2016. In 2019, 3.9 percent of GDP was allocated to education, down from 5.1 percent in 2009 (World Bank 2021). Iraq falls significantly below

the Middle Eastern and North African average of government spending on education (14 percent) and fails to meet the minimum international benchmarks for public education financing according to the Education 2030 Incheon Declaration, which calls for at least 4 percent to 6 percent of GDP or at least 15 percent to 20 percent of total expenditures to education to work toward SDG 4. The World Bank estimates that total government spending in Iraq center in 2019 was 10.8 trillion dinars.²¹

Education spending is heavily skewed toward operational expenditures, most notably wages. Of the estimated 10.8 trillion dinars spent on education in 2019, 93 percent went to wages, leaving little room for non-personnel operational expenses (4.7 percent in 2019) and capital investment (2.4 percent in 2019).²² Capital spending, such as for building and rehabilitating schools, has been chronically under-resourced in Iraq, contributing to the need for safe and learning-conducive school buildings after years of destruction and neglect. In 2019, after 2 years of virtually no investment, the MoE's capital expenditures budget was less than 24 billion dinars. The MoHESR accounted for less than 4 percent of education investment expenditure in 2018 and 2019. The governorates, the Council of Ministers, and other ministries and agencies spend most of the education investment budget (96 percent in 2018, 87 percent in 2019). Concerns expressed regarding the implementation of INES relate to the need to increase investment significantly throughout the course of implementation and to address under-execution of the investment budget, especially regarding school construction.²³ Section 6.x discusses this in more detail.

Although the Iraqi national economy, including the education sector, is likely to face financial constraints in the coming years because of the economic impacts of the COVID-19 pandemic coupled with volatile oil revenues (Iraqi GDP fell by almost 10 percent in 2020), the most recent estimates project a return to

20 School grants are projected at a unit cost of 2,000 dinars per student for primary and secondary schools and 6,000 dinars for vocational schools. The unit cost is based on number of students per school. School grants do not replace other operational non-salary expenses such as textbooks, in-service teacher training, and school meals.

21 Finance data for the macroeconomic framework relied fully on the World Bank Public Expenditure Review, given the difficulties of calculating the total amount of public spending on education. There are several reasons for this, including lack of a functional classification within the federal budget to identify and quantify education sector expenditures across all spending units. Decentralization of most public education spending to the governorate level since 2017 has increased fragmentation in the reporting of education spending across different spending units.

22 According to the 2021 World Bank Public Expenditure Review, governorates and other spending units financed 60 percent of the education wage bill, the MoE financed 17 percent, and the MoHESR financed 23 percent. The MoE and MoHESR financed 41 percent and 20 percent of non-personnel recurring expenditures, respectively, and the governorates and other units accounted for the remaining 39 percent.

23 In 2019, the execution rate of capital expenditures was 20 percent for the MoE and 13 percent for the MoHESR (World Bank 2021, 113).

growth as early as 2022, during which the economy is expected to grow at approximately 4.4 percent, increasing to 6 percent in 2023.

Within that framework, funds available for education during implementation of the INES 2022-2031 can be estimated based on three parameters:

- The evolution of GDP;²⁴ which is based on IMF projections until 2026, after which an average increase of 3.3 percent is applied up to 2031
- Government expenditures as a share of GDP, with the assumption that the rate will remain at current levels of approximately 35 percent
- Education as a share of total government expenditures, which is expected to increase from 9.7 percent in 2019 to 16 percent in 2031; operational expenditures is expected to increase from 12 percent in 2019 to 17.5 percent in 2031, and investment expenditure is expected to increase from 1 percent to around 7 percent. Expanding the share of investment expenditures in the education budget requires “increasing the investment budget allocations dedicated to the education sector in the medium term, as well as improving the efficiency of budget disbursement and utilization” (World Bank 2021:99).

Under these assumptions, total public spending for education would thus increase from around 11 trillion dinars in 2019 to around 19 trillion dinars in 2031 (in 2019 constant dinars), accounting for 5.6 percent of GDP.

Table 6.2 provides an overview of financial parameters for the INES 2022-2031.

Although the evolution of the first two parameters is external to the education system, the share of government expenditures on education relies on convincing and financially sound arguments presented by the MoE and MoHESR to the Ministry of Finance. The existence of a high-quality long-term plan should strengthen this argument, especially because the government has identified the education sector as critical to overcoming longstanding development challenges (GoI 2021). The INES 2022-2031 also includes several strategies to strengthen the financial management of the education sector by increasing engagement between the MoE, the MoHESR, and the Ministry of Finance within a wider public financial management reform.

Table 6.2.
Financial Parameters of the Iraq National Education Strategy 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
GDP (billion constant 2019 dinars)	247,294	262,206	273,814	282,650	291,988	301,635	311,600	321,894	332,529	343,514
GDP real growth rate (%)	4.4	6.0	4.4	3.2	3.3	3.3	3.3	3.3	3.3	3.3
Government expenditure as % of GDP	32.5	32.8	33.0	33.3	33.6	33.9	34.2	34.4	34.7	35.0
Education budget										
Billion constant 2019 dinars	9,320	10,375	11,360	12,280	13,269	14,321	15,440	16,629	17,894	19,237
As % of total government expenditures	11.6	12.1	12.6	13.0	13.5	14.0	14.5	15.0	15.5	16.0
As % of GDP	3.8	4.0	4.1	4.3	4.5	4.7	5.0	5.2	5.4	5.6

Source: Iraq Simulation Model 2021.

24 GDP includes Kurdistan, whereas the rest of financial and education data are specific for Iraq center.

6.4 Total Cost of the INES 2022-2031

Table 6.3 shows the projected cost of the INES 2022-2031 (~140 trillion dinars in 2019 constant values) for the duration of strategy implementation according to education level. There is a significant increase in nominal terms in pre-primary and vocational education, which reflects the planned expansion of both sub-sectors.

Closer examination of the share of operational and investment expenditures reveals that, although operational expenditures remain the largest part of education spending, investment expenditures are expected to become an increasingly important part, increasing from 7 percent of total education expenditures in 2022 to more than 10 percent in 2031 (Table 6.4), which is in line not only with current infrastructure needs, but also the quantitative expansion of the system expected in the coming years.

Table 6.3.
Projected Total Expenditures (Operational + Capital) According to Education Level, 2022-2031

Level	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
2019 constant billion dinars										
Kindergarten	137	213	245	278	314	352	394	441	495	560
Primary	5,161	5,390	5,434	5,478	5,563	5,660	5,755	5,846	5,943	6,043
Intermediate	2,161	2,382	2,514	2,693	2,846	2,982	3,108	3,242	3,375	3,509
Preparatory academic	1,382	1,461	1,488	1,503	1,484	1,471	1,458	1,456	1,445	1,416
Preparatory vocational	465	316	400	449	493	541	595	648	697	737
Higher education	2,732	2,710	2,762	2,854	2,966	3,084	3,195	3,303	3,415	3,531
Informal	87	105	122	139	156	174	191	208	225	243
TOTAL	12,125	12,579	12,964	13,394	13,823	14,264	14,696	15,144	15,594	16,039

Source: Iraq Simulation Model 2021.

Table 6.4.
Projected Operational Expenditures and Share of Operational Expenditures Out of Total Education Expenditures, 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Operational expenditures, 2019 constant billion dinars	11,084	11,274	11,621	11,961	12,312	12,652	12,966	13,267	13,556	13,829
Share of operational expenditures out of total education expenditures, %	92	90	90	90	90	90	90	90	90	89

Source: Iraq Simulation Model 2021.

Table 6.5.

Projected Intra-Sectoral Allocation of Operational Expenditures According to Education Level, 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
	%									
Kindergarten	1	1	1	2	2	2	2	2	2	2
Primary	42	42	42	42	41	40	40	39	39	39
Intermediate	18	19	19	19	20	20	21	21	22	22
Preparatory academic	11	11	11	11	11	11	11	10	10	10
Preparatory vocational	4	3	3	3	3	4	4	4	5	5
Higher education	23	22	22	22	22	21	21	21	21	21
Informal	1	1	1	1	2	2	2	2	2	2

Source: Iraq Simulation Model, 2021.

Table 6.5 shows the distribution of operational expenditures according to education level. In 2031, it is projected that pre-university education will account for 77 percent of all operational expenditures, with higher education accounting for 21 percent and informal education for 2 percent.

6.4.1 Analysis of Operational Expenditures

Within operational costs, it is projected that the share of salaries for teaching and non-teaching staff will

decrease from around 96 percent in 2022 to around 90 percent in 2031, with the most significant change in the wage bill for non-teaching staff, from almost 19 percent to less than 11 percent of total operational expenditures (Annex A). Expenditures for teacher salaries will increase slightly, from 77 percent in 2022 to 80 percent by 2031, because of the substantial increase in enrollment in all levels of education. The projected share of the wage bill by 2031 is the lowest for primary education (85 percent). The projections assume that teacher: class ratios and non-teaching staff: teacher ratios²⁵ decrease significantly. Table 6.6 summarizes these policy targets.

Table 6.6.

Projected Non-Teaching Staff: Teacher and Teacher: Class Ratios, 2018/19 and 2031, According to Education Level

	Teacher: class ratio		Non-teaching staff: teacher ratio	
	2018	2031	2018	2031
Kindergarten	2.1	1.1	62%	15%
Primary	1.7	1.3	25%	10%
Intermediate	2.0	1.5	17%	10%
Preparatory academic	2.2	1.5	28%	25%
Preparatory vocational	-	-	16%	10%
Higher education	-	-	50%	50%

Source: Iraq Simulation Model 2021.

²⁵ The ratio of non-teaching staff to teachers indicates how many staff member without teaching responsibility there are per teacher. A non-teaching staff/teacher ratio of 15 percent indicates that, for every two teachers there is one non-teaching staff member.

Table 6.7.
Distribution of Operational Expenditures, 2022-2031

	2022	2025	2028	2031
	%			
Wages, teachers	77.0	77.3	78.6	80.0
Wages, non-teaching staff	18.7	16.1	13.6	10.8
Learning materials (textbooks)	0.2	0.3	0.3	0.3
Other non-salary operational expenses	0.9	1.0	1.2	1.4
Continued teacher training	2.5	2.4	2.3	2.2
Scholarships	0.7	2.9	4.1	5.4

Source: Iraq Simulation Model, 2021.

Major teacher (and non-teaching staff) management reforms²⁶ are needed to address the current inefficient, inequitable use of funds. Any reform must address a wide range of factors, including governance, political, and economic factors, including displacement, recent decentralization reforms resulting in a lack of clarity regarding decision-making processes, lack of teachers in rural areas because teachers (especially experienced teachers) prefer to work in urban areas and the politicized nature of the recruitment of education staff, existence of ghost teachers and ghost non-teaching staff, and various education staff mismanagement practices, to succeed. If achieved, the relatively small decrease in salary expenditures by 2031 will be a significant achievement, a solid basis for future corrections. A detailed distribution of operational expenditures is shown Table 6.7.

6.4.2 New Teacher and Trainer Recruitment

Despite the planned efficiency reforms, teachers and trainers must be recruited and trained to meet increasing enrollment, with a total need of almost 681,000 new teachers and trainers over the strategy period. Figure 6.2 shows the distribution of teaching staff needs (new recruits) and projected enrollment for all education levels.

Table 6.8 shows the distribution of teacher recruitment needs according to education level and year of strategy implementation. Over the period of the strategy, the most teachers will need to be recruited for primary education (301,988), followed by preparatory academic education (191,848), intermediate education (117,433), vocational education (28,773), higher education (23,941), and pre-primary education (16,798).

26 The MoE, with support from the World Bank, is developing a teacher allocation policy that analyzes past and current teacher allocation practices (including challenges and opportunities associated with decentralization) and provides a framework for teacher allocation in line with international best practices. Once implemented, the teacher allocation policy is designed to ensure more equitable and efficient distribution and use of human resources under current budget constraints (World Bank 2021).

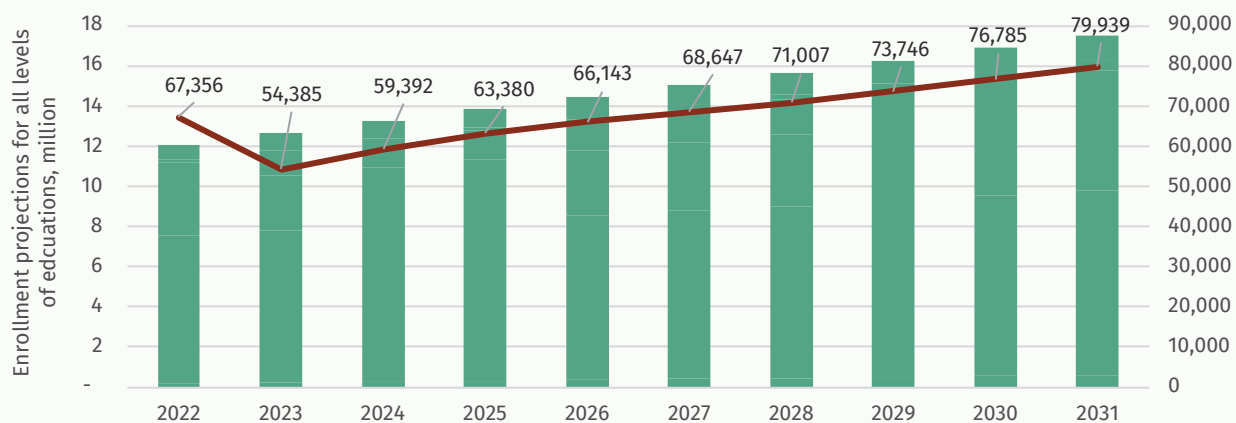


Figure 6.2. Annual Enrollment Projections and Teacher Recruitment Needs (New Recruits) for All Education Levels, 2022-2031

Source: Iraq Simulation Model 2021.

Table 6.8. Number of New Teachers Needed According to Education Level, 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Kindergarten	290	1,544	1,610	1,677	1,747	1,819	1,895	1,977	2,067	2,170
Primary	29,307	21,783	22,157	23,999	26,450	29,245	32,232	35,328	38,830	42,658
Intermediate	11,525	11,304	11,262	12,243	12,320	12,081	11,751	11,692	11,637	11,619
Preparatory	19,005	19,612	19,952	20,074	19,882	19,547	19,126	18,713	18,253	17,683
Vocational	3,291	-	2,914	3,093	3,069	3,144	3,345	3,432	3,360	3,124
Higher education	3,938	142	1,498	2,293	2,675	2,810	2,658	2,604	2,639	2,684
TOTAL	67,356	54,385	59,392	63,380	66,143	68,647	71,007	73,746	76,785	79,939

Source: Iraq Simulation Model 2021.

Other operational expenditures, such as learning and teaching materials, in-service teacher training, and equity measures including school meals for kindergarten and primary students, scholarships for preparatory and higher education, and other non-salary operational expenses (e.g., school grants for kindergarten, primary, and secondary schools) increase steadily from a cumulative 4 percent of all operational expenditures in 2022 to 9 percent in 2031.

A key driver of this development is the school meals program, with average annual growth of 46 percent for kindergarten and 28 percent for primary education. At the higher education level, scholarships drive the increase in non-salary operational expenditures, with average annual growth of 28 percent (Table 6.9). Annex A provides detailed information on the distribution of operational expenditures according to education level.

Table 6.9.

Operational Expenditures for Selected Posts According to Education Level, in, 2022 and 2031, and Average Annual Growth Rate, 2022-2031

	2022	2031	Average annual growth rate 2022-2031
	Billion dinars		%
Kindergarten			
Teaching and learning materials	0.6	1.8	13
Other non-salary recurring expenses	0.5*	1.2	12*
Continued teacher training	3.5	11.2	14
School meals	0.3	10.5	48
Primary			
Teaching and learning materials	14.3	19.5	4
Other non-salary recurring expenses	14.1	16.2	2
School meals	63.9	603.0	28
Intermediate			
Teaching and learning materials	7.5	9.9	3
Other non-salary recurring expenses	4.9	7.4	5
Preparatory academic			
Teaching and learning materials	1.7	5.0	13
Other non-salary recurring expenses	7.7	36.3	19
Preparatory vocational			
Teaching and learning materials	0.6	2.0	14
Other non-salary recurring expenses	1.3	10.5	26
Continued teacher training	12.7	17.6	4
Higher education			
Other non-salary recurring expenses	64.7	103.3	7
Scholarships	15.1	122.4	28

Source: Iraq Simulation Model 2021.

*2023 used as baseline.

6.4.3 Investment Expenditures

As mentioned in Section 6, under INES 2022-2031, investment expenditures will increase considerably by 2031 to meet urgent education infrastructure needs due to increasing enrollment, destroyed

school buildings, and the planned reduction of double shifts. It is estimated that total investment expenditures over the strategy period will be 13.4 trillion dinars, with the assumption that 50 percent of new schools are low-cost infrastructure.²⁷

²⁷ The model introduces a unit cost of 80 million dinars for one regular classroom and 48 million dinars for one low-cost classroom. Unit costs for regular classrooms were taken from the governorate education plans, developed with support from UNICEF.

Table 6.10.

Projected Investment Expenditures and Share of Investment Expenditures Out of the Total Education Expenditures, 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Investment expenditures, billion dinars	962	1,223	1,235	1,285	1,310	1,351	1,403	1,476	1,557	1,644
Share of investment expenditures out of total education expenditures, %	8	10	10	10	10	10	10	10	10	11

Source: Iraq Simulation Model 2021.

6.4.4 Classroom Needs

As indicated in section 6.x, school construction projects are underfunded, and education investment budgets are under-executed. The 2021 World Bank Public Expenditure Review notes that “A special committee was established by the Council of Ministers to aid in the implementation of large-scale education construction programs, bypassing the MOE” (World Bank 2021, 114).

Table 6.11 shows the annual need for construction of more than 166,000 new classrooms over the strategy period. To meet the pressing classroom shortage, strategies are needed to explore better use of existing infrastructure²⁸ and low-cost construction methods such as use of local construction materials and methods, which are also likely to generate local jobs.²⁹ The MoHESR requires a special investment plan to close the infrastructure gap because data is not available to estimate infrastructure and equipment needs.

Table 6.11.

Projected Number of New Classrooms Needed According to Education Level, 2022-2031

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Kindergarten	77	885	1,017	1,177	1,372	1,615	1,922	2,319	2,844	3,561
Primary	5,291	7,096	6,590	6,684	7,011	7,420	7,831	8,178	8,624	9,106
Intermediate	2,719	4,213	4,330	4,957	5,176	5,254	5,299	5,512	5,753	6,046
Preparatory	1,973	2,385	2,221	2,093	1,760	1,661	1,643	1,806	1,868	1,783
Vocational	590	-	755	806	801	830	905	955	970	944
TOTAL	10,650	14,579	14,914	15,716	16,120	16,780	17,600	18,769	20,059	21,440

Source: Iraq Simulation Model 2021.

²⁸ As part of World Bank support to the MoE, strategies are being developed to use existing infrastructure better (World Bank 2021).

²⁹ The MoE, with technical support from the World Bank, is developing a comprehensive school construction policy that will focus on developing national school construction planning guidelines and data-driven prioritization mechanisms for school construction. The policy will make recommendations for more efficient use of current school infrastructure and propose a set of norms for different types of schools, considering local needs.

6.5 Government Expenditures per Student

The Iraq Public Expenditure Review for Human Development Sectors notes that, for 2017, the most recent year for which complete enrollment and expenditure data are available, Iraq's unit cost (expenditure per student) in higher education was approximately 3,000,000 dinars, compared with 800,000 dinars for pre-university education (World Bank 2021) (Figure 6.3). Spending per student is 3.5 times as high in the university sector (World Bank 2021).

Despite better allocation of funds between higher and pre-university education since 2013, with a decreasing unit cost for higher education, public funding per student in higher education is projected

to remain significantly higher than in pre-university education (Figure 6.4). Projections for the INES 2022-2031 are based on the following constructed unit costs per education level.

Figure 6.4 shows that, even though public spending per student in higher education is projected to decrease, the gap between higher education spending and pre-university education spending will remain significant because pre-university spending per student (except for vocational education) remains flat in inflation-adjusted terms. The INES 2022-2031 is designed to address the existing efficiency challenges in education spending by increasing public spending per student in pre-university education more than in higher education in the next decade while seeking internal efficiency savings in the short term, including reducing allocations for teacher and non-teaching staff wages and improving budget execution, especially for capital expenditures, including construction and rehabilitation of schools.

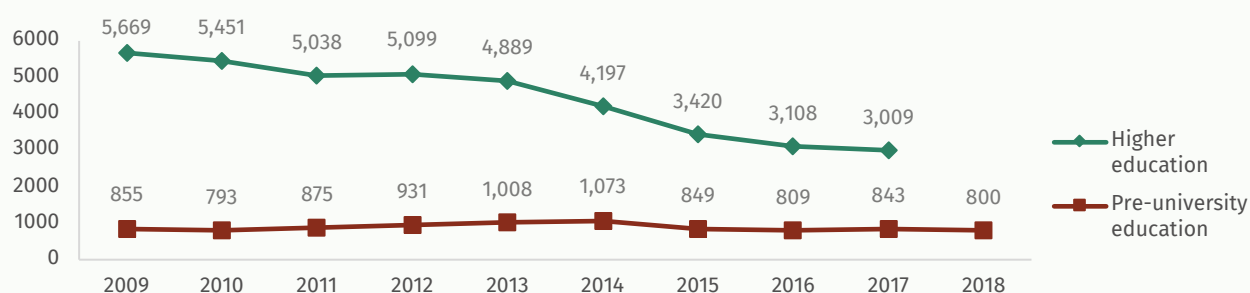


Figure 6.3. Government Expenditure per Student for Pre-University and Higher Education, 2009-2018

Source: World Bank 2021.

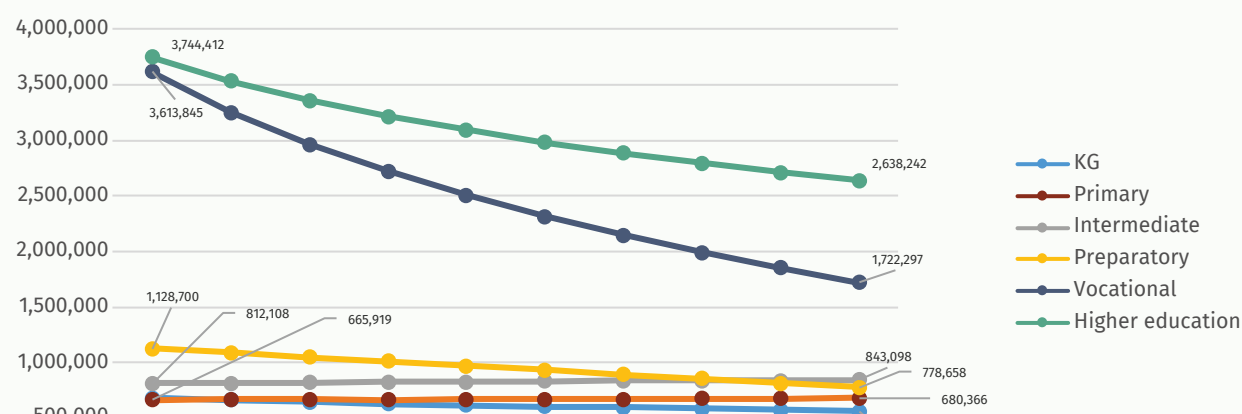


Figure 6.4. Projected Unit Costs According to Education Level According to Year, 2022-2031

Source: Iraq Simulation Model 2021.

Table 6.12.

Annual Funding Gap (Constant 2019 Billion Dinars) and Share of Funding Gap Out of Total Projected Resources, 2022-2031

Funding gap	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
	Constant 2019 billion dinars (%)									
Operational	-1,981 (-18)	-1,160 (-10)	-606 (-5)	-140 (-1)						
Investment	-824 (-96)	-1,043 (-93)	-998 (-88)	-974 (-81)	-907 (-74)	-836 (-65)	-755 (-56)	-673 (-47)	-575 (-37)	-457 (-28)
Total	-2,805	-2,203	-1,604	-1,114	-554	-836	-755	-673	-575	-457

Source: Iraq Simulation Model, 2021.

6.6 Funding Gap

Because of the unfavorable economic environment resulting from the COVID-19 pandemic combined with longstanding system inefficiencies and increasing demand for resources to accommodate a rapidly increasing number of children and youth, the funding gap is almost 12 trillion dinars (2019 constant price). By 2026, the funding gap for operational expenditures will be closed. The funding gap for investment expenditures will decrease from 2023 onward, although it will remain substantial until the end of the strategy period, with 28 percent of total investment expenditures not funded. Major efforts are needed to close the investment gap.

The INES is expected to mobilize support from all relevant stakeholders—national and international. Although no data are available on the planned level of investment, it is evident that donor support must increase throughout the INES period from the 2019 level of USD 34.2 million according to data from the OECD Official Development Assistance.³⁰ Of USD 63 million in 2019 declared for education, only USD 34.2 million was channeled to the education sector through the public sector or multilateral organizations according to the OECD nomenclature. The remaining USD 28.8 million was expenditures for experts, other technical assistance, scholarships, and student costs in donor countries. Compared with the financing gap in the first year of the INES alone, international donor financial support would cover only 2 percent of the 2022 financing gap.

Significant resource mobilization efforts with the private sector and donors are needed to fill the financing gap, especially for education infrastructure investment, to support transformative changes in education in Iraq. Planned governance reforms are expected to improve system efficiency and re-build trust among financing partners to close the financing gap. It is critical that households not be expected to take on a greater financial burden. Households in Iraq contribute approximately 0.5 percent of GDP to financing education. This is less than elsewhere in the Middle East and North Africa but more than in most OECD countries (World Bank 2021). Because household spending is neither an equitable nor sustainable way of providing education at pre-university levels, other funding sources should be explored.

6.7 A forward Look: Updated cost estimates for Higher Education as part of the annual and medium-term plans for operationalization of the Iraq National Education Strategy

It is imperative that projections are continuously updated and re-evaluated. The MoHESR and the MOED teams recognize the importance of updating the projections as part of the annual and medium-term plans for operationalization of the Iraq National Education Strategy. This exercise should allow for a more accurate estimation of detailed unit costs from which more realistic cost projections can be made for the short and medium term.

30 OECD Official Development Assistance data do not capture funding that donors provide indirectly, for example global funds such as the Global Partnership for Education and Education Cannot Wait.

Table 6.13:

Project Operation and Capital Expenditures for Ministry of Higher Education and Scientific Research, 2022-2031

	Projected operational expenditures		Projected capital expenditures	
	Billion dinars	Billion USD	Billion dinars	Billion USD
2022	3,635	2.507	563	0.389
2023	3,770	2.600	563	0.389
2024	3,905	2.693	563	0.389
2025	4,040	2.786	563	0.389
2026	4,040	2.879	563	0.389
2027	4,175	2.972	241	0.167
2028	4,309	3.065	241	0.167
2029	4,444	3.158	241	0.167
2030	4,579	3.251	241	0.167
2031	4,714	3.344	241	0.167
Total	41,611	29.255	4,024	2.775

Source: Projected cost estimates³¹ by the MoHESR.

The costing scenarios presented in this chapter were developed during a face-to-face workshop held in September 2021 in Erbil involving key technical and decision-making staff from the Ministry of Education, the Ministry of Higher Education and Scientific Research (MoHESR), and the Ministry of Planning.

MoHESR continued sub-sectoral reflections, including projecting new estimates of operational and investment cost requirements for 2022 to 2031. The table below provides an overview of the latest expenditure estimates by MoHESR. The projections include 45.6 trillion dinars for 2022 to 2031, split between 41.6 trillion dinars (91 percent) for operational expenditures and 4.0 trillion dinars for capital expenditures. Table 6.13 provides details according to year.

In addition, going forward, given the immense funding needs for the education sector in Iraq, especially higher education, it would also be useful to capitalize on all available funding, including private sector funding and user fees. Additional reflections and analyses underway at MoHESR should shed some light on this important endeavor while providing concrete action points.

Finally, in addition to efforts to mobilize additional resources for the education sector, there is an equally crucial need to increase the efficiency of the use of public funds for education in Iraq. The recent [Public Expenditure Review](#) of Human Development in Iraq provides promising avenues for efficiency gains in the Iraqi education sector.

31 *Commitments of the Ministry of Higher Education for current projects in (IQD) =2,090,560,024,000 (Construction and projects Department Letter 3923/ش in 22/11/2021).

Estimated Costs for implementation of new Universities projects according to the specifications of the designs in (IQD)= 19,320,180,000,000 (Construction and projects Department Letter 3923/ش in 22/11/2021).

Requested Investment Cost = Expected Investment Cost based on the increase in the number of students(\$)+Commitments of the Ministry for current projects(IQD)+ Estimated Costs for implementation of new Universities projects according to the specifications of the designs in (IQD).

Chapter 7

Coordination, Implementation, and M&E Framework

This chapter describes the framework for coordination, implementation, and M&E of the INES and is divided into three parts:

- Mechanisms for implementation and M&E
- M&E processes
- Key Performance Indicators (KPIs)

7.1 Mechanisms for Implementation and M&E

The mechanisms for the implementation and M&E of the INES are as follows.

7.1.1 Developing Operational Plans to Implement and Monitor Implementation of the INES

The INES is a long-term plan that provides a 10-year framework for policies, strategies, and programs. It does not and cannot describe in detail how these policies, strategies, and programs will be implemented and therefore does not provide a description of detailed activities.

To begin implementation of the strategy, a multi-year operational plan will be prepared. Operational plans provide a detailed description of how the pillars and programs of the national strategy will be implemented

by identifying program activities and implementation procedures for each activity. Operational plans also identify responsible entities per activity, the timeline for each activity, and the cost of implementing each activity. To measure progress, operational plans set output targets that are the basis for achieving the intermediate results of the programs proposed in the strategy. Figure 7.1 shows key elements of the operational plan:

Operational plans allow emerging educational challenges and recommendations resulting from M&E processes to be addressed at different levels. Annual review of the education sector, which helps ensure that implementation is directed toward achieving the outputs defined in the operational plans and consequently the results defined in the results matrix of all the pillars and programs of the INES for each of the three education sub-sectors (general education, TVET and higher education), is critical.

7.1.2 Assigning Roles and Responsibilities at the Central and Provincial Levels in Implementation and Monitoring of the INES

The operational plans provide a framework for the implementation and monitoring process. This includes a definition of roles and responsibilities of education departments at the central and decentralized levels regarding implementation of the INES.

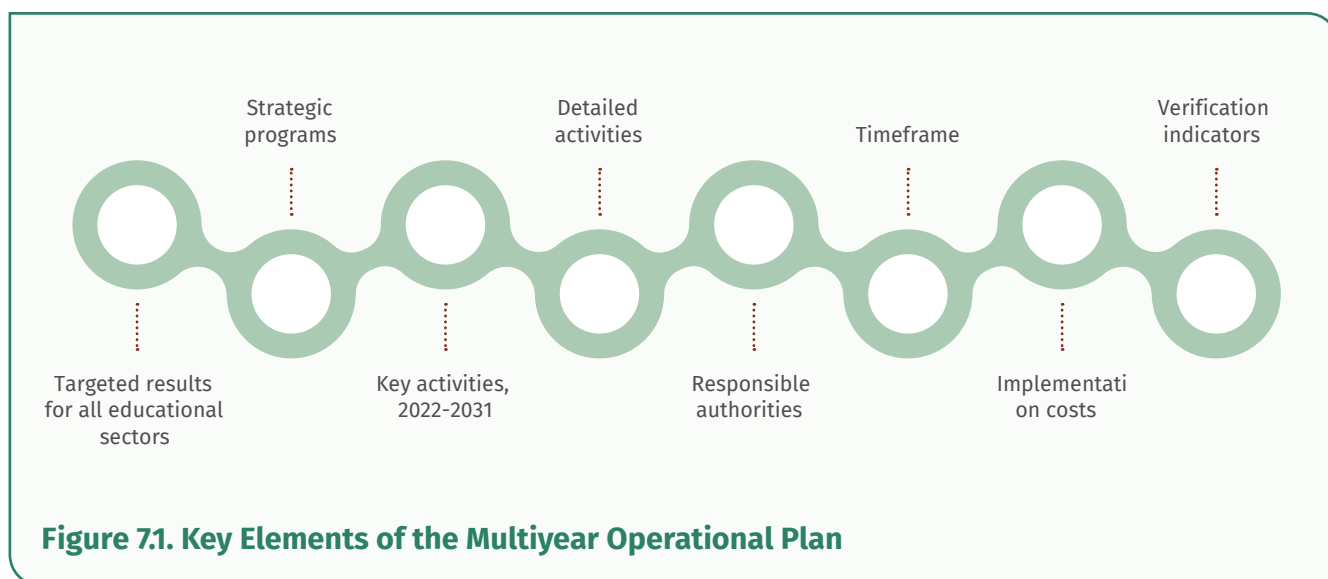


Figure 7.1. Key Elements of the Multiyear Operational Plan

ROLES AND RESPONSIBILITIES AT THE CENTRAL LEVEL

The Higher Committee of Monitoring and Evaluation supervises implementation and M&E of the INES. Two committees will be created, one each in the MoE and the MoHESR. The ministers of Education and Higher Education will chair these committees, which will include in their membership the deputy ministers of each ministry, the Director General of Financial Affairs, and the directors of the relevant bodies in each ministry. The Deputy Minister for Technical Affairs in each ministry serves as rapporteur for the respective committee. The committees will:

- Review the progress reports that the public departments in charge of implementation or the Technical Committee of Monitoring and Evaluation submit
- Identify and prompt appropriate measures to ensure the success of implementation and M&E of the INES according to the timeline specified in the operational plan
- Communicate and coordinate with other ministries and donors and partners to ensure effective implementation and alignment of activities and prevent duplication
- Consult with and provide support to the implementing bodies to ensure effective implementation and follow-up

The Technical Committee of Monitoring and Evaluation will supervise implementation and monitoring of the INES at the national level. Two M&E committees will be established, one each in the MoE and the MoHESR. The Deputy Minister of Technical Affairs of the respective ministry will chair the committee. The Director General of the Planning Directorate in each ministry will be a member of and rapporteur for the committee. The operational plan in the MoE and the MoHESR is a framework for both committees to supervise the implementation and monitoring processes. The committees are tasked with following:

- Lead and guide development of operational plans in the MoE and the MoHESR, ensure their alignment with INES pillars and programs, and collect and prepare data to monitor progress toward set targets.
- Develop criteria and principles to distribute funding among the governorates and universities according to identified targets.
- Communicate and coordinate with education partners in Iraq to ensure availability of technical and financial support to programs and activities

identified in the INES and respective operational plans.

- Develop and strengthen coordination mechanisms with the private sector to support the educational process.
- Coordinate with responsible departments to review and develop regulations and laws and issue ministerial decisions relevant to implementation and M&E of programs as outlined in the INES and associated operational plans.
- Identify and coordinate with the authorities responsible for achieving each of the targets listed in the strategy.
- Specify the data required to monitor result indicators in the INES and ensure availability of the required data in the educational information management system.
- Support and follow up on development of the educational information management system to include all data required to monitor progress toward set objectives and result indicators and ensure training of MoE and MoHESR staff to use the educational information management system for planning and M&E.
- Lead the process of designing tools for data collection and analysis and issue the annual statistical report, which provides data to monitor progress on sector objectives.
- Coordinate and provide support to other departments to collect data and monitor and evaluate results indicators that cannot be obtained from the annual statistical reports or the educational information management system.
- In cooperation with relevant departments in the MoE and MoHESR, lead preparation of annual progress reports showcasing progress made on outputs as indicated in the operational plans and progress toward INES targets for general education, TVET, and higher education.
- Prepare and oversee annual review conferences of the education sector to review progress made toward set targets and prepare development of operational plans.
- Submit M&E reports to the higher committee for the implementation of the INES.

Based on previous experience in M&E, a higher committee, chaired by the Deputy Prime Minister, will be formed to oversee implementation of the INES. The committee includes the Minister of Education, the Minister of Higher Education and Scientific Research, and ministers of line ministries, including but not limited to Finance, Planning, and Labor and Social Affairs. The committee monitors progress and achievements made in implementing the INES

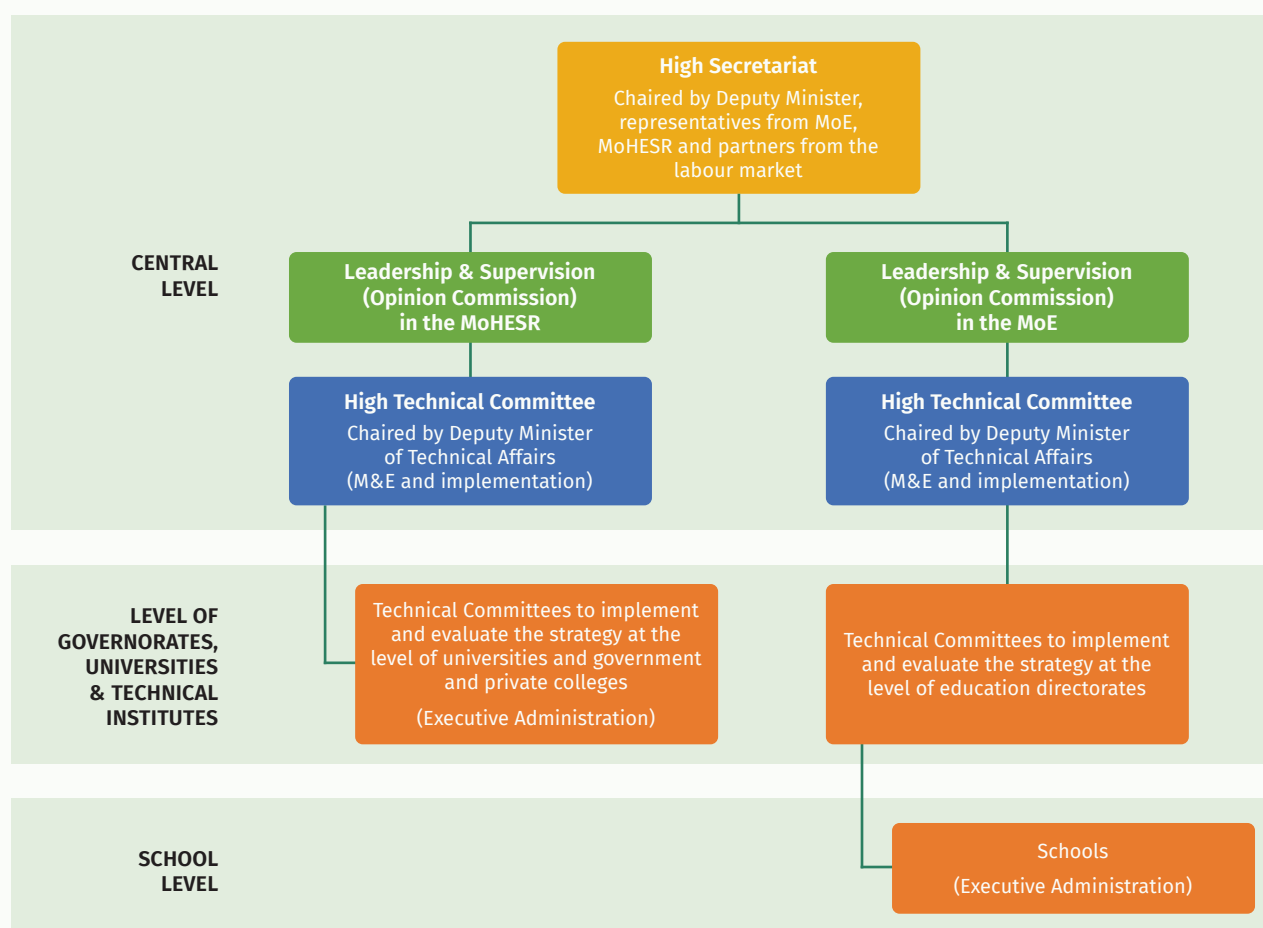


Figure 7.2. Mechanism of Implementation and Monitoring, and Evaluation (M&E) of the Iraq National Education Strategy in the Ministry of Education (MoE) and the Ministry of Higher Education and Scientific Research (MoHESR)

and works to overcome and address difficulties that obstruct or prevent implementation and support and coordinate collaboration and alignment between the relevant ministries to ensure effective implementation and provide the required resources. Figure 7.2 outlines the mechanism for implementing, monitoring, and evaluating the INES in the MoE and the MoHESR at the central and decentralized levels.

ROLES AND RESPONSIBILITIES AT THE GOVERNORATE LEVEL

At the sub-central level, operational plans are implemented, monitored, and evaluated under the supervision of the directors of the education directorates in the governorates and districts. Universities and technical institutes implement and monitor progress made toward intended results for higher education. A committee will be created to lead the implementation and monitoring process in

each education directorate in the governorates and universities, chaired and supported by the Deputy Assistant Director-General for Technical Affairs. The Manager of the Planning Department will be responsible for:

- Leading preparation of operational plans at the level of governorates or universities and coordinating implementation and monitoring
- Reviewing and managing entry of education data collected from departments, universities, districts, and schools; providing feedback regarding data quality; and identifying any contradictions in data
- Holding periodic meetings to monitor implementation of the educational programs regularly as outlined in the operational plans and developing strategies to overcome obstacles
- Analyzing education data and information related to outputs listed in operational plans at

the university level and governorate education offices

- Preparing reports for the Technical Committee of Monitoring and Evaluation in the MoE and MoHESR to inform progress reports at the national level

At the district level, department of education offices shall assume the following tasks:

- Develop and implement the operational plan
- Hold regular meetings to monitor progress of programs and activities defined in the operational plans and identify solutions to overcome obstacles in implementation
- Prepare and submit implementation reports to the Educational Planning Department in the Education Office at the governorate level
- Provide support to schools to develop and implement school development plans

School and training institute administrations assume the responsibility of developing the school or institute plan according to the framework of the school-based management program, monitoring progress, and submitting reports to the Education Office departments in the districts.

7.1.3 Assigning Roles of Partners in Implementation and Monitoring of the INES

The INES provides a common framework for all education stakeholders. It is expected that donors and partners will align available resources with the objectives outlined in the INES through strategic and technical dialogue and partnership.

A key coordination body for the education sector is the Education Sector Coordination Group, which was established in 2019. As outlined in the group's terms of reference, the group aims to:

- Coordinate interventions in the education sector to support the INES and development of the sector as a whole by aligning donor funds and government budgets to ensure their most effective use
- Oversee sector progress in meeting strategic objectives by tracking performance using an agreed-upon results framework

- Conduct regular education reviews to track implementation of the INES and related strategies and to address any challenges affecting achievement of results
- Establish a forum for technical and strategic discussions and dialogue for education policy and programs and, in doing so, share information to avoid duplication
- Leverage the experience of international and local partners to increase institutional capacity and strengthen education systems

Technical working groups will be established for each sub-sector to support the Education Sector Coordination Group. Thematic taskforces will support specific interventions and strategy development. The appropriate directorates or departments in the ministries will lead the technical working groups and thematic taskforces, which one of the partners will co-chair.

The chairs (MoE, MoHESR), the co-chairs (European Union, PMO), UNESCO, and UNICEF will lead the group. International partners will be subject to re-election after 2 years.³²

The Education Sector Coordination Group includes directorates of MoHESR and MoE, education partners and donors, and civil society representatives.

7.2 M&E Processes of the INES

M&E is a critical element in implementing the INES and helps:

- the MoE and MoHESR verify whether strategy implementation is on track with regard to inputs, activities, and results and takes appropriate corrective measures to enhance the chances of successful strategy implementation
- the MoE and MoHESR verify whether the strategy has achieved its objectives, which strategies worked, and what lessons can be learned
- build a culture of risk management responsibility and accountability, taking corrective actions, learning from experience, and making informed decisions (Figure 7.3)

³² UNICEF has hosted the group's secretariat since November 2020, although the MoE is expected to take on the role in the longer term, so capacities at the Cultural Affairs Directorate of the MoE will be increased.

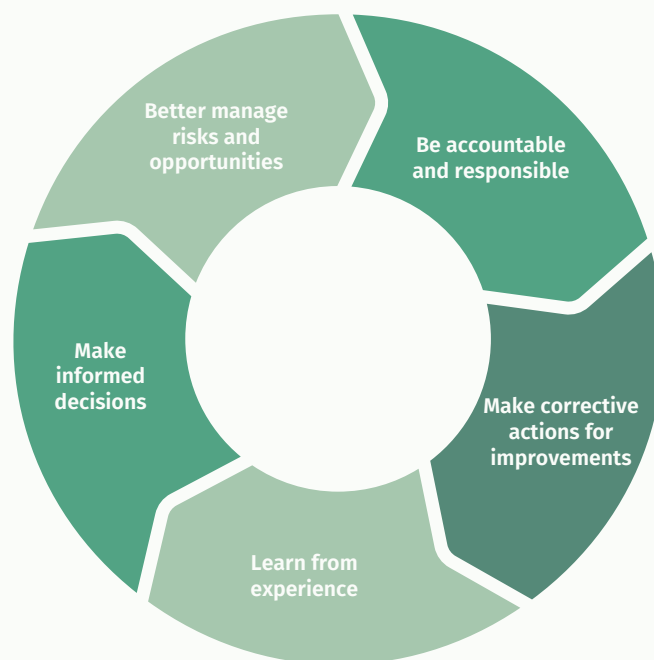


Figure 7.3. Key Functions of the Monitoring and Evaluation Process

The INES will be monitored and evaluated on four levels:

7.2.1 Regular Monitoring

Regular monitoring ensures implementation of detailed activities of annual operational plans that the INES generates at the central and decentralized levels. At the central level, the higher committees of M&E in the MoE and MoHESR hold periodic or as-needed meetings to follow up on and monitor progress made in implementation of operational plans and take the necessary steps to resolve any difficulties hindering implementation. At the decentralized level, university presidents and education office directors in the governorates and districts hold regular meetings to monitor and follow up on implementation of operational plans and develop solutions to any problems hindering implementation. The MoE and MoHESR supervisory committees of M&E will be involved if immediate solutions cannot be found.

7.2.2 Annual M&E

The technical committees of M&E in the MoE and MoHESR organize annual review meetings to measure progress made toward targets set in the INES. The

annual progress report is the basis for preparing the operational plan for the following year. The annual progress report and annual review discussions focus on:

- Reviewing progress made in the previous year and analyzing the performance of the education sector, identifying recommendations, policies, and priorities for the following year
- Reviewing financial resources provided for implementation of the plan from the government and external partners
- Reviewing the level of coordination, alignment, and harmony of support provided to implementation of the operational plans of the INES

7.2.3 Mid-Term Review

Five years into implementation of the INES, the MoE and MoHESR will conduct a mid-term review to:

- Evaluate progress made toward targets listed in the INES
- Introduce any amendments to policies and strategies outlined in the INES document for the second phase of the INES, including:
 - Changing or introducing programs and strategies to achieve objectives and targets identified in the INES

- Updating and modifying KPIs until 2031
- Updating financial data related to the cost of implementing the INES during the second phase of the INES, available resources, and the funding gap
- Proposing new policies and recommendations to ensure effective implementation of the INES by 2031

The mid-term review can be performed through annual review meetings of the sector for 2026, for example by preparing a mid-term achievement report in addition to the annual progress report.

7.2.4 Final Review

The final review of the INES will take place in the last year of strategy implementation (2031) to evaluate achievements in each sub-sector. The results inform development of the next education strategy. The achievement report determines to what extent INES targets were achieved and analyzes the performance of the MoE and the MoHESR during implementation. The evaluation highlights the most important lessons learned related to planning, implementation, and M&E over the past 10 years and guides the education sector in preparing a new strategy for after 2031.

Table 7.1 summarizes the objectives, timing, focus, and expected results of the annual, mid-term, and final reviews of the INES.

Table 7.1.

Objectives, Timing, Focus, and Expected Results of Annual, Mid-Term, and Final Reviews of the Iraq National Education Strategy (INES)

	Annual review	Mid-term review	Final review
Objective	Evaluate annual progress with the aim of following up work on the next year's operational plan	Review progress made to introduce any mid-term adjustments	Evaluate implementation of sector strategy, and use results as input to the following education strategy
Timing	Once a year, synchronized with preparation of the budget	Middle of implementation period	Last year of the INES
Focus	Review implementation of previous year's operational plan	Review trends on indicators and financial results Review suitability and effectiveness of strategies Incorporate changes related to strategy and objectives for second phase of the strategy	Evaluate success and challenges throughout implementation period
Expected result	Modify annual or multi-year operational plan	Update and modify strategy, especially the results framework	Analyze education sector and inform new education strategy

7.3 Results Matrix and Key Performance Indicators (KPIs)

Table 7.2 to 7.4 are results matrices for the INES 2022-2031 according to sub-sector. The matrices contain a limited, carefully selected set of KPIs that reflect the

sub-sector strategies and selected priority programs. KPIs, at the outcome and intermediate outcome levels, focus on the intended results and outcomes of the INES and not on inputs, activities, or specific projects. They are disaggregated to the extent possible, although further disaggregation, especially regarding regions, governorates, and districts, should be undertaken as part of the short-term operational plans. It is important to ensure that the situations of specific groups of learners are carefully monitored,

including students with special needs and displaced learners, as well as disparities in access to and completion of education and learning achievements between boys and girls and between regions.

KPIs were selected based on the following decisions:

- Selection of KPIs involves a technical, political process during which feasibility, relevance, desirability, and sustainability of selected policies is assessed.
- The number of KPIs should be limited and focused because of the cost of M&E (financial and human resources).
- The selected KPIs should be clear and easily understandable by decision makers and the wider public. Composite indicators and indicators that require substantial statistical measures should be avoided.
- The simulation model provides several KPIs, which helps link the simulation model with M&E and cost estimates.
- To the extent possible, indicators should be related to international frameworks, such as the SDGs and the 2030 Education Agenda.
- Selected KPIs are SMART indicators: Specific, Measurable, Achievable, Realistic, Time-bound.

Table 7.2.

Key Performance Indicators (KPIs) for General Education

		Baseline (2018/19 or latest year)	Target 2031 (or as indicated)
Pillar 1: Access and Equity			
Intended result: Greater enrollment and retention of learners in pre-primary, primary, and secondary education, especially girls, learners affected by conflict and displacement, students with special needs, and poor students			
KPIs (outcome-level results)			
	GER kindergarten	11%	30%
	GER primary (M/F)	129%/123%	131%/130%
	GER intermediate (M/F)	103%/85%	137%/123%
	GER preparatory (academic) (M/F)	47%/46%	68%/65%
	Completion rate, primary education (Grade 6)	84%/83%	100% (2025)
	Completion rate, intermediate education (Grade 9)	62.2%/63%	100% (2028)
	Enrollment in informal education programs		
	Youth	10,507	300,000
	Adults	72,242	170,000
	Governorate primary and intermediate GER gap	TBD	TBD
	Learners with special needs as share of total learners in primary school	0.26%	2%
Selected performance indicators (output-level results) by priority program			
Infrastructure	Share of schools operating double shifts		
	Primary	34%	14%
	Intermediate	10.3%	6.9%
	Preparatory (academic)	10.1%	3.4%
Safe and healthy learning environment	Percentage of students receiving school meals at public primary schools	5%	100%

Baseline (2018/19
or latest year)

Target 2031
(or as indicated)

Pillar 2: Quality and Relevance

Intended result: Greater school readiness of primary pupils, better teaching and learning outcomes in primary and secondary education, and harmonized educational content

KPIs (outcome-level results)

Percentage of students at end of primary school who achieve proficiency in reading and mathematics according to gender	TBD	80%
Percentage of students succeeding in international exams according to gender (e.g., Trends in International Mathematics and Science Study)	TBD in 2023	TBD
Share of budget for teaching and learning materials, school grants and teacher training (non-salary operational expenditures)	N/A	
Pre-primary		4.3%
Primary		3.6%
Intermediate		3.3%
Preparatory academic		6.2%

Selected performance indicators (output-level results) according to priority program

Attractive learning environment	Share of schools with tools to facilitate digital classes (electricity, Internet, computer labs, science labs, school libraries)	TBD	TBD
	Class size (kindergarten/primary G1-3/primary G4-6)	64/40/36	30/35/35
School leadership and community participation	Percentage of schools achieving fourth level of quality in the external assessment (part of school-based development program's external evaluation by supervisors matching school achievements with set quality criteria)	N/A	100%
	Percentage of schools included in school-based development program	10% (2,260 schools)	100%
Curriculum and assessment	Standards-based curricula developed for all general education grades integrating skills and knowledge expressed in Sustainable Development Goal 4, including sustainable development, human rights, gender equality, promoting a culture of peace and non-violence, global citizenship, and acceptance of cultural diversity	N/A	100%

Pillar 3: Governance and Educational and Financial Planning and Management

Intended result: Greater legislative, administrative, planning, and financial capacity of the education system to ensure effective, efficient education service delivery

KPIs (outcome-level results)

Percentage of schools with operational budget available to implement school improvements plans (school-based management approach) according to education level	10% (2,260 schools)	100%
Annual joint sector reviews and mid-term review of INES conducted	N/A	Yes
Educational information management system active at central and governorate levels and used for planning and monitoring and evaluation processes	N/A (Pilot starting in late 2021)	Yes
Share of departments of education that have specific plans and budgets to ensure education continuity during crises, with a total of 20 plans (19 governorates and educational districts in Baghdad + 1 center)	7 (Source: 2018 Humanitarian Development Plan)	20

Selected performance indicators (output-level results) according to priority program

Institutional management and leadership	Trainees from middle and upper management in aspects of planning, coordination, and administrative skills as percentage of total number	TBD	TBD
Financial management	Implementation rate of annual MoE current and investment budget	9.62% (2019)	100%
Crisis management	Crisis cell or unit established in MoE and departments of education, helping schools prevent and manage crises affecting education service delivery	N/A	MoE and department of education crisis cells or units established, staffed, and active
School management	School canteen law amended	N/A	1
School management	Parent-teacher council system amended	N/A	1
Partnership with private sector	Ratio of private schools to government schools		
	Kindergarten	14.5%	25%
	Primary	4.2%	15%
	Intermediate	4.5%	10%
	Preparatory	5.3%	15%
	Higher	27%	25%

		Baseline (2018/19 or latest year)	Target 2031 (or as indicated)
Pillar 4: Human resources			
Intended result: Qualified, motivated teachers and education staff recruited and trained to ensure good-quality education and effective, efficient education service delivery			
KPIs (outcome-level results)			
	Ratio of non-teaching staff to teaching staff (by governorate, TBD)		
	Kindergarten	62%	15%
	Primary	21%	10%
	Intermediate	16.7%	10%
	Preparatory academic	28%	25%
	Teacher: class ratio		
	Kindergarten	2.1:1	1.1:1
	Primary	1.7:1	1.3:1
	Intermediate	2:1	1.5:1
	Preparatory academic	2:1	1.5:1
	Student:teacher ratio according to governorate and education level	TBD	TBD
Selected performance indicators (output-level results) according to priority program			
Recruitment and training of non-teaching staff	Share of counselors and supervisors who participate in relevant training courses	TBD	TBD
Pre-and in-service teacher training	Percentage of teachers with a university degree in education	TBD	TBD

a. The government of Iraq has signed a memorandum of understanding to participate in the Trends in International Mathematics and Science Study 2023.

Note: GER, gross enrollment rate; M&E, monitoring and evaluation; MoE, Ministry of Education; N/A, not applicable; TBD, to be determined.

Table 7.3.

Key Performance Indicators (KPIs) for Technical and Vocational Education and Training (TVET)

		Baseline (2018/19)	Target 2031
Pillar 1: Equitable access to TVET			
Intended result: Greater demand for and enrollment in TVET programs through promotion of TVET offers and resulting job opportunities, making it one of the best options for young people			
KPIs (outcome-level results)			
	GER vocational schools (M/F)	7%/3% (2021)	14%/13%
	Transition rate to preparatory vocational schools (M/F)	8%/3%	19%
	Number of school dropouts who rejoin vocational training programs	TBD	TBD
	Number of students attending technical institutes (higher education)	69,000	169,000
	Transition rates from vocational and academic schools to technical colleges	TBD	TBD
	Enrollment rates of women in technical colleges	TBD	TBD
Pillar 2: Internal efficiency			
Intended result: Greater efficiency and effectiveness of internal systems in educational, training, and technical institutions to reach high quality standards			
KPIs (outcome-level results)			
	Retention rates at vocational schools	81%	90%
Selected performance indicators (output-level results)			
Infrastructure	Number of vocational school classrooms constructed or rehabilitated	N/A	7,555 between 2022 and 2031
Capacity building	Training needs for administrative staff and TVET trainers (resulting from annual performance evaluations) are met	TBD, past 2 years cannot be used as baseline	90%
Pillar 3: Quality and external efficiency			
Intended result: Greater quality and relevance of TVET programs improve education outcomes and increase employability			
KPIs (outcome-level results)			
	Share of students who are self-employed or employed 1 year after graduation	TBD	TBD
	Share of students obtaining jobs in their chosen specializations in TVET	TBD (based on tracer studies)	To be defined once labor market information system is in place

		Baseline (2018/19)	Target 2031
Selected performance indicators (output-level results)			
Curricula, training materials, and accreditation	Share of TVET courses offering students work-based training.	10%	50%
	TVET institutions are accredited based on results of national qualifications framework-linked accreditation	N/A	100%
Training of TVET trainers	Share of trainers trained in relevant fields	TBD	90%
Pillar 4: Governance, Management, and Finance			
Intended result: Better governance, financing, and management frameworks ensure better coordination and higher allocations for TVET			
KPIs (outcome-level results)			
	Higher council of TVET activated and meets, and produces recommendations regularly	N/A	Higher council of TVET endorsed by Parliament and regular meetings conducted
Selected performance indicators (output-level results) by priority program			
Governance and legislation	National qualifications framework activated	N/A	1
Strategic and operational planning	TVET committee under Higher Education Coordination Committee activated and in regular exchange	N/A	1
	Labor market information observatory established and used for policy and planning decisions	N/A	1 (2025)

Note: GER, gross enrollment rate; M&E, monitoring and evaluation; MoE, Ministry of Education; N/A, not applicable; TBD, to be determined.

Table 7.4.
Key Performance Indicators for Higher Education

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
Internal efficiency	Develop admission capacity of education institutions	Expand by creating specialized universities and colleges and expansion of existing universities	Higher education institutions with high admission capacity proportionate to population growth	Number of new universities and colleges and expansion percentage in existing universities
		Develop private education institutions through opportunities appropriate to needs of labor market and geographic distribution		Number of private education institutions relative to needs of labor market and geographic distribution
		Hold integrative partnerships of cultural exchange with developed countries to increase grants and scholarships to Iraqi students in rare postgraduate specializations.		Number of integrative partnerships of cultural exchange with developed countries that provide grants and scholarships to Iraqi students in rare postgraduate specializations
	Encourage variety in higher education patterns and programs	Adopt globally adopted educational systems for some leading government universities to evaluate and transition to universality	Flexible higher education institutions with varied patterns and programs	Number of internationally accredited educational systems in leading government universities
		Accredit and consolidate integrated electronic learning in Iraqi universities		Number of Iraqi universities that provide integrated, accredited, solid electronic education
		Support and develop continuous education centers and programs		Number of continuous education centers and programs
	Develop university admission policies	Review and amend laws governing admission of students to universities to ensure fair opportunities for all students wishing to increase and are qualified for admission to appropriate specializations and laws related to study systems	Modified university admission policies that ensure fair enrollment	Number of laws governing student admission to universities that have been reviewed and amended
		Sustain and increase percentage of students enrolled in undergraduate and postgraduate studies		Percentage of students enrolled in undergraduate and postgraduate studies

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
Governance of higher education institutions	Enable higher education institutions to achieve their goals efficiently and effectively	Develop management system	Higher education institutions that achieve their goals efficiently and effectively	Extent of development in administrative leadership system in higher education institutions
		Develop organizational structures of higher education institutions		Number of organizational structures developed
		Activate strategic planning for higher education institutions		Extent to which strategic planning role of higher education institutions is activated
		Enhance stakeholder participation in decision-making process		Extent of stakeholder participation in decision-making process in Iraqi institutions
		Increase integrity, transparency, and accountability		Extent to which integrity, transparency, and accountability are increased
	Achieve integration between relevant parties in higher education institutions	Keep abreast of technological developments in e-governance	Integrated higher education institutions	Number of advanced and technologically advanced electronic governance applications
		Implement electronic governance to achieve flexibility and facilitate procedures through electronic linkage of procedures and services in higher education institutions		Percentage of procedures and services to which electronic governance has been applied

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
Quality and accreditation	Achieve national and international academic accreditation of higher education institutions	Complete construction of national accreditation system for all major specializations	Higher education institutions with national and international accreditation	Extent of achievement of national accreditation system for main specializations
		Higher education institutions obtain national and international institutional accreditation		Number of higher education institutions with national and international institutional accreditation
		Academic programs in scientific departments obtain program accreditation		Number of academic programs with program accreditation
	Enable entry of higher education institutions into local and international rankings and taking advanced positions in them	Enable higher education institutions to enter important accredited international classifications	Advanced higher education institutions with advanced global ranking	Number of higher education institutions included in important, accredited international classifications
		Create spirit of competition and promote culture of excellence in universities		Number of universities competing to enter national and international rankings
	Provide quality assurance and greater effectiveness of higher education institutions	Develop curricula, student skills, teaching and learning methods, evaluation and assessment	Greater effectiveness of higher education institutions	Number of curricula and methods of teaching, learning, assessment, and evaluation developed and updated
		Improve quality of e-learning		Quality of electronic learning
		Improve quality of educational laboratories		Number of educational laboratories whose quality has been improved
		Distinguished trained and developed human resources		Number of human resources trained and developed
		Provide quality control of learning process according to inputs and outputs		Extent to which educational process is controlled according to inputs and outputs

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
Scientific research and innovation	Develop skills of researchers in Iraqi universities and their high-quality research productivity	Provide environment that supports scientific research in Iraqi universities	High-quality research productivity of researchers in Iraqi universities	Extent to which environment supports scientific research in Iraqi universities
		Develop mechanisms of cooperation in scientific research between Iraqi universities and their counterparts outside Iraq		Number of Iraqi universities that have developed cooperation with their counterparts outside Iraq
		Develop researchers' skills		Number of researchers whose research skills have been developed
		Develop work of research centers and mechanisms to provide research services to all institutions		Number of centers developed to provide research services to other institutions
		Strengthen requirements of scientific research		Extent to which requirements of scientific research are enhanced in higher education institutions
	Fund scientific research to advance international publishing in highly rated global journals	Adopt and develop scientific research methodology in Iraqi universities	Scientific research that supports innovation	Extent to which scientific research methodology has been developed in Iraqi universities
		Increase funding rates for scientific research		Scientific research funding rates
		Allocate specific percentage of annual budget for scientific research		Percentage of annual budget allocated for scientific research
		Establish research academies and centers in various scientific specializations		Number of academies and research centers established in various scientific specializations
		Encourage solid international publishing		Number of research papers published in solid scientific journals
		Help Iraqi scientific journals obtain accreditation in reputable journals		Number of Iraqi scientific journals with accreditation in reputable journals

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
	Protect intellectual property rights in Iraqi universities	Apply legal procedures related to copyright protection	Establishment of system to protect intellectual property for scientific research	Extent of application of legal procedures related to copyright protection
		Reduce violation of intellectual property rights and plagiarism		Number of violations of intellectual property rights and plagiarism
	Attract Iraqi scholars from abroad	Target rare and modern scientific specializations	More Iraqi scholars returning from abroad	Number of Iraqi scholars in rare and modern scientific specializations returning from abroad
		Determine material and moral privileges for returning scholars		Material and moral privileges that have been allocated for Iraqi scholars returning from abroad
		Develop regulations and instructions to guarantee rights of returning scholars		Issuance of regulations and instructions to guarantee rights of Iraqi scholars returning from abroad
		Determine budget for work of returning scholars		Extent of allocating a financing budget for work of Iraqi scholars returning from abroad

Strategic issues	Strategic objectives	Strategic options	Required output	Measurement indicators
	Develop solid academic institutions capable of building sustainable strategic partnerships with the business sector and industry	Create scientific and applied links between higher education institutions and various business entities	More higher education institutions capable of building sustainable strategic partnerships with business sector and industry	Number of higher education institutions that could create scientific and applied links with business entities
		Develop and amend laws to ensure effective communication of higher education institutions in the labor market		Number of laws developed and amended to ensure effective communication between higher education institutions and the labor market
		Develop organizational structures in universities to interact with business sector (labor market)		Number of organizational structures developed in higher education institutions to interact with business sector (labor market)
		Meet needs of economic and social development plans		Number of needs of economic and social development plans that have been met
External efficiency	Develop professors who keep pace with scientific development	Promote field skills of university professors	University professors with field skills that keep pace with scientific developments in field of specialization	Number of university professors whose field skills have been developed
	Distinguished generation that is active in society and meets requirements of labor market	Ensure that graduates understand and can manage dynamics of labor market and entrepreneurship	Graduates with specialized and general skills that meet requirements of labor market and play an active role in development of society	Number of graduates with ability to understand and manage dynamics of labor market and entrepreneurship Number of graduate initiatives that serve and develop society for the better

Annex 1.

Distribution of Share of Operational Expenditures per Education Level, 2022–2031

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Kindergarten											
Wage bill, teachers	76.9	78.9	96.0	113.7	131.8	151.9	172.9	194.9	217.9	242.1	267.6
Wage bill, non-teaching staff	47.6	48.8	54.4	58.5	61.0	62.3	62.0	59.7	55.4	48.9	40.1
Learning materials	-	0.6	0.7	0.8	1.0	1.1	1.2	1.4	1.5	1.6	1.8
Other non-salary recurring expenses	-	-	0.5	0.6	0.6	0.7	0.8	0.9	1.0	1.1	1.2
Continued teacher training	-	3.5	4.3	5.1	5.9	6.7	7.5	8.4	9.3	10.2	11.2
School meals	-	0.3	0.8	1.5	2.3	3.2	4.4	5.7	7.1	8.7	10.5
Total operational expenditures	124	132	157	180	203	226	249	271	292	313	332
Primary											
Wage bill, teachers	3,610.0	3,755.6	3,801.3	3,823.1	3,838.0	3,890.0	3,943.0	3,996.4	4,048.5	4,101.0	4,154.4
Wage bill, non-teaching staff	680.7	674.6	644.1	609.1	572.7	541.5	509.6	477.0	443.5	409.5	375.1
Learning materials (textbooks)	13.1	14.3	47.3	8.7	21.6	17.6	19.2	18.9	19.2	19.4	19.5
Other non-salary recurring expenses	13.6	14.1	14.5	14.7	15.0	15.2	15.4	15.6	15.8	16.0	16.2
Continued teacher training	159.0	165.4	166.2	165.9	165.1	164.2	163.1	161.9	160.4	158.8	157.0
School meals	32.0	63.9	127.9	255.8	302.9	350.9	399.8	449.6	500.1	551.2	603.0
Total operational expenditures	4,508	4,688	4,801	4,877	4,915	4,979	5,050	5,119	5,188	5,256	5,325
Intermediate											
Wage bill, teachers	1,505.9	1,626.0	1,742.5	1,858.0	1,988.4	2,119.8	2,247.1	2,368.9	2,489.8	2,609.9	2,729.9
Wage bill, non-teaching staff	251.4	271.4	277.9	282.5	287.5	290.8	291.5	289.7	286.0	280.4	273.0
Learning materials (textbooks)	16.8	7.5	7.0	7.5	8.0	8.3	8.6	8.9	9.2	9.5	9.9
Other non-salary recurring expenses	4.6	4.9	5.1	5.4	5.7	6.0	6.3	6.6	6.8	7.1	7.4
Continued teacher training	57.3	60.7	63.7	66.6	69.9	73.1	75.9	78.5	80.9	83.1	85.2
School meals	-	-	-	-	-	-	-	-	-	-	-
Total operational expenditures	1,836	1,970	2,096	2,220	2,360	2,498	2,629	2,753	2,873	2,990	3,105

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Preparatory											
Wage bill, teachers	870.2	933.2	973.8	1001.9	1019.6	1021.6	1016.1	1006.1	996.2	983.5	964.5
Wage bill, non-teaching staff	241.6	259.2	267.4	272.1	273.7	271.1	266.5	260.8	255.2	248.9	241.1
Learning materials (textbooks)	1.2	1.7	2.0	2.4	2.7	3.0	3.4	3.7	4.2	4.6	5.0
Other non-salary recurring expenses	5.5	7.7	10.3	13.1	16.1	19.1	22.2	25.4	28.9	32.6	36.3
Continued teacher training	34.5	37.0	38.6	39.7	40.4	40.5	40.2	39.8	39.4	38.9	38.2
School meals	-	-	-	-	-	-	-	-	-	-	-
Total operational expenditures	1,153	1,239	1,292	1,329	1,353	1,355	1,348	1,336	1,324	1,308	12,85
Vocational											
Wage bill, teachers	323.2	364.0	265.0	303.8	345.2	385.9	427.7	472.8	519.1	564.0	604.3
Wage bill, non-teaching staff	51.3	57.8	40.4	44.3	48.0	51.2	53.9	56.5	58.7	60.1	60.4
Learning materials (textbooks)	0.5	0.6	0.5	0.6	0.7	0.9	1.1	1.3	1.5	1.8	2.0
Other non-salary recurring expenses	1.0	1.3	1.2	1.7	2.4	3.2	4.2	5.4	6.9	8.6	10.5
Continued teacher training	11.5	12.7	9.1	10.2	11.3	12.4	13.5	14.6	15.8	16.8	17.6
School meals	-	-	-	-	-	-	-	-	-	-	-
Total operational expenditures	388	436	316	361	408	454	500	551	602	651	695
Higher Education											
Wage bill, teachers	1,695.7	1,742.4	1,745.5	1,784.1	1,841.9	1,906.5	1,972.1	2,028.4	2,079.7	2,129.6	2,180.1
Wage bill, non-teaching staff	706.6	726.0	686.9	660.8	639.6	617.9	593.5	563.5	529.5	493.0	454.2
Learning materials (textbooks)	-	-	-	-	-	-	-	-	-	-	-
Other non-salary recurring expenses	62.9	64.7	68.0	72.2	77.0	82.0	86.9	91.3	95.4	99.4	103.3
Continued teacher training	-	-	-	-	-	-	-	-	-	-	-
Scholarships	14.7	15.1	23.0	32.1	42.4	53.8	66.2	79.2	92.9	107.3	122.4
Total operational expenditures	2,480	2,548	2,523	2,549	2,601	2,660	2,719	2,762	2,798	2,829	2,860

Annex 2.

Matrix of compatibility of the programs of the National Strategy for Education with developmental orientations (general education)

Development Trends		National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
Strategic programs																												
N	General Education Programs																											
1	Schools building	*		*			*										*							*				*
2	Rehabilitation of schools										*						*						*					
3	Adding school facilities for people with special needs				*												*			*			*					
4	Transfer to single-shift schools		*		*												*						*					
5	School feeding and health education																*					*	*					
6	Conditional financial allocations for elementary and middle schools		*	*																						*		
7	School transportation and safety procedures				*																		*					
8	Accelerated learning and catch-up																*				*		*					
9	Adolescent education																*				*		*					
10	Evening schools and literacy programs									*							*				*		*					
11	Digital education and distance education			*													*						*		*			
12	Teaching for people with special needs and talented learners				*												*			*			*					
13	Science and computer laboratories and equipment for ICT		*	*													*						*		*			

Development Trends		Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
			Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07					
14	Sports halls and stadiums				*													*							*				
15	School libraries				*													*							*				
16	School gardens																	*							*				
17	Updating preparatory vocational education specializations (curriculum development, training of teachers and supervisors, overlapping with technical and vocational education and training)																												
18	Updating the national learning assessment methods					*				*															*				
19	Participation in international tests									*																			
20	Complete and implement the general framework for equal or inclusive education for students with special needs				*													*			*				*				
21	Increasing the number of school days to 182 days and increasing the duration of the study period																	*							*				
22	Enhancing and diversifying evaluation methods for the teaching staff																	*							*				
23	Providing new books for the first four grades of the primary stage																	*							*				
24	Developing, implementing and monitoring school development plans				*																				*				
25	Improving the relationship between communities, school leadership and local councils				*																								
26	Ensuring the participation of parent–teacher councils in the school feeding program				*																								
27	Introducing a new system for parent–teacher councils in school feeding programs				*																								

Development Trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
28	Introducing and amending education laws to meet quality standards							*										*							*			
29	Strengthen inter-ministerial coordination to ensure implementation and monitoring of progress in the National Education Strategy							*																				
30	Strengthen coordination with international partners to ensure harmonization, implementation and monitoring of progress in the national strategy							*																				
31	Strengthening the oversight function by reviewing tasks and advancing oversight reform linked to the decentralization effort			*														*							*			
32	Supporting the deployment of administrative and electronic educational systems, including EMIS and Kobo tools			*														*							*		*	
33	Determining the powers between the Ministry's office and the General Directorates of Education to meet the requirements of the educational process							*																				
34	Developing an information management system																											*
35	Building capacities and motivating financial officers to plan for providing cash liquidity to implement programs and projects			*																								
36	Developing the diversification of educational financing sources (investment)			*																								
37	Providing accurate financial data and information with a high level of transparency			*																								

Development Trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
38	Providing schools with an operational budget to implement school development plans			*																								
39	Include all schools in the school-based management program and improve performance to achieve goals			*																								
40	Amending school canteen system to integrate it into the school's operational budget			*																								
41	Developing the school management selection system			*																								
42	Promoting human rights awards in schools				*																							
43	Expanding the school-based management program to include all schools			*																								
44	Strengthening the role of the Crisis Cell in the Ministry of Education and the directorates in the governorates in risk management			*																								
45	Training workers in the Crisis Cell, the Ministry of Education and the General Directorates of Education in the governorates to respond to crises to ensure the continuation of education during crises			*																								
46	Amending the private school system							*																				
47	Amending legislation for the purpose of encouraging private sector investment in education							*																				

Development Trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
48	Improving the selection and employment of teachers and the diversity of their qualifications, and increasing admission to the Open College of Education and Colleges of Education in the Ministry of Education and Scientific Research			*														*							*			
49	Strengthening the teacher management system to ensure equitable distribution of teachers according to needs, through regulations, effective implementation and monitoring of progress			*																					*			
50	Developing a new rating system for teachers that links performance evaluations to professional development			*																								
51	Provide incentives for teachers in rural areas			*																								
52	Diversifying the qualifications of teachers through organized admission to the Open College of Education and Colleges of Education in the Ministry of Education and Scientific Research according to the needs of the subject.			*																								
53	Training teachers on updated curricula			*																					*			
54	Training teachers on child-centered teaching			*														*						*	*			
55	Training teachers on digital and technology literacy / online learning			*						*								*							*		*	
56	Training teachers to teach people with special needs			*	*													*						*	*			
57	Employment and training: 1. Educational counselors (for psychological and social support) 2. Education staff in schools and directorates 3. Other planning and administrative staff			*				*										*						*				

Annex 3.

Matrix of compatibility of the programs of the National Strategy for Education with developmental orientations (vocational and technical education)

Development Trends		Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
			Objectives																SDG 4: Inclusive, Equitable and Quality Education										
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
N	Vocational and technical education																												
1	Expanding the educational paths of technical, vocational and education training (TVET) institutions and facilitating the movement of students between these institutions					*															*				*				
2	Designing and creating new majors and educational and training programs to enhance the enrollment of girls, people with special needs, and talented students					*															*				*				
3	Developing procedures for admitting dropouts from TVET institutions and including them in training programs					*																			*				
4	Establishing systems to admit people with special needs in majors that allow them to obtain suitable job opportunities																*	*			*				*				
5	Developing guidance and counseling services at all stages, especially in the basic education stages, to introduce TVET institutions and jobs coming from such institutions					*		*																					

Development Trends		National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Strategic programs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06				
6	Develop career development centers					*																	*		*			
7	Building small, income-generating loan programs for graduates of TVET institutions																								*			
8	Enhancing community awareness of the importance of TVET and the role of women in the labor market through awareness campaigns																		*		*				*			
9	Enhancing central license concerned with the reconstruction, maintenance and rehabilitation of the infrastructure of all vocational and TVET institutions (competence treatment)		*				*																	*				
10	Establishing and equipping centers of excellence in TVET institutions with the participation of business owners to serve strategic sectors																							*		*		
11	Update job descriptions for workers in administrative and coordination positions and match them with existing employee profiles to identify gaps, training needs, etc.			*																								
12	Implementing the National Qualifications Framework																								*			
13	Establishing a program to build the capacity of trainers in distance education and digitizing educational and training content			*			*																*		*	*		
14	Digitizing materials and content, development of distance learning platforms and mechanism						*											*						*		*		
15	Develop a competency-based curriculum in vocational schools and provide job-based training				*													*						*				

Development Trends		National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
16	Review calendar procedures								*																			
17	Developing the scientific research system with a view to developing TVET			*													*							*				
18	Providing sound training, plus in-service training, for TVET teachers and trainers			*																								
19	Training curriculum developers				*																							
20	Training of trainers on distance learning and digital educational and training content																								*			
21	Review criteria for selecting trainers (human resources)																*											
22	Develop bodies responsible for vocational and technical training		*	*	*												*											
23	Conducting evaluative tracking studies (following supply and demand respectively) to determine the compatibility of programs with the needs of the labor market				*																			*				
24	Develop work-based training programs (delivery, trainers, and agreements with entrepreneurs)				*																			*				
25	Collectively updating occupational standards and job descriptions with the private sector and business owners (taking into account Arab and international classifications) through sectoral skills councils			*																								
26	Issue a unified law for the governance of TVET; and activate the national qualifications framework				*			*																				
27	Allowing private sector funding for TVET through legislation (covering tax deduction, special allocations)				*			*																				

Development Trends		Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
			Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07					
28	Issuing legislation for private sector participation in TVET (establishing sectoral skills councils)			*				*																					
29	Establishing a labor market observatory in cooperation with the Ministry of Higher Education and Scientific Research, the Ministry of Labor and Social Affairs, other selected ministries, representatives of the private sector and other partners			*																									
30	Establishing a TVET council to ensure effective coordination, policy development, and coordinated implementation across sectoral ministries as stipulated in the proposed law for that council			*				*																					
31	Planning financial resources directed to TVET institutions according to annual financial plans			*																									
32	Applying a program and performance budget rather than item-based budget			*																									
33	Establishing a fund to finance and support TVET institutions to respond quickly to the needs of the labor market			*																									
34	Diversifying funding sources			*																									

Annex 4.

Matrix of compatibility of the programs of the National Strategy for Education with development trends (higher education)

Development trends Strategic programs		National Development Plan 2018 – 2022 Objectives																2030 Agenda for Sustainable Development SDG 4: Inclusive, Equitable and Quality Education							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
N	Higher Education Programs																											
1	Establishing specialized scientific universities that are interested in technological, health and applied sciences that meet the requirements														*										*	*	*	*
2	Establishing technical colleges/institutes affiliated with technical universities in large cities and districts														*					*	*				*	*		
3	Increasing the capacity of all existing universities vertically and horizontally																			*					*			
4	Completing projects prepared for new / young universities.														*										*		*	*
5	Complete the process of rehabilitation and reconstruction of educational institutions damaged by terrorist operations.																*			*					*			

Development trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
6	Developing scientific laboratories, specialized workshops, and research centers in line with the planned increases in the number of students admitted annually and according to the needs of the labor market and the development of specializations.													*											*			
7	Coordinating with the relevant authorities to find solutions for owning and allocating real estate for the Ministry of Higher Education.		*																							*	*	
8	Establish cooperation and partnerships with local government institutions (such as Ministry of Health, Ministry of Agriculture, Ministry of Oil) to support related disciplines and the labor market.																							*		*		
9	Coordinate with international donors (governments and institutions) and sources of national support for the work of the Higher Education Support Fund on the “Establishment of the Iraqi University Educational City”																							*				
10	Preparing a program for the efficient use of the resources available to universities, including classrooms, laboratories and human resources in terms of timings and study hours.																	*						*		*		
11	Providing a supportive environment for scientific research in Iraqi universities												*	*										*				

Development trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
12	Developing cooperation mechanisms in the field of scientific research between Iraqi universities and their counterparts outside Iraq													*											*			
13	Developing the skills of researchers														*										*			
14	Developing the work of research centers and mechanisms for providing research services to all institutions													*	*										*			
15	Enhancing the requirements of scientific research													*	*										*			
16	Adopting and developing scientific research methodology in Iraqi universities														*										*			
17	Increasing funding rates for scientific research													*											*			
18	Applying internationally approved systems at the level of majors (specializations) or universities													*											*			
19	Allocating a specific percentage of the annual financial budget for scientific research													*											*			
20	Establishing research groups and centers in various scientific disciplines													*	*										*			
21	Encouraging international publishing													*	*										*			
22	Supporting Iraqi scientific journals to obtain robust accreditation													*	*										*			

Development trends	Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
23	Applying legal procedures related to copyright protection													*														
24	Reducing plagiarism and infringement of intellectual property rights													*														
25	Targeting rare and modern scientific disciplines														*													
26	Determine material and moral privileges													*														
27	Determine the budget for the work of returning scholars													*														
28	Complete the plan and standards of the national accreditation councils																		*						*			
29	Describe the system of accreditation procedures and set indicators and criteria for evaluation and measurement and its mechanisms																		*						*			
30	Qualifying national residents within the measurement and evaluation system														*									*	*			
31	Approval of the national accreditation plan														*													
32	Create the independent national authority for quality and academic accreditation														*													
33	Applying national institutional accreditation standards in higher education institutions														*													
34	Applying specialized accreditation standards for all scientific departments and their academic programs in all Iraqi public and private universities														*													

Development trends		Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
			Objectives																SDG 4: Inclusive, Equitable and Quality Education										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07					
35	Develop administrative and financial skills														*									*	*				
36	Developing electronic management skills														*									*	*				
37	Promote a culture of participation in the decision-making process													*	*														
38	Modernizing mechanisms for selecting university and administrative leaders														*														
39	Reviewing and evaluating the educational structures of educational institutions in accordance with criteria of excellence, efficiency and effectiveness														*														
40	Developing human resources in line with the change in educational structures														*										*				
41	University independence														*														
42	Establish a culture of strategic thinking and planning in higher education institutions														*														
43	Strategic planning for the development of higher education institutions in the field of electronic knowledge														*														
44	Strategic planning for the role of higher education institutions in achieving sustainable development														*									*					
45	Reviewing the strategic plans of universities and aligning them with the strategic plans of higher education																							*					
46	Strategic planning for crisis management														*									*					

Development trends		Strategic programs	National Development Plan 2018 – 2022																2030 Agenda for Sustainable Development							Iraq Vision 2030	Government reform document (white paper)	Response document and crisis recovery plan	Future development directions for the budget strategy
			Objectives																SDG 4: Inclusive, Equitable and Quality Education										
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	04:01	04:02	04:03	04:04	04:05	04:06	04:07				
47	Creating business incubators in universities															*													
48	Entrepreneurship projects															*													
49	A mutually beneficial mechanism for cooperation between the university and the business sector through continuous training and skills development for staff.															*	*												
50	Periodic review of programs, curricula, teaching methods and assessment and evaluation mechanisms to ensure their development in a manner that is impactful															*													
51	Developing specialized courses with internationally accredited certificates that are provided by the private sector for the benefit of the university through agreements and memoranda of understanding with the private sector																*												
52	Reviewing legislation and regulations that govern the relationship between the educational institution and the labor market																*												

Annex 5.

Committees and Sectoral Working Groups

The Supreme Committee for Strategy:

The Supreme Committee, charged with reviewing and evaluating the projects of the National Strategy for Education and Higher Education in Iraq (2012–2022) and preparing a new strategy, was formed pursuant to Cabinet Order 343 issued by the Prime Minister's Office in the letter ref. M. R. W / 59/14560 on 10/24/2018, headed by Prof. Dr. Hamid Khalaf Ahmed, Vice Chairman of the Prime Ministers Advisory Commission (PMAC), and comprising the following membership:

1. Prof. Dr. Musa Jawad Al-Musawi, Advisor to the Ministry of Higher Education and Scientific Research;
2. Dr. Majeed Sadiq Al-Alaq, Director General of Curricula, Ministry of Education;
3. Uqood Hussein Salman, Director General of Human Development, Ministry of Planning;
4. Dr. Kashani Rasan Heljan, Acting Director General of the Economic Department, Ministry of Finance; and
5. Prof. Dr. Tariq Ali Jassim, Director of the Office of Education and Culture, PMAC (member and coordinator of the Committee).

Members were replaced and others were added to the committee according to Cabinet Order ref. M.R.W/D 7/59/3850 on 4/3/2021 issued by the Prime Minister's Office, as follows:

1. Dr. Aadil Najeh Al-Busaisi, Director General of Curricula at the Ministry of Education, replacing Dr. Majeed Sadiq Al-Alaq, Former Director General of Curricula, due to his retirement;
2. Mrs. Najah Abbas Ali, Director General of the Economic Department at the Ministry of Finance, replacing Dr. Kashani Rasan Heljan, former Director General of the Economic Department due to his retirement;
3. Dr. Ihab Naji Abbas, Director General of Studies, Planning and Follow-up at the Ministry of Higher Education and Scientific Research;

4. Assist. Prof. Dr. Maha Abdul-Karim Hammoud, Director General of Human Development at the Ministry of Planning, replacing Ms. Uqood Hussein Salman due to her retirement (according to the Cabinet Order ref. M.R.M / D 7/7604 on 22/10/2020); and
5. Dr. Abeer Abdel-Amir Abdel-Nabi, Assistant Director General of the Economic Department at the Ministry of Finance, replacing Mrs. Najah Abbas Ali, (according to the letter of the Ministry of Finance ref. 7568 dated 29/10/2021).

Strategy Drafting Team:

This team was formed pursuant to Administrative Order No. 36 issued by the PMAC in its letter ref. A/648 on 12/5/2019, as follows:

1. Prof. Dr. Jamal Al-Dabbagh, Emiretus Professor;
2. Prof. Dr. Waleed Ameen Al-Johar, Assistant Head of the Scientific Supervision and Evaluation Agency, Ministry of Higher Education and Scientific Research;
3. Prof. Dr. Haider Nima Ghali, Al-Mustansiriya University;
4. Assist. Prof. Dr. Kefaa Ali Issa Abu Ragheef, Al-Mustansiriya University;
5. Tariq Muhammad Shukr, Director of Planning, Ministry of Education; and
6. Assist. Prof. Dr. Raad Khalaf Attiya, Director of the Department of Quality Management and Institutional Development, Ministry of Education.

Higher Education and Scientific Research Team

#	Name	Job Title
1	Prof. Dr. Walid Amin Al-Johar	Scientific Supervision and Evaluation Department
2	Prof. Dr. Khaled Shanawa Zeyara	Scientific Supervision and Evaluation Department
3	Dr. Ahmed Kanaan Salman	Director of the Quality Assurance Department, at the Supervision and Scientific Evaluation Department
4	Dr. Haider Mashali	Director of the Studies Department, at the Planning, Studies and Follow-up Department
5	Dr. Hala Salman Farhan	Director of the Statistics Department, at the Planning, Studies and Follow-up Department
6	Assist. Prof. Dr. Youssef Al-Dinenawi	Director of the Pioneering Projects Department, at the Research and Development Department
7	Enas Zaki Hadi	Head of the Strategy Division at the Department of Planning, Studies and Follow-up

Note: The Ministry of Higher Education and Scientific Research, according to Ministerial Order ref. T.M.3/A.S./ 2432 on 8/3/2021, formed a Higher Education Strategy Review Committee headed by Dr. Ehab Naji Abbas, Director General of Studies, Planning and Follow-up, with 46 members divided into six specialized work teams as follows:

Team One: Structure and Formulation Review

#	Name	Work Place / Job Title	
1	Prof. Dr. Mostafa Mounir Ismail	University of Baghdad, College of Administration and Economics, Assistant Dean for Postgraduate Studies	Member
2	Assist. Prof. Dr. Muhammad Khalil Ibrahim	Maysan University, Dean of the College of Administration and Economics	Member
3	Prof. Dr. Wafa Jaafar Amin Al-Mahdawi	Al-Mustansiriya University, College of Administration and Economics	Member
4	Assist. Prof. Dr. Hadeel Kazem Saeed	University of Baghdad, College of Administration and Economics	Member
5	Assist. Prof. Dr. Yarub Adnan Hussein	University of Baghdad, College of Administration and Economics, Head of Business Administration Department	Member
6	Assist. Eng. Enas Zaki Hadi	Department of Studies, Planning and Follow-up	Member

Team Two: Availability

#	Name	Workplace / Job Title	
1	Prof. Dr. Salahuddin Awad Al-Kubaisi	University of Baghdad, College of Administration and Economics	Member
2	Prof. Dr. Ali Naeem Jassim	Administrative and Financial Department, Director of the Department of Budgets and University Affairs	Member
3	Assist. Prof. Dr. Haider Mohsin Salman	Thi-Qar University, College of Education for Pure Sciences	Member
4	Assist. Prof. Dr. Saate Hikmat Abboud Al-Najjar	Iraqiya University, College of Engineering	Member
5	Assist. Prof. Dr. Hatem Ali Abdullah	Tikrit University, College of Administration and Economics	Member
6	Dr. Haider Hassan Mashali	Studies, Planning and Follow-up Department, Director of the Studies and Planning Department	Member and Coordinator
7	Dr. Ammar Salem Kazem	University of Baghdad, College of Dentistry	Member and Coordinator
8	Dr. Mohammed Abbas	Department of Private University Education, Director of Student Affairs Department	Member
9	Dr. Abbas Farhan Aliwi	Department of Scholarships and Cultural Relations	Member
10	Ismail Khalil Murad	Department of Reconstruction and Projects, Director of the Department of reconstruction and Projects DepartmentReconstruction Department of Universities	Member

Team Three: Quality Assurance

#	Name	Work Place / Job Title	
1	Prof. Dr. Yusra Mohammed Abdullah	University of Baghdad, Director of the Quality Assurance Department	Member
2	Prof. Dr. Sabeeh Abbas Al-Mashhadani	University of Baghdad, College of Medicine	Member
3	Assist. Prof. Dr. Khaled Shanawa Zeyara	Scientific Supervision and Evaluation Dept., Director of the Quality Assurance and Academic Accreditation Department	Member
4	Assist. Prof. Dr. Abdul Salam Ali Hussein	Anbar University	Member
5	Dr. Ahmed Kanaan Suleiman	Scientific Supervision and Evaluation Dept., Director of the Quality Assurance Department	Member and Coordinator
6	Dr. Hayder Mahdi Baqer	Middle Technical University, Director of the Quality Assurance Department	Member
7	Dr. Ali Jalil Majali	Al-Mustansiriya University, Director of the Quality Assurance Department	Member
8	Ahmed Abdel Mahmoud	Department of Studies, Planning and Follow-up	Member and Coordinator

Team Four: Governance

#	Name	Workplace / Job Title	
1	Prof. Dr. Fadel Abd Ali Kharmit	Wasit University, College of Administration and Economics	Member
2	Assist. Prof. Dr. Bashar Jahem Ajami	Legal Department	Member
3	Assist. Prof. Dr. Yousif Khalaf Yousif	Research and Development Department, Director of Pioneering Projects Department	Member
4	Assist. Prof. Dr. Omer Abdul Karim Jassim	University of Fallujah, Assistant Professor Department of Quality Assurance and Academic Accreditation	Member
5	Assist. Prof. Dr. Esraa Saleh Hassoun	Wasit University, College of Pure Education	Member and Coordinator
6	Dr. Ehsan Salman Jassim	Diyala University, College of Basic Education	Member
7	Dr. Zaid Hassan Ali	Karbala University, Computer Science and Information Technology	Member
8	Dr. Firas Ibrahim Hazaa	Department of Studies, Planning and Follow-up, Department of Statistics	Member
9	Dr. Amer Saleem Al-Amir	University of Information and Communication Technology	Member
10	Zaid Ahmed Ali	Studies, Planning and Follow-up Department, Studies and Planning Department	Member and Coordinator
11	Ruaa Adnan Abdul Reda	Administrative and Financial Department, Human Resources Department	Member

Team Five: Scientific Research

#	Name	Workplace / Job Title	
1	Mr. Dr. Bashar Abbas Hussein	University of Babylon, College of Administration and Economics	Member
2	Assist. Prof. Dr. Rafi Abbas Hassan	Al-Karkh University of Science, Director of the Department of Quality Assurance and University Performance	Member and Coordinator
3	Assist. Prof. Dr. Naseer Ali Hussain	Wasit University, College of Education for Pure Sciences	Member
4	Assist. Prof. Dr. Salam Jassem Abdullah Al-Ezzi	University of Diyala,	Member
5	Assist. Prof. Dr. Wissam Abd Mohamed Mahmoud	Central Technical University	Member
6	Dr. Ali Badr Roumi	Al Ain Private University, Associate Dean of the College of Medical and Health Technologies	Member
7	Dr. Sinan Hossam Mahdi	Research and Development Department, Director of the Scientific Affairs Department	Member and Coordinator

Team Six: Linking Universities to the Labor Market

#	Name	Workplace / Job Title	
1	Prof. Dr. Aseel Ali Mazhar Suleiman	Al-Qadisiyah University, College of Administration and Economics	Member
2	Assist. Prof. Dr. Sami Ali Neema	Middle Technical University, Head of Rehabilitation, Employment and Follow-up Division	Member
3	Dr. Mostafa Hamid Farhan	Al-Nahrain University, Rehabilitation, Employment and Follow-up Division	Member and Coordinator
4	Dr. Azhar Jassim Ali	Studies, Planning and Follow-up Department, Director of Development and Human Development Department	Member and Coordinator
5	Dr. Lateef Abdel-Reda Attia	Al-Qadisiyah University, College of Administration and Economics	Member
6	Ali Abdul-Jabbar Muhammad Jawad	Southern Technical University, Computers Center	Member

Vocational and Technical Education Team

#	Name	Job Title
1	Prof. Dr. Abdul Mohsen Naji Al-Muhaisen.	Chairman of the Technical Education Authority
2	Prof. Dr. Farid Majeed	Assistant President of the Central Technical University for Administrative Affairs
3	Prof. Dr. Waddah Amer Hatem	Assistant President of the Central Technical University for Scientific Affairs
4	Saad Ibrahim Abdel-Rahim	Former Director General of Vocational Education (Ministry of Education)
5	Mahmoud Al-Maliki	Director General of Vocational Education (Ministry of Education)
6	Thayer Mizher	Director of Planning, Directorate of Vocational Education
7	Assist. Prof. Dr. Abdul-Raheem Maktouf Al-Taie	Senior Supervisor, Vocational Supervision

Note: Some members have been replaced as a result of retirement or change of work locations; that applies in particular to the Director General of Vocational Education.

Pre-University Education Team (Ministry of Education)

#	Name	Job Title
1	Dr. Majeed Sadiq Al-Alaq	Curriculum Director
2	Tariq Muhammad Shukr	Planning Manager
3	Dr. Raad Khalaf Attia	Director of the Total Quality Management Department
4	Hussain Hindi Hussain	Assistant Director General of Educational Supervision
5	Safaa Laftah Younis	Director General of Educational Planning
6	Nayef Thamer Hussein	Director General of the Finance Department

Note: Dr. Adel Al-Busaisi was assigned to replace Dr. Majeed Al-Alaq, as of March 2021, due to his retirement, and some members were replaced as a result of retirement or change of work locations.

Ministry of Planning Team

#	Name	Job Title
1	Dr. Maha Abdul-Kareem Hammoud	Director General of Human Development
2	Nahida Ahmed Abdel-Jabbar	Senior Research Head, Department of Human Development.
3	Sawsan Abdel-Razzaq Ibrahim	Director of Educational Statistics
4	Zainab Ali Hussain	Chief Statistician at the Central Statistical Organization
5	Moataz Abdul-Sattar Abdul-Jabbar	Department of Human Development
6	Taif Makki Abdul-Khaleq	Senior Researcher at the Department of Human Development
7	Eman Abdel-Wahhab Abdel-Razzaq	Central Statistical Organization

Annex 6.

Ministry of Higher Education Strategies and Programs

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (1-3)	Developing the absorptive capacity of educational institutions	
Strategic option (1-3)	Expansion through the creation of specialized universities and colleges and the expansion of existing universities	
T	Programs	Measurement indicators
1	Establishing specialized scientific universities interested in technological, health and applied sciences that are suitable for the requirements of the labor market.	The number of new specialized universities
2	Establishing technical colleges or institutes affiliated with technical universities in large cities and districts according to the regulations.	Number of technical colleges and institutes established
3	Increasing the capacity of all existing universities vertically and horizontally.	The volume of the increase in absorptive capacity
4	Completing projects prepared for new or young universities.	The number of completed projects
5	Complete the process of rehabilitation and reconstruction of educational institutions damaged by terrorist operations.	The number of completed projects
6	Developing scientific laboratories, specialized workshops and research centers to keep pace with the planned increases in the annual number of students accepted according to the needs of the labor market as well as technical and conceptual progress.	The number of laboratories and workshops developed
7	Coordination with the relevant authorities to resolve issues of land allocation to the Ministry of Higher Education and associated ownership.	The number of land-titles allotted held
8	Cooperation and partnerships with local government institutions (Ministry of Health, Ministry of Agriculture, Ministry of Oil, and others) to support the relevant specializations and the labor market.	The number of cooperation programs
9	Coordination with international donors (governments and institutions) and national support to establish a support fund for higher education under the title "Establishment of Iraqi Educational University City".	The number of international donors, establishment of the Higher Education Support Fund and completion rate
10	Preparation of programs for efficient use of the resources available to universities, including study rooms, laboratories and human resources in terms of scheduling and study hours.	Rate of program preparation

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (1-3)	Developing the absorptive capacity of educational institutions	
Strategic option (2-3)	Developing private educational institutions through opportunities which meet labor market needs and offer rational geographical and spatial distribution	
T	Programs	Measurement indicators
1	Creation of private colleges and universities through announced opportunities that meet the needs of the labor market and offer rational geographical and spatial distribution.	The number of new institutions
2	Opening branches of high-profile foreign universities in high-ranking scientific disciplines.	The number of new branches
3	Expanding and developing the infrastructure of existing private universities and colleges so that they comply with the applicable law and the standards drawn up.	The scale of expansion and development
4	Preparing a program for efficient use of the resources available to private universities and colleges, including classrooms, laboratories, and human resources in terms of scheduling and hours of study.	Rate of program preparation

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (1-3)	Developing the absorptive capacity of educational institutions	
Strategic option (3-3)	Set up and run integrated partnerships for cultural exchange with developed countries to increase university scholarships and fellowships for Iraqi students in rare specializations for postgraduate studies	
T	programs	measurement indicators
1	Determining rare specializations.	Determining specializations
2	Concluding memoranda of cooperation with international organizations to establish integrative scientific partnerships that allocate study chairs between the two parties.	The number of posts created for both parties
3	Holding conferences and workshops between Iraqi universities and high-profile international universities to present educational systems, specialized departments and scientific research papers.	Number of conferences and workshops
4	Developing the scholarship system for gifted students (graduates of gifted schools) by allocating places for them to study at prestigious international universities for bachelor's, master's, and doctoral degrees.	Special scholarship system The number of gifted students on scholarships

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (2-3)	Developing the absorptive capacity of educational institutions	
Strategic option (1-3)	Adopting internationally accredited educational systems for some of the leading public universities for the purpose of evaluation and universality	
T	programs	measurement indicators
1	Applying internationally approved systems at the level of majors or universities	Number of systems applied
2	Digitization of curricula for all majors	The number of digitized curricula
3	Developing a digital student platform for university admission	Preparing the platform

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (2-3)	Diversity in the patterns and programs of higher education	
Strategic option (2-3)	Adopting and consolidating electronic and integrated education in Iraqi universities	
T	Programs	Measurement indicators
1	Issuing governing legislation that allows the adoption of electronic education and blended learning.	Issuing legislation
2	Establishing a national center of electronic education in selected specialized universities for performance evaluation, impact analysis and development.	Number of centers
3	Establishing advanced virtual lecture halls in public and private universities.	Number of virtual units
4	Adopting a contemporary teaching policy and electronic education and preparing teachers to implement it.	Percentage of teachers who can deliver modern e-learning
5	Creating departments for integrated electronic education in the continuous education centers in all Iraqi universities.	The number of departments created
6	Preparing electronic platforms for Iraqi universities, compliant with international norms and standards, for the delivery of electronic university education.	Number of electronic platforms prepared

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (2-3)	Diversity in the patterns and programs of higher education	
Strategic option (3-3)	Support for and development of continuing education centers and programs	
T	Programs	Measurement indicators
1	Developing a program to develop and support continuous education centers to impart technical and vocational skills.	Developing programs
2	Coordinating with ministries to organize real experiences and field visits that make students aware of real practical and skill aspects and the labor market.	Number of visits

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (3-3)	Develop university admission policies	
Strategic option (1-2)	Reviewing and amending legislation, including that governing university admission policy, in a way that ensures the provision of equitable opportunities to all prospective students qualified for admittance to suitable specializations	
T	Programs	Measurement indicators
1	Legislating for new instructions for the admission of students with higher degrees to colleges of the medical group according to the real absorptive capacities and needs of the labor market.	Issuing instructions
2	Reviewing and amending the health criteria for admission to universities.	Review system; amend the regulations
3	Reviewing and amending the evening study instructions for consistency with the requirements of the labor market and economic feasibility.	Review system Amend the regulations
4	Reviewing and amending examination instructions.	Review system; amend the regulations
5	Review and amend summer training instructions.	Review system; amend the regulations
6	Reviewing and amending the legislation, instructions, regulations and systems which the Ministry of Higher Education and Ministry of Education are currently working on to simplify the procedures for students and graduates to obtain documents.	Review instructions or legislation Amend instructions or legislation

The first strategic issue		Internal efficiency
Objective	Improving internal capability of higher education institutions and providing higher education opportunities for all who are eligible and interested	
Strategic objective (3-3)	Develop university admission policies	
Strategic option (2-2)	Increasing the percentage of students enrolled in primary and higher studies	
T	Programs	Measurement indicators
1	Increasing students enrolled in technical specializations in institutes.	The amount of increase in net enrollment ratio
2	Sustaining and increasing the net enrollment rate in higher education for undergraduate studies.	The amount of increase in net enrollment ratio
3	Sustaining and increasing the net enrollment rate in higher education for postgraduate studies.	The amount of increase in net enrollment ratio
4	Adopting the goals of the 2030 Sustainable Development Agenda in developing enrollment and admission policies.	Advanced enrollment and admission policies
5	Adopting the Technical and Vocational Qualifications Framework System approved by Council of Ministers Resolution No. 99 of 2021 to develop the policy of admission, qualification and development of technical education.	Developed admission and qualification policy
6	Harmonizing educational institutions (in terms of infrastructure and human resources) which take into account the needs of persons with special needs in a manner consistent with their situation and health needs according to the approved standards.	Developing standards that take into account people with special needs, the quantity of material and human resources for people with disabilities, and the number of buildings designed and constructed to accommodate people with disabilities
7	Responding to the needs of other vulnerable groups (illiteracy, adolescent girls, rural women, displaced persons, refugees, families below the poverty line, orphans, child labor).	The number of students benefiting from undergraduate study. The number of students benefiting from postgraduate study. Annually adopted controls.

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (1-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (1-5)	Developing the administrative leadership system	
T	Programs	Measurement indicators
1	Developing administrative and financial skills.	Passing advanced training programs
2	Developing legal skills.	Passing advanced training programs
3	Developing specialized skills.	Passing advanced training programs
4	Developing electronic management skills.	Passing advanced training programs
5	Enhancing the culture of participation in decision-making.	Number of training programmes, the percentage of participants, distribution of a - questionnaire designed to find out the extent to which the culture of participation is promoted
6	Updating the mechanisms for selecting administrative and academic university leadership.	Updated selection mechanisms

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (1-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (2-5)	Developing the organizational structures of HEIs	
T	Programs	Measurement indicators
1	Reviewing, assessing and improving the organizational structures of education institutions according to criteria of distinction, efficiency and effectiveness.	Study the state of the structures—prepare an improvement plan implementation rate
2	Developing human resources consistently with change in organizational structures.	Conduct a study that determines training needs; Number of implemented training programs
3	Independence of universities.	Number of independent universities

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (1-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (3-5)	Activating the role of strategic planning of higher education institutions	
T	Programs	Measurement indicators
1	Developing a culture of strategic thinking and planning in higher education institutions.	Number of training programmes- The percentage of participants- A questionnaire designed to find out the extent to which culture is entrenched
2	Strategic planning to develop higher education institutions in the field of electronic knowledge.	The existence of strategic plan; Plan implementation rate
3	Strategic planning of the role of higher education institutions in achieving sustainable development.	Having a strategic plan; Plan implementation rate
4	Reviewing the strategic plans of universities and aligning them with the strategic plans for higher education.	The number of reviews completed
5	Strategic planning for crisis management.	Having a strategic plan Plan implementation rate

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (1-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (4-5)	Enhancing the participation of stakeholders in decision-making	
T	Programs	Measurement indicators
1	Activating the role of internal and external stakeholders.	Representatives of stakeholders in the consultative board and university and college councils
2	Developing and advancing the capabilities of stakeholders.	Number of training programmes

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (1-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (5-5)	Enhancing procedures underpinned by integrity, transparency and accountability	
T	Programs	Measurement indicators
1	Raising awareness about the culture of integrity, transparency and accountability.	Number of training programs- The percentage of participants- A questionnaire designed to find out the extent to which culture is entrenched
2	Planning to improve procedures underpinned by integrity, transparency and accountability according to corruption perception index.	Preparing an improvement plan
3	Mutual auditing among higher education institutions.	Mutual audit schedule audit completion
4	Developing a system a system that is immune to external interference in the institution's decisions, that prevents pragmatist effects and ensures the independence of administration.	Existence of a system that guarantees the independence of the decision

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (2-2)	Enabling higher education institutions to achieve their goals efficiently and effectively	
Strategic option (1-2)	Keeping pace with modern technological developments in the field of electronic governance	
T	Programs	Measurement indicators
1	Using information technology in administrative, financial and educational systems to establish educational institutions that are capable of coping with challenges.	Used IT
2	Developing computer centers in universities and upgrading their tasks to include new training and robust research activity according to global criteria.	Preparing an improvement plan Completion rate

The second strategic issue		Governance of Higher Education Institutions
Objective	Higher education institutions that are able to achieve their objectives efficiently and effectively and improve their performance by following effective strategic plans through the application of rational management that is based on integrity, transparency, accountability and participation in decision-making by all concerned parties	
Strategic objective (2-2)	Achieving integration among the parties relating to higher education institutions	
Strategic option (2-2)	Applying electronic governance to achieve flexibility and facilitate procedures through electronic connection to all procedures and services in higher education institutions	
T	Programs	Measurement indicators
1	Developing a system to present all procedures and tasks explicitly, show all works clearly, and permanently disclose the works to seek to achieve the objectives of the institution according to the principle of transparency.	Existence of an advanced system
2	Re-engineering academic and administrative operations, the accurate distribution of tasks; and the measurement of time and movement in performance according to the principles of governance.	Perform re-engineering process

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (1-3)	Achieving national and international academic accreditation by higher education institutions	
Strategic option (3-3)	Obtaining program accreditation by academic programs in scientific departments	
T	Programs	Measurement indicators
1	Applying the criteria of specialized accreditation for all the scientific departments and their academic programs in all Iraqi public and private universities.	From five to 10 academic programs obtain accreditation for each university every three years, and the same for private education

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (2-3)	Entry of higher education institutions into university classifications and achieving advanced positions in them	
Strategic option (1-2)	The entry of higher education institutions into important and approved world classifications	
T	Programs	Measurement indicators
1	Developing the data collection and analysis skills needed for the evaluation of educational quality.	Develop 2-5 teams per year
2	The entry of Iraqi universities into the global classifications of universities (Shanghai, Thames, QS, Green Metrics, URAP, Webometrics) and promoting their ranking.	Helping universities to rise through the international rankings

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs consistently with the best international standards in order to achieve excellence	
Strategic objective (2-3)	Entry of higher education institutions into university classifications and achieving advanced positions in them	
Strategic option (2-2)	Creating a spirit of competition and promoting a culture of distinction among universities	
T	Programs	Measurement indicators
1	The national classification of Iraqi universities.	Developing a comprehensive website for classification management -Announcing the results of the classification on a specific annual date
2	Benchmarking against world-class universities. 10% of the Iraqi university annual	10% of the Iraqi university annual
3	Spreading the culture of quality and practicing total quality management.	Obtaining an internal and external audit certificate for quality institutions at the university (five universities annually))

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (3-3)	Quality assurance and improving the effectiveness of higher education institutions	
Strategic option (1-5)	Developing curricula, students' skills, methods of teaching and learning, assessment and evaluation	
T	Programs	Measurement indicators
1	Describing the academic program.	30% of the academic programs for each university to be completed by the end of the year 2025
2	Assessing academic programs according to external evaluation methodology.	Evaluating five programs for each university during the first three years, and doubling the number in subsequent years provided that the actual implementation of the program begins in 2023
3	Adopting contemporary teaching methods that enhance the process of teaching and learning, developing the intellectual and creative research of students.	Increasing the number of teachers enrolled in advanced developmental courses from 15% to 33% for one scientific department in each university, starting from 2022 and renewed three years after the first session
4	Developing undergraduate and postgraduate curricula according to world criteria	Updating educational programs by more than 5% every year according to changes in the scientific specialization on the curricula

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (3-3)	Quality assurance and improvement of the effectiveness of higher education institutions	
Strategic option (2-5)	Upgrading the quality of electronic education	
T	Programs	Measurement indicators
1	Applying obligatory quality criteria to electronic education.	Preparing 40% annually of the standards requirements for the years 2021–2023
2	Applying blended learning, with appropriate systems.	Voluntary shift towards blended learning by 40% of educational programs for each university annually
3	Developing the terms of reference of electronic education and blended education on the basis of globally approved mechanisms and experiences to monitor and develop the quality of electronic education.	The number of approved mechanisms and experiences

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (3-3)	Quality assurance and improvement of the effectiveness of higher education institutions	
Strategic option (3-5)	Improving the quality of educational laboratories	
T	Programs	Measurement indicators
1	Improving the quality of teaching laboratories, ensuring that safety and security requirements are met.	10% of educational labs in each university annually

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (3-3)	Quality assurance and improvement of the effectiveness of higher education institutions	
Strategic option (4-5)	Distinguished, trained and developed human resources	
T	Programs	Measurement indicators
1	Developing teaching capabilities.	High levels of performance according to the annual performance appraisal. Number of trainees increased compared to the plan
2	Developing training packages for courses.	Number of training packages developed according to scientific specialization
3	Developing systems of incentives, rewards and awards for excellence.	Increasing the number of incentive systems, rewards and excellence awards
4	Developing criteria for assessing general performance of employees.	Periodically updating the general performance appraisal formats

The third strategic issue		Quality and accreditation
Objective	Ensuring and enhancing educational quality and controlling its outputs in conformity to the best international standards in order to achieve excellence	
Strategic objective (3-3)	Quality assurance and improvement of the effectiveness of higher education institutions	
Strategic option (5-5)	Controlling the quality of the educational process according to inputs and outputs	
T	Programs	Measurement indicators
1	Establishing university admission examinations for each scientific specialization.	Issuing a ministerial order for the admission exam
2	System of measurement and assessment.	Standard exam results analysis

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (1-6)	Providing an environment that supports scientific research in Iraqi universities	
T	Programs	Measurement indicators
1	Providing the necessary equipment and requirements for scientific research.	Increasing the amount of equipment and scientific research supplies
2	Providing a suitable research environment and allocating a suitable number of hours for scientific research according to the nature of research.	Researchers' satisfaction ascertained through an electronic questionnaire
3	Adopting international criteria in the keeping of dangerous materials and disposal of hazardous chemical and biological waste.	No accidents

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (2-6)	Developing cooperation mechanisms in the field of scientific research between Iraqi universities and their peers outside Iraq	
T	Programs	Measurement indicators
1	Developing memoranda of understanding and mechanisms of cooperation and pairing with high-profile foreign universities and research institutions.	Increasing the powers granted to universities
2	Developing the "visiting researcher" program to host foreign researchers.	Increasing the number of foreign researchers in universities
3	Establishing research academies within the joint specializations between Iraqi and foreign universities.	The number of foreign researchers in research groups

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (3-6)	Developing the skills of researchers	
T	Programs	Measurement indicators
1	Conducting programs involving accredited courses for instructors by experts and institutions specialized in scientific research methods and publication of research papers.	Number of courses held by foreign experts
2	Supporting the participation of researchers in scientific conferences held outside Iraq.	Number of reports submitted
3	Appreciating the efforts exerted by researchers by allocating annual awards to Iraqi universities entitled "Award of Distinction for Evaluators of Research Papers, Award of Distinction for The Most-Cited Research Paper, Award of Distinction for Authored Books, Award of Distinction for Patents, and Award of Distinction for Evaluator of Research Papers".	Number of awards granted
4	Requiring postgraduate students to study the following topics, as separate additional but obligatory course components: scientific research methodology and ethics, and methods of research writing and publication.	Add items to the study plan
5	Adding scientific research methodology principles to the undergraduate curriculum as a separate subject or as items within the obligatory final-year research project.	Add items to the study plan
6	Developing programs for "Full-Time Scientific Study Outside Iraq, Post-Doctorate Study, and Research Fellowships" by increasing funding, facilitating the procedures and reducing conditions and obligations.	Increase the number of beneficiaries
7	Re-activating the program of "Research Scholarships" for postgraduate students to complete their papers in foreign universities.	Increase the number of beneficiaries

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (4-6)	Developing the work of research centers and mechanisms for providing research services to all institutions	
T	Programs	Measurement indicators
1	Developing the infrastructure of research centers and providing them with all required laboratory equipment and supplies relating to scientific research.	Increasing the number and capabilities of research laboratories
2	Developing the policies and programs for marketing scientific research papers.	Number of research projects with commercial application
3	Developing and planning for the adoption of scientific research applicable to solving problems encountered in other ministries and sectors.	Number of research projects used to solve a problem in the ministries -Number of research projects applicable to other sectors
4	Seeking the help of experienced retired professors in research centers.	The number of retired professors

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (5-6)	Enhancing the requirements of scientific research	
T	Programs	Measurement indicators
1	Enhancing the financial allocations of university libraries to upgrade the tasks assigned to them.	Increase financial allocations
2	Adopting electronic systems in the provision of services in university libraries in a way that would achieve the requirements of electronic education.	Availability electronically of services in university libraries
3	Developing the Iraqi virtual library by increasing subscription to high-profile fee-paying journals and databases (such as Scopus and Clarivate) and including all researchers in their services.	Increase the number of journal subscriptions
4	Applying the UNESCO recommendations on open learning by establishing a unified electronic portal to archive all the types of scientific products of universities and making them electronically accessible.	Increase the volume of archived material

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (1-4)	Developing the skills of researchers in Iraqi universities and increasing their high-quality research productivity	
Strategic option (6-6)	Adopting and developing scientific research methodology in Iraqi universities	
T	Programs	Measurement indicators
1	Adopting harmonized requirements for academic references applicable to research methodology in both scientific and humanitarian specializations in undergraduate and postgraduate studies.	Naming scientific references
2	Preparing a guide to international criteria of scientific research methodology for the medical group specializations (such as the application of the Helsinki Principles).	Completing the guide

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (2-4)	Funding scientific research to upgrade international publication in high-profile world databases	
Strategic option (1-5)	Increasing the funding of scientific research	
T	Programs	Measurement indicators
1	Amending the instructions governing the Higher Education Fund to increase the proportion of funding earmarked for scientific research.	Modifying the instructions of the Education Fund so that there is a sufficient percentage for the Scientific Research Fund
2	Obtaining a decision or legislation that increases the allocations to Iraqi universities for scientific research.	Cabinet decision amending the Federal Budget Law
3	The contribution of the private sector to funding of scientific research.	Funding ratio at the level of each university; the number of accredited programs at the level of each university

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (2-4)	Funding scientific research to upgrade international publication in high-profile world databases	
Strategic option (2-5)	Allocating a certain proportion of the annual financial budget for scientific research	
T	Programs	Measurement indicators
1	Determining a specified proportion of the estimated budget sent to the Ministry of Finance for the purposes of scientific research.	The percentage is proportional to the annual increase in the number of researchers in universities
2	Determining a specified proportion of the estimated budget sent to the Ministry of Finance for the research papers of postgraduate students.	The percentage is proportional to the annual increase in the number of postgraduate students

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (2-4)	Funding scientific research to upgrade international publication in high-profile world databases	
Strategic option (3-5)	Establishing research academies and centers in various scientific specializations	
T	Programs	Measurement indicators
1	Increasing the number of research units and centers in various scientific specializations.	Increasing the number of research units and centers
2	Encouraging the participation of foreign researchers within research academies in Iraqi universities and providing the financial support and necessary facilitation.	Increasing the number of foreign researchers in research groups

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (2-4)	Funding scientific research to upgrade international publication in high-profile world databases	
Strategic option (4-5)	Encouraging authentic international publication	
T	Programs	Measurement indicators
1	Issuing an annually updated guide to high-profile scientific journals.	Issuing the guide
2	Encouraging postgraduate students to publish in high-profile scientific journals and awarding marks for each published paper.	Increasing publication in respected journals
3	Directing researchers to write the name of the university correctly in published papers.	Reducing failure to publish research in journals or erroneous publication
4	Encouraging researchers to publish scientific books in international publishing houses and to obtain international and approved patents in high-profile databases by providing financial and moral support.	Increase the publication of books in international publishing houses
5	Updating the protocol for professional development, linking promotion to quality of published research, especially for attainment of professorship.	Increasing research published in reputable journals with a high impact factor

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (2-4)	Funding scientific research to upgrade international publication in high-profile world databases	
Strategic option (5-5)	Supporting Iraqi scientific journals to obtain accreditation in high-profile databases	
T	Programs	Measurement indicators
1	Developing a plan for Iraqi journals to become international journals and obtain accreditation in high-profile databases.	Increasing the number of Iraqi journals approved in high-profile databases
2	Establishing an Iraqi publishing house for the publication scientific journals or to facilitate membership of Iraqi journals in high-profile world publishing houses.	Increasing the number of foreign readers of Iraqi journals

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (3-4)	Protecting intellectual property rights in Iraqi universities	
Strategic option (1-2)	Applying the legal procedures relating to the protection of the rights of publication and authoring	
T	Programs	Measurement indicators
1	Updating the legal provisions relating to protection the rights of intellectual property, authorship and publication to meet the relevant international criteria.	Updating legal articles
2	Issuing a list of penalties relating to violation of the rights of intellectual property and plagiarism.	Issuing a list of penalties
3	Setting up standing committees specialized in considering complaints about violation of the rights of intellectual property and plagiarism.	Formation of committees

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (3-4)	Protecting intellectual property rights in Iraqi universities	
Strategic option (2-2)	Reducing violation of the rights of intellectual property and plagiarism	
T	Programs	Measurement indicators
1	Issuing annually updated guidelines that define the rights of intellectual property and plagiarism.	Issuing the guidelines
2	Holding introductory workshops about the rights of intellectual property and plagiarism.	Training all researchers and postgraduate students
3	Developing the mechanisms relating to the inspection of similarity.	Training researchers and postgraduate students Providing appropriate funding for students and teachers

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (4-4)	Attracting Iraqi scientists from abroad	
Strategic option (1-4)	Targeting rare and new scientific specializations	
T	Programs	Measurement indicators
1	Preparing questionnaires to explore the universities' need of Iraqi scientists abroad according to their specializations and scientific output.	Completion of questionnaires
2	Establishing an electronic portal and database for Iraqi scientists outside Iraq according to their specializations and scientific achievements.	Creating the database
3	Adopting criteria to identify Iraqi scientists targeted for recruitment.	Issuing the required standards
4	Exploring the possibility of benefiting from the expertise of Iraqi scientists who do not wish to return through their participation on the editorial boards of journals, supervising and discussing theses and dissertations, training on surgical operations, and so forth, by using electronic means of communication.	Preparing scientists working in universities

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (4-4)	Attracting Iraqi scientists from abroad	
Strategic option (2-4)	Determining material and moral privileges for returning scientists.	
T	Programs	Measurement indicators
1	Legislating to ensure exceptional privileges and salaries to the returning scientists according to their scientific output.	Satisfaction of returning scholars with financial benefits
2	Giving the returnees job positions in research and development directorates in universities and research centers in proportion to their scientific output.	Returning scholars taking up posts

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (4-4)	Attracting Iraqi scientists from abroad	
Strategic option (3-4)	Legislating for regulations and instructions to ensure the rights of returning scientists	
T	Programs	Measurement indicators
1	Legislating to attract returning scientists by facilitating travel with retention of their acquired nationalities.	Issuing relevant laws and instructions
2	Legislate to guarantee that returning scientists will retain their intellectual property rights to their scientific products and scientific titles.	Issuing relevant laws and instructions
3	Legislate in a way that would allow the returning scientists to market their scientific products inside and outside Iraq.	Issuing laws guaranteeing rights

The fourth strategic issue		Scientific research and innovation
Objective	High-quality scientific research and innovation that contributes to driving the economy and achieves the goals of sustainable development	
Strategic objective (4-4)	Attracting Iraqi scientists from abroad	
Strategic option (4-4)	Determining the funding budget for the work of returning scientists	
T	Programs	Measurement indicators
1	Developing mechanisms to calculate annual financial needs for the work of returning scientists.	Issuing mechanisms and controls
2	Developing guarantees to provide the appropriate financial benefit to the returning scientists in a way that would guarantee the continuity of their stay in Iraq.	Returning scientists continue to work for extended periods
3	Adopting central funding from the ministry to cover the remunerative, scientific and financial requirements for the work of distinguished returning scientists.	Paying financial dues to returning scholars

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (1-3)	A robust academic institution capable of building sustainable strategic partnerships with entrepreneurs and industry	
Strategic option (1-4)	Creating scientific and practical links among higher education institutions and the various stakeholders	
T	Programs	Measurement indicators
1	Creating business incubators in universities.	Number of business incubators
2	Entrepreneurship projects.	The number of entrepreneurship projects completed
3	Cooperation mechanism between the university and entrepreneurial business involving continuous training and skill development to the mutual benefit of both parties.	-Develop a mechanism for cooperation -Number of beneficiaries from the business sector - Number of beneficiaries from the university
4	Periodic review of programs, curricula, teaching methods, assessment and evaluation mechanisms and any consequent modification that consolidates the educational process while keeping pace with new developments.	Carrying out the periodic review The number of programs and curricula developed
5	Developing specialized courses with internationally accredited certificates that are provided by the private sector in favor of the university through agreements and memoranda of understanding with the private sector.	-Signing an agreement with private sector institutions -Number of supporting institutions - Number of beneficiary students

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (1-3)	A robust academic institution capable of building sustainable strategic partnerships with entrepreneurs and industry	
Strategic option (2-4)	Developing and amending legislation in a way that would ensure effective communication between higher education institutions and the labor market	
T	Programs	Measurement indicators
1	Reviewing the laws, regulations and instructions which organize the relationship between the educational institution and the labor market.	-Completion of the review -Number of amended laws

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (1-3)	A robust academic institution capable of building sustainable strategic partnerships with entrepreneurs and industry	
Strategic option (3-4)	Developing the organizational structure in universities to interact with the business sector (the labor market)	
T	Programs	Measurement indicators
1	Empowering and transferring rehabilitation, recruitment and follow-up departments to vocational development centers for graduates.	- Preparing a plan or program for transfer; Percentage of implementation of the plan or program
2	Establishing an electronic platform that links universities with the labor market through rehabilitation and employment departments.	Establishing the electronic platform during the first three years of the plan; Update the data periodically
3	Establishing an observatory and database for the needs and requirements of the labor market through a comprehensive field study conducted by the rehabilitation, employment and follow-up departments to survey the labor market.	-Establishment of the observatory and database -Determine the requirements of the labor market -Directory of majors required annually in the business sector

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (1-3)	A robust academic institution capable of building sustainable strategic partnerships with entrepreneurs and industry	
Strategic option (4-4)	Meeting the needs of the economic and social development plans	
T	Programs	Measurement indicators
1	Aligning the national qualifications framework of vocational and technical education with labor market requirements.	Achieving the national qualifications framework that is appropriate to the requirements of the labor market
2	Linking the educational policy of higher education institutions with human development needs.	Linking of educational policy achieved
3	Identifying and reducing the gap between the output of the higher education institutions and labor market requirements.	Determine the gap; Number of seminars and workshops -Beneficiaries
4	Preparing a vocations guide containing accurate descriptions of current and future occupations.	Issuing the guide

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (2-3)	An academic instructor who keeps pace with global development	
Strategic option (1-1)	Upgrading the field skills of the university instructor	
T	Programs	Measurement indicators
1	Continuous training of the teaching staff on modern training curricula according to labor market requirements.	-Develop a modern training program -Program implementation rates
2	Activating field practice and coexistence of teachers in local and international entrepreneurship to enhance the skills of teachers and their knowledge about labor market changes.	Setting controls for field practice and coexistence; the number of beneficiary teachers

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (3-3)	A distinguished and effective generation which meets labor market requirements	
Strategic option (1-2)	A graduate who is capable of understanding and dealing with the dynamics of the labor market and entrepreneurship	
T	Programs	Measurement indicators
1	Activating the work of consultative committees in every university according to the scientific specializations of the university.	The number of activated advisory committees
2	Introducing a program of pioneering graduation projects for students with an investment aspect, and focus on bringing the private sector on board to invest in these projects.	Existence of a mechanism to encourage students to submit pilot projects Number of entrepreneur graduation projects Number of projects invested in by the private sector
3	Motivating students to be creative and innovative by holding scientific competitions and contests.	Number of annual competitions
4	Investing in the capabilities and resources of universities by providing workshops and productive laboratories which are open throughout the year to enhance collective work, productivity, and applied specialized skills, enabling students to use their time positively.	Develop a plan or programme; Percentage of implementation of the plan or program
5	Keeping pace with global trends by using modern information technology in the process of training the students and developing and advancing their skills.	Number of training courses Number of beneficiary students Number of beneficiary graduates
6	Developing the skills of students, including their experiences and applied skills, through local, regional and international training of the students and internship programs according to specialization.	The existence of an agreement with regional and international companies Number of courses Number of beneficiary students
7	Transformative training of graduates.	Number of courses Number of beneficiaries
8	Mobile training of students and graduates.	Number of courses Number of beneficiaries
9	Measuring the quality of outputs in light of specific indicators for the purpose of reviewing and improving the teaching process.	Preparing an output quality scale and updating it annually

The fifth strategic issue		External efficiency
Objective	Higher education institutions that enhance students' applied skills and qualify them to compete in the labor market	
Strategic objective (3-3)	A distinguished and effective generation which meets labor market requirements	
Strategic option (2-2)	A responsible and proactive graduate who serves society and works toward positive change	
T	Programs	Measurement indicators
1	Enhancing student public service initiatives and incorporating them into university requirements according to the nature of every specialization.	Setting controls for initiatives as a requirement for the academic stage; establishing a mechanism for evaluating students
2	2. Encouraging specialized, cultural and humanitarian extracurricular activities.	Students' participation in at least one extracurricular activity per academic year
3	3. Activating the role of educational counselling to entrench the concepts of good citizenship and good governance.	Number of training courses Number of trained teachers

